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The East and South Asian Bathroom Product Markets

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Overview

BRG Building Solutions is a specialised market research consultancy serving the global building technologies industry. Combining unparalleled industry experience with expert research and analysis, our team delivers lasting results to hundreds of companies worldwide.

Our core expertise lies in our coverage of the bathroom, kitchen, HVAC, and other building products/technologies. Throughout a history spanning over 40 years, we have assisted industry players in identifying and realising growth opportunities through a combination of regular market monitoring and ad-hoc consultancy services.

BRG's unique Enterprise Database is at the centre of our offering. This is a comprehensive tool and very detailed source of historic, current and forecast market trends and market share data (by holding and brand) on a country comparable basis. The database provides an essential tool to our clients as a platform to support their strategic decision making.

BRG Building Solutions is a genuinely global company, staffed with individuals from varied backgrounds and nationalities, working collaboratively from four offices across four countries. Combining a high level of industry experience with expert research and analysis, our team delivers lasting results to hundreds of companies worldwide.

Our market coverage is also global, reflecting the nature of the industries we serve. Currently, 68 individual markets are covered annually in detail across the following key geographical regions (and number of countries):

- Africa (3)
- Australasia (2)
- East & Southeast Asia (6)
- Central America (4)
- Central Asia (3)
- Europe (31)
- South Asia (1)
- Middle East (7)
- North America (3)
- South America (5)
- Sub-Saharan Africa (3)

BRG has its own in-house dedicated modelling and forecasting unit. In cooperation with clients, we have since extended our offering to include market sizing for countries outside the traditional geographic reach of our main research

programme, via global market models

Product Coverage

Since the 1980s, BRG's Bathroom & Kitchen programme has provided industry players with insightful analysis and evidence-based information. In addition to detailed data on trends, values, end use, technical segmentations, market shares and distribution, our programme also includes perceptive text commentary, enabling our clients to easily identify and understand market characteristics and the latest industry developments.

BRG's programme has evolved steadily in terms of both geographic scope and product coverage and continues to do so each year. Each country report follows a uniform structure, to ensure that the data can be readily compared across regions and countries.

The product coverage has also expanded to cover virtually all bathroom, sanitary and kitchen related products, used in both residential and non-residential buildings, including:

Baths and Shower Trays

- Baths: cast iron, steel and synthetic
- Hydromassage baths
- Shower Trays: ceramic, steel/ cast iron and synthetic
- Shower systems (NAM): shower modules, integrated tub showers

Sanitary Ware

- WCs: wall hung, floor standing 1-piece and squat
- Washbasins: hand washbasins and countertops
- Other CSW: bidets, kitchen sinks, pedestals, shower trays, urinals and cisterns
- Plastic Cisterns: concealed and exposed.

Taps and Mixers ('Faucets'), both sanitary and kitchen:

- By application: basin/sink, bath, bidet, shower, kitchen sinks
- By type: pillar, two head, one head, thermostatic, self-closing, electronic
- By type (Americas): 4in centre set, 8in widespread, electronic, hand spray, pressure, single handle/one-head, self-closing, thermostatic, three handle.

Shower Wall

- Bath screens, conventional shower cubicles, hydrotherapy cubicles, bath & shower surrounds (NAM)

Bathroom Furniture

- Base units, mirror / mirror cabinets and separate cabinets.

Bathroom Accessories

- Soap holders and dispensers
- Towel holders, towel rings and towel bars
- Toilet paper holders and toilet brushes
- Towel and bathrobe hooks
- Toothbrush holders and tumblers
- Shelves, wastepaper bins, shower storage units and tissue holders.

Shower Accessories

- Shower heads, flexible hoses, shower supports, shower panels and shower pipes.

Kitchen Sinks

- Cast Iron, ceramic, enamelled steel, stainless steel and synthetic.

Definitions

Please note that all volumes given in the report refer to the number of units sold, where not indicated otherwise. In-depth product definitions are included in a separate document and available upon request.

All prices, both manufacturer and retail, exclude value-added-tax (VAT) and installation costs.

Methodology

The methodology behind the data remains true to our origins in that the industry participates in the research process, in many cases by providing BRG Building Solutions with annualised volume sales by product category per country and contacts for detailed expert interviews in all countries covered.

Our approach follows a multi-pronged strategy, gathering data inputs and information from many different sources. A further USP of BRG is the fact that we live and breathe the markets in which we specialise (bathroom, kitchen, plumbing & HVAC) and are in constant contact with manufacturers, distributors and other key influencers all of the time during our repeat annual research programme. The markets are built via a bottom-up approach. Extensive primary and secondary research complement the picture and provide all necessary insight for many

additional segmentations shown in the reports, like e.g. price segmentations, technical and end-use segmentations.

The reports are constantly evolving and refined based on industry feedback, and always reflect the latest developments in the industries.

All research is conducted in-house with a team of around 30 individuals who speak 20 different languages between them. As a result, we can transfer that knowledge and experience into services designed to assist stakeholders in understanding the opportunities available to them, at a fraction of that charged by the larger and less nimble global consultancies. BRG operates a policy of direct management of the team required to conduct the regular research work and avoids subcontracting its market studies to outside agencies.

The analysis has grown in complexity as the number of drivers affecting the markets grows, like e.g. economic uncertainty, construction cycles, new technologies, legislation. As a result, BRG Building Solutions considers all drivers through a holistic approach, recognising that products, end use segments and countries are all inter-linked.

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CHINA

BACKGROUND

THE EAST AND SOUTH EAST ASIAN BATHROOM PRODUCT MARKETS



Geography and Demographic

China, officially the People's Republic of China (PRC), is a country located in eastern Asia along the coastline of the Pacific Ocean. China is the world's third-largest country, covering a land area of 9.6 million km², and extends about 4,000 km from north to south and 4,800 km from east to west. It has land frontiers extending a total of 20,000 km and shares borders with North Korea, Mongolia, Russia, Kazakhstan, Kyrgyzstan, Tajikistan, Afghanistan, Pakistan, India, Nepal, Bhutan, Myanmar, Laos, and Vietnam.

Administratively, China is divided into 23 provinces, five autonomous regions and four directly controlled municipalities (Beijing, Tianjin, Shanghai and Chongqing) in six regions: North, Northeast, East, Mid-South, Southwest and Northwest; and two special administrative regions (SAR): Hong Kong and Macau. The Republic of China (ROC) now governs Taiwan, as a province of China. Mainland China refers to the six administrative regions and excludes Hong Kong SAR, Macao SAR, and Taiwan province.

Provinces, autonomous regions, and municipalities are further divided into prefectures and counties. Cities and prefectures are at the same administrative level. There were 653 cities, including four municipalities and 293 prefecture-level cities, and 387 county-level cities, in China by the end of 2019, which are graded tier one to tier six by population, GDP and importance (excluding Hong Kong, Macao and cities in Taiwan). In 2019, there were 2 counties updated into cities, representing the continuation of urbanization. The number of prefecture-level cities remained the same as it was in 2018. There are no official statistics on the city by tier/grade in China. In 2019, YICAI (No. 1 Business Network) published a list and ranking of 337 cities by tier one to tier five cities according to its own survey, including Beijing, Shanghai, Shenzhen and Guangzhou as tier one cities; 15 new tier one cities such as Chengdu, Hangzhou, Chongqing, Wuhan, Xi'an, Suzhou, Tianjin, Nanjing, Changsha, Zhengzhou, Dongguan, Qingdao, Shenyang, Ningbo, and Kunming; 30 tier two cities, 70 tier three cities, 90 tier four cities, and 128 tier five cities. The remaining 316 cities are either tier six or non-tier cities. Eastern China is the region with the greatest number of cities and comprises most of China's large cities, while most of the smaller Chinese cities are concentrated in the northwest. Guangdong is the province with the most cities, and Tibet is the region with the fewest.

China is a multi-ethnic country, with Han Chinese as the majority people and 55 other ethnic groups as minority people. Standard Chinese or Mandarin (Putonghua) is the official language, with dialects of Yue (Cantonese), Wu (Shanghainese), Minbei (Fuzhou), Minnan (Hokkien-Taiwanese), Xiang (Hunan), Gan (Jiangxi), Hakka, and other local dialects also officially recognized. Several other languages are also spoken amongst China's minority populations, including Zhuang, Mongolian, Uighur, Kyrgyz, Tibetan, and Korean. Folk religion, Buddhism, Christianity, and Islam are the four main religions in China, although the majority of

the Chinese population are unaffiliated with religion.

China has the largest population in the world, with 1.4 billion persons residing in mainland China in 2019 – an increase of 4.67 million compared to 2018, due to the relaxation of policies on birth control. In 2019, the urban population accounted for 60.6% of the total population, with an increase of 1.02% over 2018, reflecting the continuous urbanization trend.

Residential Dwelling Stock

Overview

There are no official statistics on the Chinese dwelling stock due to the imperfect housing registration systems in China, so the National Bureau of Statistics of China (NBSC) cannot provide such data every year. The government releases some data on the dwellings stock based on the population census every ten years, but the statistics only cover the number of people that participated in the census rather than the whole population, and the latest census was conducted in 2010, therefore, the data from the population census in 2010 cannot reflect the recent development of dwelling stock in China. Recently, the Ministry of Natural Resource and the Ministry of Housing and Urban-Rural Development jointly launched a real estate registration system in 2015, however, the pilot implementation was limited to certain cities such as Beijing, Qingdao, Xiamen, Guangzhou, Nanjing, Shenyang, etc. The government claimed the system has covered 335 prefecture-level cities and 2,808 counties and districts and released 14 million certificates for both residential and commercial real estate; most of these are new buildings, which is far lower than the actual dwelling stock in China. It will take several years to complete a full coverage of real estate systems in China due to the large size of the population and complicated regional and historical reasons.

BRG has based its data on dwelling stock on NBSC's statistics on population, sampling survey statistics on number of family households and population of family households, and average size of dwellings, as well as the statistics and information on average number of dwellings owned by each family household, government public and private renting housings, secondary housings, and floor space of commodity housings unsold from various secondary sources such as Chinese Academy of Social Sciences, the People's Bank of China, and Industry Information (Chanye Xinxì).

In 2019, the overall housing stock of existing dwellings in China was estimated to be about 436 million, of which about 61% are in urban areas and 39% in rural areas due to a larger population in the urban areas than rural areas, and the smaller size of dwellings in urban areas as opposed to in rural areas.

The average size of dwellings is about 110 m² in China, according to the statistics on the number and floor space of dwellings completed by real estate companies in 2018 according to NBSC. The average size of dwellings in rural areas is larger than the size of dwellings in urban areas.

Dwelling Stock by Age

There are no up to date statistics on the age of dwellings in China and the only data available are from the 6th population census in 2010. According to the census, about two thirds of the dwellings were aged less than 30 years: 34% were built after 2000, and 33% were built during 1990 to 1999, representing a young dwelling stock in China comparing with the other countries due to the boom of construction during 1990s to 2010s. The remaining dwellings are considered old and built before the 1980s: 21% were built in the 1980s, 7% were built in the 1970s, 2% were built in the 1960s, 1% were built in the 1950s, and 1.2% were built before 1949 when the People's Republic of China was established. Many of the old dwellings were gradually demolished and replaced by new flats or apartments; therefore, the number of old dwellings should be much smaller now.

Guangdong, Sichuan, Henan, and Shandong have more dwellings built after the 1990s due to the rapid growth of new housing construction and large population. The new housing construction in Shenzhen is believed to have been the fastest in the 1990s and the 2000s because it is a new city with many migrants.

Dwelling Stock Type

The type of dwellings in China is complicated due to the diversified culture by region and is simply classified into low block/building *Ping Fang*, two- to three-story blocks, four- to six-story blocks, seven- to nine-story blocks, and blocks with 10 or more stores, according to the population census in 2010. About 47.2% of the dwellings are *Ping Fang*, followed by two- to three-story blocks (25.4%), four- to six-story blocks (18.3%), seven- to nine-story blocks (6.1%), and 10 or more story blocks (2.8%) according to the census.

However, this situation has changed greatly now due to the widespread demolishing of *Ping Fang* and low story blocks in urban areas, which were replaced by multi-story blocks and high rising blocks in large residential complex during the 1990s to the 2010s, with the growth of population, boom of housing construction, transformation of urban infrastructure, urbanization, and improvement of living standard and income.

Most of the dwellings in urban areas are apartments or flats in multi-story blocks or high-rise buildings. These multi-story blocks are normally former Soviet Union-style buildings, while many high-rises are between 10 and 30 floors high. Most of China's multi-story blocks were built from the 1970s to the 1990s. They are presently either facing demolition for new construction or are transformed with energy-conserving heating systems and insulation materials. High-rise residential buildings were mostly built throughout the 2000s and 2010s. There is a tendency to build more and more very large complexes with more than 10 blocks over 10 to 30-stories (or even 40-stories) in cities for the last decade, indicating a very high density of inhabitants per complex. The largest residential complex is in Huaguo Garden in Guiyang, where there are 200 blocks of 40-storey high-rise buildings and

inhabited with more than 420,000 people. The other large complexes include Qifu New Village in Guangzhou, Kang Cheng City in Shanghai, Tiantongyuan in Beijing, etc.

The number of houses is small in cities, such as villas or detached houses, semi-detached houses, and townhouses constructed after the 1990s. The number of old houses remained very limited because most of them were demolished during the Cultural Revolution period in the 1960s and 1970s, and only a limited number of them survived as listed buildings. There is a relatively larger stock of houses in rural areas, and many of them are *Ping Fangs* or bungalows without proper heating systems. The number of two- to three-story cottages, detached or semi-detached houses with large courtyards has been growing in some affluent rural areas for the last two decades with the improvement of economy and income. There has been growth in construction of residential blocks in rural areas for the last decade, mainly in areas near counties or cities with increasing housing demand from the younger population. However, the number of flats in rural areas has remained very small, as most young people raised in rural areas migrate to urban areas to pursue better jobs and standards of living.

Apartments are also divided into residential, commercial, and dual commercial/residential areas in China depending on the nature of land used for the development. The residential apartments are constructed on land permitted for residential purposes, while the latter two are built on the land permitted for commercial purposes. Therefore, the lease and utility tariffs are different for these types of apartments, and the residential lands are not used for commercial registration. The latter two types of apartments are also considered to be commercial buildings, and not limited by the governments' controlling policies on property purchasing. Therefore, the construction of apartments still grows quickly in large cities with an increasing number of consumers which are not allowed to buy pure residential flats or apartments with the implementation of controlling policies since 2010. Most of the apartments are in urban areas, especially in the large cities, due to government controlling policies on real estate.

Dwelling Stock by Tenure

There are no consistent official statistics on the ownership of dwellings in China due to the imperfect system of real estate registration and some complicated historical reasons. There are no freehold properties in China and the residents only have the right of living in the property without owning the land, similar with leased property in Europe.

The average length of a lease on residential property is normally 70 years and the lease for commercial property normally ranges from 30 to 50 years. Some of the commercially leased apartments can be also used as residential or dual-used as both commercial and residential purposes. There are also many collectively owned properties, such as military, central government and rural village-owned properties, which also belong to the private property category but cannot be sold at normal market price since there is no individual certificate of property but a transaction

agreement with the owner. The main properties sold in the real estate market are called commodity properties which normally have a 70-year lease and are under the government controlling policies in tier one and tier two cities currently. However, there are also many indemnificatory housings including low rent housings, resettlement housings, and price set housings invested by governments in the market, which also have a lease of 70 years but are restricted to be sold in certain years. The shared leased hold property is a new model launched by the government in recent years. However, all these properties can be purchased and owned by consumers in terms of the right of using, subject to meeting certain policies and requirements.

BRG based its data on the ownership of dwellings in China according to the floor space of dwellings unsold, average size of commodity dwellings from NBSC, and number of government low renting housings from the Ministry of Housing and Urban-Rural Development of China, and then gave the number of private owned dwellings. The information from other sources such as the Ministry of Housing and Urban-Rural Development of China, Chinese Academy of Social Science (CASS) and others are also considered.

The private ownership of dwellings is high in China with the ending of free housing allocation systems in the 1990s and the starting of the real estate transaction in the late 1980s. With the rapid growth of housing construction and expansion of privatization of housings, the private ownership of dwellings has been growing for the last two decades. The private ownership of dwellings has exceeded over 80% and reached nearly 90% in recent years. Public or government-owned dwellings stocks are still small in China, although the number of such dwellings recently grew with the government's encouragement of construction via the public/low-rent housing program. There are a small number of dwellings, mainly commodity housings, unsold by the real estate companies, which belongs to vacant dwellings, due to the rapid growth of housing construction in urban areas, especially in large cities. However, the number of vacant dwellings started to decline after 2015 with the governments' efforts on de-capacity of housings supplies in tier one cities, and there is still a large park of vacant dwellings which needs to be reduced in the tier three to tier four cities.

Most of the dwellings are primary owned in China, and there is also an increasing number of secondary dwellings with the growth of investments in properties, improvement of income and increased number of rich and middle-class people for the last decade. However, with the improvement of real estate registration systems and implementation of property tax in the future, the number of secondary dwellings is expected to decline in the future.

The number of primary renting homes remained low in general in China comparing with other countries in the west, however, the number of rented dwellings increased very quickly for the last few years (especially the last two years) with the government's controlling policies on property purchasing and policy of encouraging home renting in both private and public sectors. The share of rented dwellings in tier one cities such as Shanghai, Beijing, Guangzhou, and Shenzhen are much

higher than other cities due to the increasing number of young graduates and migrants looking for more job opportunities.

Non-Residential Building Stock

Recently, the stock of non-residential buildings has grown quickly in the accommodation, food and beverage, retail and wholesaling, education, medical and civil air transportation sectors in China.

The following analysis is based on the latest official statistics:

- There were 20,614 and 26,258 enterprises in the accommodation and food and beverage retail businesses respectively in 2018 (latest data available) in China, up by 4.2% and 1.4% respectively compared to 2017. The number of guest rooms was 3.9 million in 2018 (latest data available), up by 0.4% over 2017. The business floor space of accommodation enterprises and food and beverage retail enterprises in 2018 was 102.8 million square meters (latest data available), up by 7.2% compared to 2017. The number of star hotels was 9,566 in 2017, compared with 11,685 in 2016 (latest data available).
- There were 512,581 schools in total in 2019, up by 2.7% over 2018.
- There were 1,007,545 health institutions in 2019, compared with 997,433 in 2018.

Macroeconomic Overview

China has gone through an extraordinary course after the civil war 70 years ago. The economy has experienced a series of up and down, affected by both internal and external factors, such as reconstruction of economic and industrial systems, sharp declining of economy caused by serious natural disasters during 1960 to 1962; ten years stagnation due to the cultural revolution from 1967 to 1976; recovery in the 1980s and 1990s due to the reform and opening and the interruption by political turmoil in 1989. However, the economy has been growing rapidly since the 2000s driven by construction boom, joining WTO, Beijing Olympic Games 2008, and Shanghai World Expo 2010, but also affected by the Asia financial turmoil in 1997 and the global crisis in 2008. The GDP has reached to RMB 100 billion in 2000 and RMB 990 billion in 2019, comparing with only RMB 68 billion in 1952. The average growth rate of GDP from 2000 to 2019 is about 9.6%, and the GDP growth rate in 2019 was 6.1% comparing with 6.7% in 2018.

The Chinese people's income and living standard has improved quickly along with the political stability and rapid economic development in the country. The per capita GDP exceeded RMB 10,000 in 2003 and reached to RMB 66,006 in 2018, comparing with only a RMB 119 in 1952.

The consumer price index has been low in general for decades in China except certain years such as 1961, 1988/1989, and 1993 to 1995, due to a shortage of good supplies caused by natural disasters, political unrest, rising raw material

price, high bank interests, and poor management of large to medium state owned enterprise. However, the rate of inflation has been at a medium to low level in subsequent years and has been maintained in a range of 2% to 3 % in the recent seven to eight years. In 2019, the consumer price index was 2.9% comparing with 2.1% in 2018.

The urban unemployment rate has been in a low level in China, which is around 3-4% per year since the 2000s, and even slightly lower before that.

China is an exporting-oriented economy with a total trade value of US\$ 4,622.4 billion in 2018, of which 53.8% are for exports and 46.2% are for imports. The trade has been increased dramatically for the last two decades especially after China joining the WTO in 2001, with an interruption in 2009, 2015 and 2016 affected by the global financial crisis, slowing down of world economy and trade friction between China and the major destination countries. The trade started to recover and grew quickly since 2017 thanks to the promotion of the Belt and Road policies and trade to these countries. In 2019, the China-US trade friction accelerated, and the trade war has affected the trade.

Foreign investment has grown rapidly since the 1980s due to the opening up and generally stable political situation and favorable conditions for foreign joint ventures and exclusive investments, such as tax and infrastructure, which has contributed to the rapid development of the manufacturing industries and economy in China. Foreign direct investment utilization grew dramatically during the 1992 to 1993, possibly because of the transformation of the former large to medium state-owned enterprises to foreign joint ventures, as well as the large depreciation of RMB against US\$. In 2018, the actual foreign direct investment utilized reach US\$ 138.3 billion, up 151 times the investments of 1983, with an average growth rate of 15.4% per year. China has become the second largest foreign invested country and has been the largest developing country with foreign investment inflow for the last twenty-seven years.

The exchange rate of RMB against US\$ has experienced two major depreciations since 1990, with the largest depreciation happening in 1994. The exchange rate of US\$ against RMB increased from the previous 567 to 862 in that year, and such range has remained for the more than ten years before a ten years appreciation till 2014. The RMB started to weaken from 2015 apart from 2018. In 2019, the exchange rate of the RMB against the US\$ was 689.85, a 4.2% increase compared with 2018.

With the effects of COVID-19 pandemic, the Chinese GDP declined by 6.8% in Q1 2020, compared with Q1 2019, of which agriculture declined by 3.2%, industry declined by 9.6% and services declined by 5.2%. Per capita income and consumption declined by 3.9% and 13.5% for urban people, and the per capita income and consumption in rural areas also declined by 4.7% and 10.7% respectively due to the lockdown and stopping of working activities. The price index of agricultural goods in Q1 of 2020 increased by 39% comparing with a decline of 1.7% in Q1 2019. The retailing business was attacked seriously. The sales of

department stores and specialty stores declined by 34.9% and 24.7% respectively in Q1 2020, when compared with the same period in 2019, while the sales of supermarkets only grew by 1.9% comparing with 7.5% in the same period in 2019. With the ease of pandemics and controlling measures, the business activities started gradually, and the economy started to recover from Q2. In the face of serious decline of private consumption, the government issued consumption coupons in many regions to compensate the loss of income and stimulated the consumption. In addition, the investments on infrastructures such as railways, roads, harbors, bridges, airports, water dams, power, etc. will continue to grow in 2020 and the coming years to stimulate the economic recovery. The government also adopted financial stimulus measures by reducing bank interests and relax the lending regulations, as well as issue treasuries to invest on the infrastructure, especially the so called “ new infrastructures” such as 5G networks, green manufacturing infrastructures, intelligent industries, data centers, biomedical and medical equipment. According to IMF’s latest forecast, the Chinese economy is expected to grow mildly by 1.2% in 2020, representing an extremely serious recession since the financial crisis, although the IMF forecasts the Chinese economy will recover quickly and grow by 9.2% in 2021. However, such forecasts are still not assured because of the uncertainty regarding the COVID-19 pandemic.

Construction Overview

BRG's construction figures are based on NBSC’s statistics on completed floor space of housings. The most recent figures are based on the statistics published in the construction section of NBSC. Up until 2011, the fixed assets and real estate development figures are used.

All floor space figures in m² are then converted into dwelling completions based on NBSC’s statistics on the number of dwellings and floor space of dwellings sold by real estate companies in the fixed assets section. The historical data on size of dwellings in m² used for the years pre-2004 are estimated since there is no such data published by NBSC.

New Housing Construction

Chinese new housing construction is in line with the evolution of governments policies on the development of the real estate market. The market was initiated in 1981 with the pilot development of commodity housings in Guangzhou and Shenzhen. Before that, the housings were developed by work units, institutions, and governments; and allocated to their staff free as welfare; instead of sold on the market. The real estate market started to grow dramatically due to the development in the large cities in Guangdong and Hainan during 1991, and then the government started adjustment measures such as stopping loans to land developers to control the overheated real estate development in Hainan in 1993. At the same time, the welfare system of allocation of housings were stopped, which led to the steadily growth of new housing construction due to the rapid growth of procurements from work units and institutions. The private purchases of housings also started in 1993 and began to grow more quickly from the end of the 1990s. Therefore, new housing

construction started to grow very quickly from 1999 to 2003. Then the government launched some controlling measures mainly for the developers such as limitations on bank loans and an increased interest rate, which led to the slumping of new housing construction in the 10th Five Year Plan period (2001 – 2005), especially in the south. However, the real estate market started to grow quickly in the north due to the Beijing Olympic Games in 2008, which led to the construction of new housing to reach a peak in 2008 and 2009 in terms of both transaction volume and housing price, despite the increased mortgage deposits and interest rate. Then the government launched strict policies and increased the deposit up to 60% (compared to 20% to 30% previously) and forbade the purchasing of a third property, which caused a rapid slowing down of the real estate market and new housing construction. In 2011, the government further strengthened the controls on the property market and stopped the purchasing of second properties in tier one cities and increased the mortgage interest rate by 1.1 times the basic interest rate while keeping the deposit at 60%, hence, the real estate market and construction of new commodity housings cooled down. However, the construction of the government's economic affordable housings started to grow to ease the high price of commodity housings and solve the housing problems for ordinary people, which meant that construction of new housing still grew steadily during the 12th Five Year Plan period (2011 – 2015). With the strict controlling policies on real estate market in tier one cities, slowing down of the economic growth rate, crashing stock market, and postponement of new projects approved; the new housing construction declined in 2015. With the large vacant housing stocks, the government started to relax its controlling policies from the second half of 2015 in order to de-capacity of housings in tier one cities, and the new housing completion recovered a bit in 2016. However, the market remained stable due to the contraction of policies on property purchasing (including the stricter regional controlling policies in Tongzhou, Beijing and other areas) from the fourth quarter of 2016, such as increasing the deposit, mortgage interests and barriers for second home investors.

In 2017, the government resumed the controlling policies on property purchasing in tier one cities, which led to the slight decline of housing completions. However, the construction of new housing in the tier three to tier four cities continued to grow because no controlling policies have been imposed yet, rising costs of living in tier one cities and a move of people from tier one cities to tier three to tier four cities. In addition, the government reinforced the construction of housing for low income people, by increasing the investments in construction of indemnificatory housing, such as low rent, resettlement, price and dwelling size set housing, economical housing, low renting, and subsidies rent housings; instead of the previous economical affordable housings, which prevented the sharp decline of the construction of new housings despite the slow economic growth rate and contraction of real estate market in the tier one cities. In addition, the increased demands for secondary homes driven by improvement of income and governments' relaxation of policies on family planning (one child one family) and continuous acceleration of urbanization, as well as the Beijing Winter Olympic Games 2022, construction of the Beijing New Administration Centre, Xiong'an New Administration Zone, the development of the West and revitalization of the Northeast, will continue to encourage the construction of new housing in the next

few years.

In 2018, completions of fully decorated housing continued to expand from tier one cities to tier two cities and the construction of fully decorated housing in tier three and tier four cities started to grow, with the implementation of regional governments' policies from the previous 11 cities and provinces to 18 cities and provinces. In 2018, Hubei, Henan, Shanxi, Liaoning, Sichuan (Chengdu), Tianjin and Jiangsu governments launched new policies on the construction of fully decorated housing, of which Henan and Tianjin governments requested that all new government indemnificatory housings and commodity housing is fully decorated from 2018. According to the monitor of AVC, there are 1,103 fully decorated housing projects started in 2018, including 1.214 million dwellings, up by 9.9% over 2017. With the continuous expansion of fully decorated housing into tier two to tier four cities, it is expected that construction of fully decorated housing will grow very quickly, which will provide some large opportunities for the main heating suppliers in the forecast term.

New housing completions in 2019 continued to contract by -0.7% due to the tightening of controlling policy on the property market. The central government is focused on the financial risks of real estate and does not use real estate as a short-term means to stimulate the economy. It tightened the supervision of the funds and limited the approval of land permits to the real estate industry in order to keep the real estate market stable. Therefore, the new housing market continued to cool in 2019.

However, the fully renovated new dwelling completions continued to grow in 2019. In the past, the new dwellings are sold unfinished or semi-finished in a "shell" form with no basic facilities such as paints, internal doors, floorings, lighting, bathroom, and kitchen facilities. In April 2017, the central government launched the 13th Five Year Plan for Construction, which required the ratio of fully renovated housings to reach 30% by 2020, which encouraged some regional governments to launch policies on fully renovated housings in the following few years. By 2019, there were about 83 policies on fully renovated housings covering 27 provinces and cities in China, which has encouraged the construction of fully renovated housings in recent years. In 2018, with general declining of the overall new housing completions, the completion of new fully renovated housing increased very quickly by about 60% and reached to over 2.5 million dwellings. This number has also further improved to 3.4 million dwellings in 2019. It is reported that the penetration ratio of fully renovated housing in the tier 1 and tier 2 cities is high, but very low at around 8% in tier 3 or tier 4 below cities, which means there is still a large potential growth of some building materials going into the new fully renovated dwellings in the future.

In 2019, the overall construction of new villas declined due to the tightening of policies on villa construction. However, the demand for large houses and villas in suburbs is still high especially in the large cities where the flat prices are expensive in city centers, therefore, the number of villa completions in some areas is still growing. However, with government's regulation on controlling the approval on lands used for villa construction, it is expected that the completion of new villas will

continue to decline in the next few years.

The outbreak of the COVID-19 pandemic and lockdown in the beginning of 2020 affected construction in China. Housing completions declined by 25% in the first quarter. With the ease of the lockdown and gradual resuming of works, the construction of new dwellings improved gradually in April and May. With continuous works and restarting of the postponed existing projects, it is expected that the new dwelling construction will start to progress quickly from the second quarter but will still decline in 2020.

With the recovery of the economy and fully operational construction from the second half of 2020, it is expected that housing construction will grow from 2021. However, there is still no sign that the government will relax its controlling policies on the property market, therefore, the new housing construction will continue to be stable or grow at a very slow pace in the next few years. Fortunately, the penetration of fully-renovated dwellings has improved very quickly over the last few years and it is expected that the ratio of fully-renovated dwellings will continue to improve in the next few years with the government's regulations on environmental protection, energy efficiency and increasing income of consumers. These will benefit building materials sales in the new housing sector.

Home Repair, Maintenance, and Improvement (RMI)

Most dwellings delivered in China are not fully completed and do not contain the necessary heating systems, kitchen and bathroom facilities, internal doors and paints, which are termed as 'rough cast housings or shell housings. Therefore, the consumers must renovate their new homes, either themselves or through contractors. Therefore, there is a significant new home renovation market in China. In addition, there is also a growing redecoration market in China with the growth of income, gradual ending lifespans of interiors and heating systems, as well as the governments' schemes on transformation of urban infrastructure and housing conditions. In addition, the home renovation in affluent rural areas also grew in recent years with the improvement of rural income, modernization of heating systems and pursuit of comfort.

The residential RMI market continues to grow quickly, driven by the improvement of income, the ending of lifecycles of existing home renovation, and governmental incentive projects for urban and rural housing refurbishment. In 2019, the total output of residential building decoration reached RMB 2,240 billion, up by 10% over 2018, according to the CHINA BUILDING DECORATION ASSOCIATION (CBDA). This was mainly driven by the growth of housing stocks, increasing transactions of second-hand homes, the government's large old housing transformation programs, and growth in demand for upgraded decoration. The re-decoration market continued to grow quickly in 2019 and has become the main sector for home renovation.

There is also a large government program on the updating and transformation of subsidized buildings, urban shelter housings, urban infrastructures, rural housings,

and infrastructures.

The COVID-19 pandemic and lockdown affected the home decoration industry in the first quarter of 2020 and postponed the decoration seasons, however, it should start to grow from the second half of the year with the returning of works, production, and logistics after April and May.

Non-Housing Construction

The non-housing construction activities in terms of completion of commercial and service buildings declined by 4.8% in 2019 according to NBSC, due to sharp declining floor spaces completed for research institutes, offices, hotels, malls, exhibition centers, culture, sports and entertainment and mild declining completion of schools. The completion of food and beverage buildings, hospitals and industrial workshops and warehouses grew in 2019. This is mainly because of slow growth of the economy, advanced completion in the past years, and over supplies of some non-housing buildings such as offices, exhibition centers and hotels in large cities. The government's regulations on fire security, environmental protection, and limitation on building luxury entertaining and accommodation buildings also affected the completion of such commercial buildings. The completion of industrial factories continued to grow in the middle and west of China, with relocation of factories and expansion of production capacity.

However, the market for non-housing RMI – classified as renovation and decoration for commercial, public, and institutional buildings – has been growing quickly over the last decade. According to the recent statistics of the CHINA BUILDING DECORATION ASSOCIATION (CBDA), the output of non-housing RMI activities was RMB 2,360 billion – up by 8.3% – in 2019, driven by the refurbishment of commercial and public building in urban areas under governmental subsidies and policies, as well as fast-growing investment in transportation, utilities and other public infrastructures. It is expected that the non-housing RMI market will grow more quickly in the forecast term, due to the acceleration of urbanization, the upgrading of public and commercial infrastructures, and several large projects such as the Beijing Winter Olympic Games 2022, Hangzhou Asian Games 2022, Xiong'an New Administration Zone, and the Beijing New Administration Centre.

II.2 Key Figures

CN - DEMOGRAPHICS AND HOUSEHOLDS

Value		Year
Segmentation	Category	2019
Demographics	Population (% Rural)	39.75
	Population (% Urban)	60.25
	Population (Million)	1 403.39
Households	Average Size (Persons h/h)	3.00
	Total ('000)	465 695.00

Source: BRG based on Country Statistics

II.3 Macroeconomic Trends and Forecasts

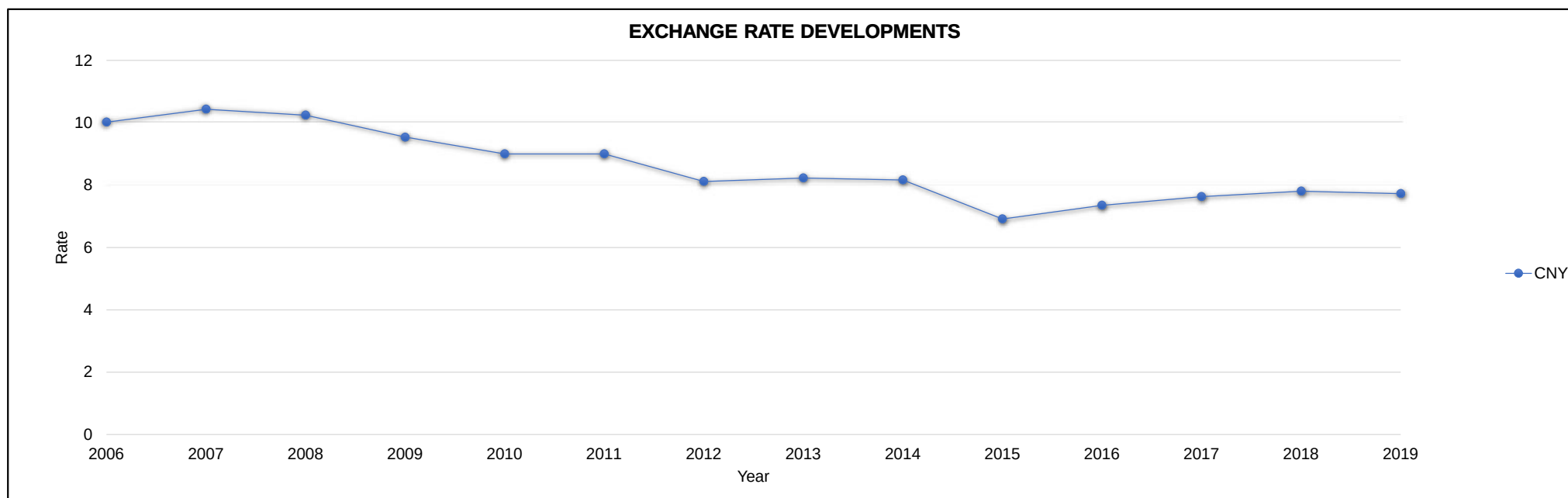
CN - KEY MACROECONOMIC INDICATORS

Value	Year																			
Category	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Consumer Price Index (% change)	- 0.80	1.20	3.90	1.80	1.50	4.80	5.90	- 0.70	3.30	5.40	2.65	2.62	1.99	1.44	2.00	1.56	2.11	2.90	3.05	2.55
GDP (% change)	9.10	10.02	10.15	11.37	12.72	14.25	9.65	9.40	10.56	9.50	7.90	7.80	7.30	6.90	6.85	6.95	6.75	6.11	1.18	9.21
Government Balance (% of GDP)	- 2.86	- 2.39	- 1.51	- 1.40	- 1.14	0.06	- 0.03	- 1.75	- 0.36	- 0.10	- 0.30	- 0.83	- 0.91	- 2.82	- 3.70	- 3.84	- 4.66	- 6.37	- 11.23	- 9.57
Unemployment Rate (%)	4.00	4.30	4.20	4.20	4.10	4.00	4.20	4.30	4.14	4.09	4.09	4.05	4.09	4.05	4.02	3.90	3.80	3.62	4.30	3.80

Source : IMF

N.B. No data available beyond 2021 at the time of publication.

II.4 Historical and Current Exchange Rates
CN - EXCHANGE RATES : HISTORICAL AND CURRENT

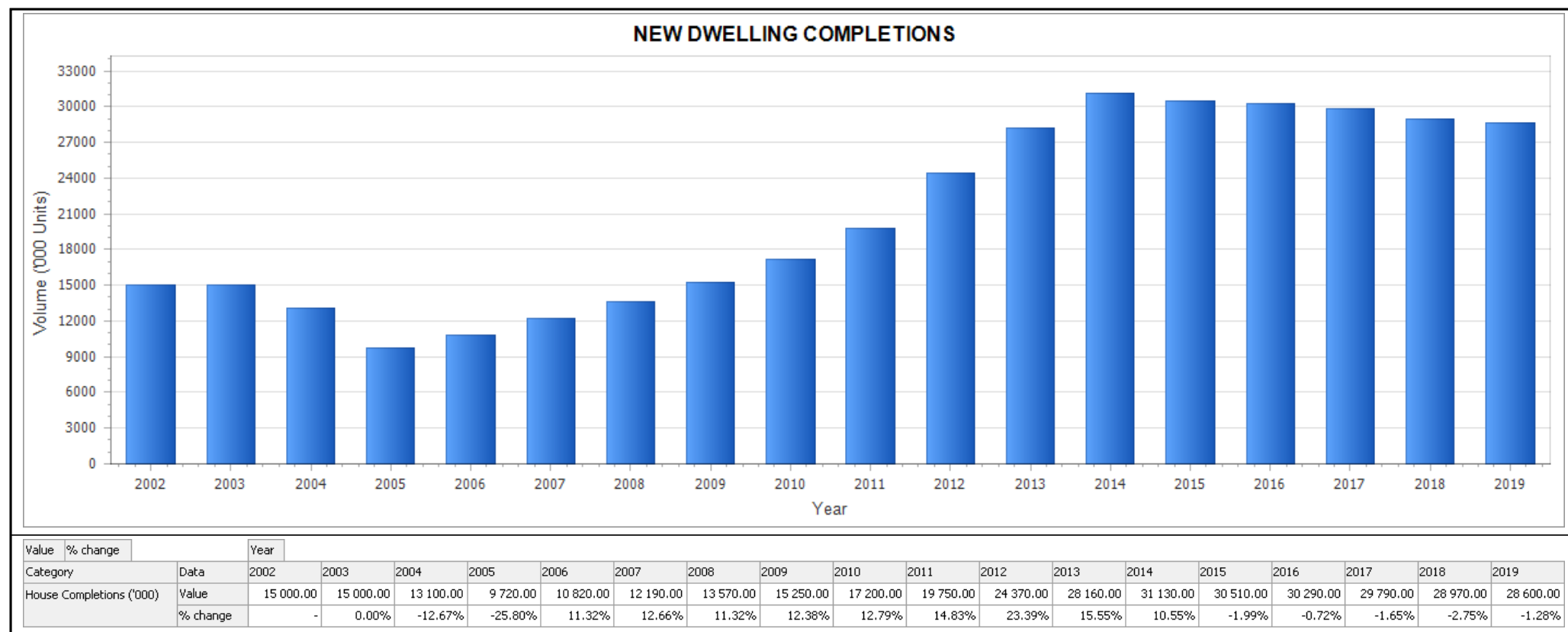


Local Currency	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
EUR / CNY	10.02	10.43	10.25	9.54	9.00	9.00	8.12	8.23	8.17	6.91	7.35	7.63	7.81	7.73
% change	-	4.09%	-1.79%	-6.91%	-5.66%	0.03%	-9.86%	1.37%	-0.70%	-15.35%	6.31%	3.78%	2.31%	-0.97%

Source: OANDA
The Chinese Yuan Renminbi (CNY) is the Currency of China. The Exchange Rates Shown are the CNY to the EUR.

II.5 New Dwelling Construction

CN - NEW DWELLING COMPLETIONS : HISTORICAL AND FORECASTS ('000 UNITS)



Source: BRG based on Country Statistics.

N.B. Due to the current uncertainties regarding the economy and construction, BRG is unable to provide an outlook for residential completions.

II.6 Dwelling Stock by Type and Tenure
CN - DWELLING STOCK BY TYPE AND TENURE ('000 UNITS)

Value		Year
Segmentation	Category	2018
Dwelling Stock by Type	Houses : Rural (%)	39.35
	Houses : Rural ('000)	170 000.00
	Houses : Urban (%)	60.65
	Houses : Urban ('000)	262 000.00
Dwelling Stock by Tenure	Primary Owned (%)	59.42
	Primary Owned ('000)	256 680.00
	Primary Rented : Private (%)	12.49
	Primary Rented : Private ('000)	53 960.00
	Primary Rented : Public (%)	8.59
	Primary Rented : Public ('000)	37 128.00
	Secondary (%)	16.09
	Secondary ('000)	69 501.00
	Vacant (%)	3.41
	Vacant ('000)	14 731.00

Latest available data. Source: BRG based on Country Statistics.

II.7 Overview of Impact of COVID-19 on Forecasts
CN – OVERVIEW OF IMPACT OF COVID-19 TO DATE

Start of the lockdown [expected start of easing]	The official lockdown in the Hubei Province started on 23 rd January 2020 with surrounding cities and provinces also being affected shortly after. Since then gradual lifting of restrictions followed in the last few weeks, however, in mid-June parts of the country re-imposed lockdown measures as a new wave of COVID-19 cases emerged in the capital city of Beijing.
Number of officially confirmed cases	84,491 cases [4,634 confirmed deaths] as of 5 th August 2020 / population 1.394 billion (2019).
General measures in place to support the economy	China's central and local governments passed several policies to alleviate the pressure on citizens and businesses. These policies include support for small and medium enterprises, foreign-invested companies, tax and fee reductions, rent waivers and social security.
Statistics on affected employees	Unemployment rate rose to 6% in April 2020 (latest available data) with further growth anticipated. Low paid workers are especially vulnerable and will need extra support from the government.
Measures in place to mitigate economic impact on INDIVIDUALS	For small and micro firms, if 15 percent of their employees are newly hired, they can apply for guaranteed startup loans; for companies with a workforce of more than 100 persons, the ratio of newly hired employees can be lowered to 8 percent. Eligible individual borrowers can borrow up to RMB 200,000 of guaranteed start-up loans. The payment of income tax on small and micro businesses and individual businesses will be postponed to next year. Epidemic-hit SMEs and micro firms, including small business owners and individual household businesses, can make applications with banks to defer repayment of principal and interest expenses payable from January 25 to June 30, 2020. Overdue loan repayments in the period will not be subject to penalties.
Statistics on affected companies (% of total forced to suspend activities)	Public records suggest that at least half a million firms were dissolved in the first quarter of 2020 and more are likely to close.
Specific measures affecting BUILDING CONSTRUCTION and RMI	Strict safety guidelines need to be adhered to at construction sites. Enterprises provide workers with standard protective supplies as well as training sessions on occupational safety and health. Overall private consumption dropped by 9% in the first quarter of 2020.
Measures in place to mitigate economic impact on COMPANIES	Implementation Opinions about Promoting Consumption and Developing a Powerful Domestic Market Measures include reducing import tariffs, shifting consumption tax on selected products and launching more duty-free shops in order to boost consumption and help businesses. ICBC will offer RMB 10 billion (USD 1.4 billion) of credit loans to Hubei private companies, especially those in the advanced manufacturing industry. Starting from 1 st June, the People's Bank of China (PBC) will use a special re-lending quota of RMB 400 billion to buy 40% of credit loans granted to small and micro companies by local banks. Corporate borrowers are able to extend repayment period of loans up to March 2021 with no punitive expenses.
Support measures from international organisations (EU, IMF, World Bank, other)	N/A

Sources:

- <https://www.theguardian.com/world/2020/jun/13/beijing-china-new-covid-19-cases-linked-to-food-market>
- <https://www.euro.who.int/en/home>
- <https://www.whitecase.com/publications/alert/covid-19-chinese-government-financial-assistance-measures>
- <https://www.china-briefing.com/news/china-covid-19-policy-tracker-benefiting-business-enterprises-comprehensive-updated-list/>

CHINA

DISTRIBUTION

THE EAST AND SOUTH EAST ASIAN BATHROOM PRODUCT MARKETS



Overview

In China, the distribution of bathroom products is generally divided between the project and retailing markets. The project or B2B market is defined by sales of bathroom products directly from manufacturers or wholesalers to developers, contractors, and installers, while the retailing market refers to sales to individual customers via intermediates including wholesalers and retailers. The project market is characterized by large bulk of purchasing and low price, while the retail market is comprised of small purchases of higher-priced products, with many intermediates involved.

The intermediates in the distribution of bathroom products in China are agents, distributors, dealers, and retailers. An agent is only responsible for selling the products, but not involved in the delivery, installation or after-sales services, and is also called the manufacturer's sales representative. The agent does not need to buy products from the manufacturers but pays a certain amount of deposit for getting a certain quantity of products for selling and earned commission. A distributor is a company that purchases products directly from manufacturers and distributes them in the market and is also responsible for installation or after-sales services. The distributor resells products and earns profit from the margin between the manufacturer selling price and the market price. A distributor may sell to several dealers, while a dealer usually sells directly to retailers or consumers. Sometimes, the agent and the distributor are the same company in China.

The distribution of bathroom products tends to be flat, with less intermediates in terms of product selling and services provided in the mature and competitive markets. In the key markets, the manufacturers tend to provide direct services to their customers by setting up regional technical and service centers (sometimes in the form of showrooms or fleet shops without selling products), and the distributors tend to cut the number of dealers to cover the end use channels through setting up their own sales offices or shops.

In the project market, the majority of the bathroom products are distributed through agents and distributors, due to their connection with customers and governments. The relationship is very important for project sales, especially in large government, land developer and contractor projects, since many bids are internally decided before they are open. Therefore, experienced and well-connected distributors and agents are highly demanded by the bathroom product manufacturers in the key markets in China. There are some manufacturers who are selling products directly to the project markets via their in-house sales forces.

Chinese installers can be grouped into large construction contractors, specialized bathroom product installers, home decorators, property maintenance companies and individual installers. The land developers normally employ installers managed by their own engineering divisions. The majority of project sales go to land developers and large construction contractors. The specialized bathroom product

installers mostly act as distributors serving both regional projects and retail markets. Home contractors and property maintenance companies employ qualified installers to serve individual customers for home improvement and maintenance works. Individual installers, many of them experienced but without qualifications, work for small distributors and dealers.

In the retail market, bathroom products are mostly sold via distributors and dealers to retailers, such as building material superstores (DIY stores), building material malls and markets, specialty stores and other retailers.

Online stores are becoming a very popular channel for the distribution of bathroom products in China, with constant promotion and improvement of services of logistics, installation and after sales services. New media, such as Wechat (micro-chat), QQ, and Weibo (micro-blog) become more important methods for manufacturers to market their products and increase sales. E-commerce sales have tended to move from the traditional B2C and B2B to O2O (online to offline), to bring potential online customers to shops or showrooms to experience the actual products and help them to make purchasing decision, which make the manufacturers to get more customers online and customers to select product and services, and pay online.

Wholesale Level

The wholesale distribution of bathroom products is very regional and fragmented in China, due to the large size of the country. The bathroom product manufacturers normally adopt a system of agents, distributors, and dealers as wholesalers to distribute their bathroom products to retailers and end users in different regions. The agents are divided into grade 1 to grade 2, depending on the size of regions. The manufacturers normally acquire grade 1 agents who are in charge of one region or city, who sell products directly (or via grade 2 agents) to distributors and dealers. There are rarely national wholesalers (agents and distributors), except those manufacturers who are inexperienced in the market and must rely largely on the wholesalers. However, as long as the manufacturers are familiar with the market, they move to using regional agents or distributors to sell their products – since they have the ability to control them. The agents and distributors are normally large bathroom material and building material traders, who are experienced in the distribution and installation of bathroom products in regional markets. They represent or distribute bathroom products for one or several manufacturers.

With the acceleration of urbanization, increasing competition, rising distribution costs and the expansion of bathroom product market from tier 1 and tier 2 cities to tier 3 and tier 4 cities and even rural areas, the previous pyramid form of the overall sales structure, from manufacturer to grade 1 and grade 2 agents, lost influence. Instead, the manufacturers tended to sink their distribution channels down to the tier 3 to tier 4 cities and rural market by either acquiring more local distributors directly, or help their regional agents and distributors to establish networks and promote products, or just setting up their own local sales branches acted as wholesalers to cover the end users.

However, some manufacturers (especially those with new or small brands, who are willing to enter the market and want to earn money quickly) just acquire as many distributors as possible without classifying them by regional grade. This has made the market unorganized and very competitive.

There is a tendency of wholesalers to move away from distributing single bathroom products towards providing full bathroom solutions such as sanitary ceramic ware, faucets, baths, shower products, shower wall products, hydrotherapy products, bathroom furniture and accessories to the customers in China. The large grade-one bathroom distributors are in the main markets, depending on products and manufacturers' regional focuses.

There are no buying group companies in China. However, group buying or collective buying (namely *Tuangou*) is increasingly used for purchasing bathroom products in the country. The group buying platforms are often e-commerce or Internet media, including large national comprehensive ones such as SINA JIAJU, FOCUS JIAJU, FANG, JD and ALIBABA, which has become one of the most important wholesale channels for many bathroom manufacturers.

The distribution of bathroom products through large building material traders is small, since they normally trade large building materials such as cement, steel and equipment.

Retail Level

DIY Retailers

DIY has never been a hobby for Chinese consumers, though the concept was introduced to China in the late 1990s. DIY is converted to BIY (Buy-It-Yourself) in China, since the country's consumers usually just buy heating products in DIY stores and then pay the contractors to install them. DIY stores are characterized by the advantages of clearly marked prices, plus a convenient, full range of products and better shopping environment; but disadvantages include having smaller sizes of displaying areas for manufacturers, and mainly targeting small groups of young consumers and the middle class.

In 1999, KINGFISHER opened its first B&Q store in Shanghai, representing the start of DIY in China. DIY experienced a short boom until the mid-2000s, due to the stopping of favorable treatment to foreign retailers and competition from the building material malls with merchant systems. KINGFISHER opened 63 outlets covering 25 cities in China, but this was reduced to 39 outlets and downsized the business areas in its 17 stores in 2008. In 2014, KINGFISHER sold its DIY business to Chinese retailers WUMART, representing the ending of KINGFISHER's story in China. Other DIY chain stores such as LEROY MERLIN (France), OBI (Germany), HOME DEPOT (USA), and ORIENTAL HOME (China) also experienced an up-and-down during the 2000s, and all of them have disappeared now.

With the changing trends of consumer purchasing habits from the conventional only through stores to online shopping, and rising of purchasing power of the consumer in general, DIY retailers also tried to reform their selling methods, such as O2O, by setting up their own online shops via the main e-commerce channels, to attract more young customers. DIY retailers also tended to sink their channels from the tier one and tier two cities down to tier three to tier four cities and even rural areas recently with the acceleration of urbanization and maturing tier one and tier two city market. In 2019, B&Q continued its diversified sales models by providing instore consumers building products and decoration services; B2B sales to small retailers and contractors; and online B2C sales to consumers, which helped its sales in 2019 increase by 2.7%.

However, with the increased costs of selling products in large DIY stores, manufacturers and distributors tend to look for alternative retail channels, such as building material malls/markets and specialist stores, which have lower costs involved but offer large space and convenient locations.

Specialist/ Other Retailers

Building Material Mall

The building and home decorative material malls were transformed from the traditional building material markets in cities. The mall lends shop spaces to distributors, dealers, manufacturers, and individual businessmen, to collect lease and maintenance fees. The mall is in charge of property management, unified promotion and after services. The tenants can have smaller to very large shopping spaces in the mall (depending on the budget and availability), and the product prices are cheaper and negotiable in the malls. The customers sometimes get wholesale prices in the manufacturers and distributors shops. Therefore, the merchant malls have become increasingly more popular since the end of 2000s, and have become the main retail channels for consumers, contractors, and installers in large cities. These shopping malls are both nationally and regionally distributed in China. The main national companies include RED STAR GROUP, and BAILIAN GROUP owned many RED STAR MACALLINE, EASYHOME (JURAN) and HOMEMART stores in the main tier 1 to tier 2 cities. The large regional malls include: JIMEI, AIKA (AIJIAJIAJU), and CHENGWAICHENG in Beijing; YIJIALI, HOME900 (JIUBAI), JINSHENG HAOLAIFU, and JISHENG WEIBANG in Shanghai; MEIJU CENTRE, WEIJIASI PLAZA, and XILONG INT'L DÉCOR in Guangzhou; and XINGFU HOME WORLD, DASHIJIE, HUANAN HOME, and JINSANJIAO in Dalian. More recently, the national retailers have tended to expand their presence into the tier 3 cities, which has increased the competition. In addition, they also started to 'touch net' to set up online shops or e-commerce division, such as RED STAR MACALLINE and EASYHOME (JURAN).

Building Material Market

This is the traditional wholesale and retail channel for bathroom products and other building materials in China. They are still very popular in the suburbs of large cities, and small towns. This is the main channel for ordinary consumers and contractors to purchase medium to low end bathroom products – including ceramic sanitary wares, faucets, shower enclosures, baths, shower heads and pipes, kitchen sinks, bathroom furniture, bathroom accessories, as well as large building materials such as wood, cement, sand and steel, and decoration material such as lights, paints and curtains. These are also the venues with the high concentration of many individual and small installers providing quick and economic installation services. Many of the large Chinese brands started to grow from this channel and moved up to national brands. With the upgrading urban infrastructure, many of these traditional building material markets tended to move to outside of large cities or refurbished and transformed to building material and home product malls. For example, some large building material markets in Beijing such as Shilihe and Baliqiao will be relocated to Hebei, and there are many large markets will be transformed such as Yuqiao, etc. As a result, there will be more opportunities for medium-to-high end HVAC brands, including international and large Chinese brands to expand their presences in large cities in the next few years; however, an increased number of small Chinese brands will have to move outside of the tier 1 markets in the future.

Specialist and Other Retailers

Manufacturers and distributors tend to set up specialty stores in the main regional markets, to sell bathroom products and increase awareness of products and brands. The specialty stores are also linked with the distributors and manufacturers' regional offices or service centers, in order to provide convenient solutions to their customers. Sometimes, the specialty stores also act as a training center for installers. This has been an effective approach for manufacturers to promote their products and educate the markets. Recently, an increased number of bathroom manufacturers are interested in setting up their own specialist stores or help their distributors to set up the specialist stores and showrooms in tier 2 to tier 4 cities and even rural areas to increase the awareness and acquire more local distributors in the regional and local markets with the distribution channels further down to the tier 2 to tier 4 markets. The other retail channels, such as superstores and hardware shops, are for a certain type of bathroom products, such as faucets.

Internet Retailers

Sales through the Internet continued to grow rapidly in China with the growth of a new generation of consumers, changes of lifestyle, and the fast development of an Internet infrastructure. Online stores and sales through social media such as WeChat have become a normal way for consumers to purchase goods in China due to the popularity of electronic payment methods and the broad penetration of the Internet. In 2019, live broadcasting became a popular method for selling products, which attracted millions of people to watch.

CHINA

BATHS AND SHOWER TRAYS

THE EAST AND SOUTH EAST ASIAN BATHROOM PRODUCT MARKETS



Market Characteristics and Product Mix

The Chinese baths market grew rapidly during the 1990's, following the first boom in housing construction and renovation. However, the growth of the bath market has slowed since 2000 due to an increase in the popularity of showering. Sales were further hampered by the average dwelling size in China, which is relatively small in general terms, with the housing market being characterized by a prevalence of small apartments. Furthermore, there is a lack of bathing culture in China in general, which makes the installation of baths in ordinary Chinese dwellings less frequent. Baths are mainly installed in dwellings with large bathrooms or those which have several bathrooms, such as villas and large apartments. The strict environmental regulations, since 2014, also forced many bathtub factories near large cities to close or suspend their production, which led to a decline in both the production and sales of baths to some regions. There are a considerable number of baths installed in hotels since baths are perceived as a mandatory facility in luxury hotels. There are also some baths, mainly hydromassage products, installed in other commercial facilities such as sauna centers, gyms, swimming pools, and beauty salons. Overall, the slowing down of housing construction due to the Chinese government controlling policies on the property market, and the popularity of shower enclosures, have affected the growth of the residential baths market. However, sales of baths in the non-housing sectors continued to grow steadily, supported by a booming tourism sector, new commercial establishment completions and the refurbishment of existing hotels.

The shower tray market grew fairly quickly from the late 1990s through to the beginning of 2000s when the standard shower enclosure was introduced to China, however, sales started to decline after 2004, linked to an increased penetration of wet rooms. Shower trays are considered as an unnecessary product in China since they are not abrasion resistant, difficult to clean, require extra cost and space, and there are no standard drains designed for shower tray installation in most Chinese bathrooms. In addition, most bathrooms are tiled and therefore, shower trays are easily replaced by wet rooms.

For the product mix, the synthetic bath is the major product in the market, due to its light weight, versatile design, aesthetic and cheaper comparative price. The size of a bathtub tends to be smaller in comparison to other markets, due to the small size of bathrooms in China and the product design tends to be versatile, including a focus on color, smart integration of digital control panels, leather headrests, waterproof TVs and audio equipment, showers, mixers and efficient drainage, in order to appeal to the needs of young consumers and people with high incomes. However, the growth in sales of shower enclosures still restricted the development of sales of synthetic baths in recent years.

The share of cast iron baths is small because of their weight, inconvenience of transportation, expense and limited design. Cast iron baths are mainly installed in hotels and high-end houses, where the emphasis is on the ease to clean, durability,

and long service life.

Enamel steel baths are generally considered obsolete products due to their poor thermal insulation, lack of durability (enamel sheds easily) and noise when filling. These are seldom used in residential housing and mainly found in economy hotels.

The share of hydromassage baths is small in China since they are mainly sold to a small number of high-end dwellings such as villas, large apartments and non-residential applications, such as luxury hotels and recreational facilities, including sauna centers, swimming pools and beauty salons. The poor water quality and lack of water treatment facilities in most residential buildings make the installation of hydromassage baths infrequent in China. The anti-corruption movement and decline in the completion of new luxury hotels and recreational facilities has also contributed to a slowing growth trend.

For shower trays, most products are synthetic models, sold together with shower enclosures. Ceramic and cast iron/steel shower trays are rarely used in China. The majority of the shower trays produced in China are destined for export markets.

In terms of product innovation, baths in China tend to be small in size and colorful, to appeal to the needs of young, urban consumers living in small dwellings. Embedded baths are more popular in ordinary housings, and the baths with sitting areas inside are popular in small apartments. Sales of free-standing baths, both synthetic and cast iron, are small and only sold in high end housing such as luxury apartments, villas, hotels and spas due to their high price. Sales of smart bathtubs with digital controls and multi-functions are very small due to their expensive implied price and only appeal to consumers with high levels of disposable income. There is less product development activity and innovation for shower trays due to the small and declining market.

Sales of baths, and manufacturer focus, are increasingly moving towards tier 2 to 3 city markets, due to product saturation in the tier 1 cities. The tier 1 city market is concentrated and dominated by international brands such as KOHLER and TOTO, specifically within the high star-rated hotels and high-end housing. This has forced the Chinese bath manufacturers to go to the tier 2 to tier 3 cities, where there is need for affordably priced baths. Demand for baths in the rural market remains limited in China, due to concerns surrounding water conservation and also low levels of sanitation.

The simpler processing technology and smaller investment required for bath production, compared to ceramic sanitary ware, makes it easy for many Chinese companies to enter the bath market. They have capacity to provide OEM services to the large domestic and international brands, as well as exports. There are fewer regulations and standards on bath production; therefore, some small manufacturers produce defect or imitated baths, which damage the product image amongst the Chinese consumers.

Cast iron baths are mainly for export in China; many manufacturers, especially companies in Hebei province, have been aggressively exporting cast iron baths for decades. However, anti-dumping regulations and increased taxes have made many Chinese manufacturers turn to the domestic market, which has led to a steady increase of cast iron bath sales to hotels and high-end housing developments in the last few years. In 2017, the EU started to levy anti-dumping tax on the imports of cast iron articles from China, and the strict environmental regulations in Hebei province made many cast iron bath companies either reduce their supplies or fully cease production.

Recent Market Trends

In general, in 2019 the bath market grew marginally after a slight decrease in 2018. Growth was spurred by sales of cast iron and synthetic products. The popularity of minimalist interior decoration, the small size of bathrooms in general and the increased penetration of shower enclosures continued to affect the growth of the bath market.

Sales of synthetic baths still constitute the majority of the market due to their price and versatile design. The market grew in 2019 due to buoyant sales to hotels but also to increased sales in the North East of the country and also to increased renovation activity. Bathing in the northern areas has become a cultural habit but also the weather is colder, and bathing is being promoted to aid circulation and good health.

The market for cast iron baths grew in 2019 to sales into the high end hotel market specifically into the southern regions of the country. Styles and colors of cast iron baths have become much more diverse in the last couple of years which is helping promote sales.

The enamel bath market continued to contract in 2019 due to the obsolete nature of the product and competition from other types of bath. The baths constructed are also of limited quality and often get damaged during distribution.

Sales of hydromassage baths declined in 2019, due to less installations in the new housing segment. The sales to elite housings such as villas remained stable but the volume of sales to that segment remained small. The sales of hydromassage baths grew well in the non-housing segment, particularly in the south, to hotels, sauna centers, swimming pools and gyms. There has been a small up tick in sales in the North Eastern area as smaller and cheaper jacuzzi style tubs have been promoted in the same way as the synthetic soaker baths.

The shower tray market continued to contract in 2019 affected by the popularity of wet rooms in China. Despite of a large production of shower trays in China, most of the products are for exports and the domestic demand is very limited. The majority of shower trays are installed in conventional shower cubicles and hydromassage shower cubicles for non-housing and elite luxury housing in China, and the sales of separate shower trays are small. Most of the separate shower trays are synthetic

products due to the cheaper price, and the demand for ceramic and cast iron/steel shower trays remains very small due to heavy weight of the products and obsolete design.

Prices and Market Values

The average MSP of baths grew marginally in 2019 after several years of more extensive price rises due to raw material price increases and the conversion of manufacturing plants from coal fired to gas fired electricity. With all these infrastructure changes in place by the end of 2018, prices stabilized.

The prices in the shower trays sector grew, albeit at a very low rate.

New Housing

The sales of baths to the new housing segment declined in 2019 because of the decrease in dwelling size and bathroom size and competition from shower enclosure installations. There are less and less families opting to install baths in their new homes due to the lack of a bathing culture and the small size of bathrooms.

The majority of the baths sold to the new housing segment are synthetic due to cheaper prices and more versatile design, and the sales of cast iron and hydromassage baths to the new housings are small due to expensive price ranges and size of space needed for installation. The share of enamel steel baths in new housing segment is limited.

The sales of shower trays to the new housing market continued to decline due to substitution by wet rooms. As the majority of shower areas are now wet rooms in dwellings in China and therefore the share of all types of shower trays remains very limited; however synthetic shower trays have a relatively larger market share than ceramic and cast iron/steel shower trays due to cheaper prices. It is expected that the of share wet rooms will continue to grow. Cast iron/steel shower trays are expected to disappear in a few years due to lack of demand.

Housing RMI

The sales of baths in housing RMI continued to grow in 2019 due to the boom of home renovation and growing demands for comfort and innovated baths amongst young consumers. The sales of cast iron baths are more for bathroom improvement and replacement for existing steel or synthetic baths, as there are very few like-for-like replacements for cast iron baths due to the long lifespan. There are some like-for-like replacements for synthetic baths installed ten to twenty years ago, but mainly limited in the tier 1 cities where there are more bath installations. There is also some installation of stylish and smart baths in the home improvement sectors especially for young consumers pursuing a more stylish and comfortable lifestyle. The sales of steel baths and hydromassage baths in housing RMI segments are declining due to the obsolete features of the products, large

space occupation or water quality issues in the Chinese bathrooms. The shower tray market continued the declining trend due to the penetration of wet rooms.

It is expected that sales of baths in the housing RMI segment will remain stable or decline slightly in the next few years due to the popularity of shower enclosures. The shower tray market will continue to decline quickly due to the popularity of wet rooms.

Non-Housing

The non-housing segment for baths has been growing reasonably in 2019 due to the recovery of new hotel construction driven by the urbanization and growth of tourism, mainly attributed by the growth of the demands for synthetic and cast iron bath in star hotels, however, the installation of enamel steel baths and hydromassage baths declined due to the substitution by cast iron and synthetic baths or shower enclosures which are more durable, less noisy, easy to clean and more simple modern design.

Sales of shower trays to non-housing segments such as small beauty salons, gyms, etc., are small. There are nearly no shower trays sold to the hotel sectors due to popularity of wet rooms and the inconvenience to maintain units. The number of shower trays sold to the non-housing segments is expected to continue to decline.

Supply Structure

The Chinese bath and shower tray market is very fragmented with thousands of small manufacturers spread across the country and due to the low entry barriers in terms of regulations, investments and production technology, there is a large “Others” in the market share to represent this.

Foshan, Xiaoshan and Changge are the three main regions for bath production, where a large number of bath and shower tray suppliers in China are concentrated. Foshan suppliers are specialized in synthetic bath manufacture due to the rich PMMA raw material supplies. It is estimated that there are over 3,000 bathroom product manufacturers in Foshan, of which the majority are able to produce synthetic baths and shower trays. Xiaoshan and Changge are populated by many small to medium suppliers of low end baths. Many of these manufacturers do not have own brands but provide OEM services to foreign or local brands.

About 90% of the baths (mainly medium to low end products) are supplied by Chinese companies, led by ARROW, FAENZA, LAGUNA, DONGPENG, JOMOO, HEGII, and APOLLO, while only 10% of the market are occupied by international companies, such as KOHLER, TOTO, AMERICAL STADNARD, ROCA, VILLEROY &BOCH, DURAVIT, KALDEWEI, etc. ARROW is the overall leader in the bath market in 2019, followed by KOHLER, TOTO, AMERICAN STANDARD, FAENZA, LAGUNA, DONGPENG, JOMOO, HEGII, LAUFEN, and APOLLO.

In cast iron bath market, ROCA, KOHLER, TOTO, and AMERICAN STANDARD have dominated the medium to high end market for many years with the rest of the market consisting of tens of Chinese suppliers, of which only SOUTHSEA has sales over 10,000 pieces and most of them had smaller sales in 2019.

The steel bath market is small with only a few players active, and they are mainly Chinese suppliers led by LUXING ENAMEL. KALDEWEI is the leading international manufacturer with others like such as ROCA and VILLEROY & BOCH.

The synthetic bath market is fairly fragmented and there are hundreds of local players dominating the economy/lower segment. The top five players are ARROW, KOHLER, TOTO, AMERICAN STANDARD and LAGUNA. The international manufacturers supply the medium to high end of the market.

The hydromassage bath market is also dominated by the Chinese suppliers mainly in Guangdong provinces, which accounted for about 90% of the market. The market is very fragmented since there are many local manufacturers supplying small volumes of products. ARROW led the hydromassage bath market in 2019, followed by KOHLER, JOMOO, FAENZA, and DONGPENG. The specialized suppliers such as SHENGYA, SSWW, LUYISI, ORANS, and APPOLO saw their shares decline in 2019 due to the slump in demand.

There are only a few suppliers in the shower tray market. The suppliers are mainly large sanitary ware manufacturers, such as HUIDA, IMEX, HCG, HANGBIAO, DONGPENG, DURAVIT, SWELL, and ROCA. Most suppliers only provide synthetic shower trays, while steel / cast iron shower trays are supplied by cast iron bath manufacturers such as SHUNSHINE BATH, YUNFENG and SOUTHSEA BATH. The ceramic shower tray market is small and supplied by several Chinese sanitary ware companies such as TANG TAO, TAITAO, etc.

The major activities include:

GOLDMAN stopped production and supplies of shower trays in 2019.

YINGJILI turned its sales of hydromassage baths from only to overseas market to the domestic market in 2019.

TOTO claimed to build a bath factory in Nanjing in July 2019 with an investment of JPY 7.5 billion. The factory is to start the operation in 2021 and the annual capacity is expected to be 110,000 baths.

Forecasts

Given the exceptional circumstances related to the spread of the COVID-19 virus and measures in place for its containment, forecasts for 2020 and beyond are based on the latest information available at the time of publication. BRG has endeavored to factor in the latest developments – see Macroeconomic data in

Table A.3 and supporting summary table at the end of the Background section.

The COVID-19 pandemic has widely affected the sales of all baths and shower trays in China due to the cessation of both production and construction in the first quarter. With the resumption of production and construction/RMI activities, it is estimated that the bath market will decline by 14% and the shower tray market by over 27% (this being said, the shower tray market was already in decline) in 2020.

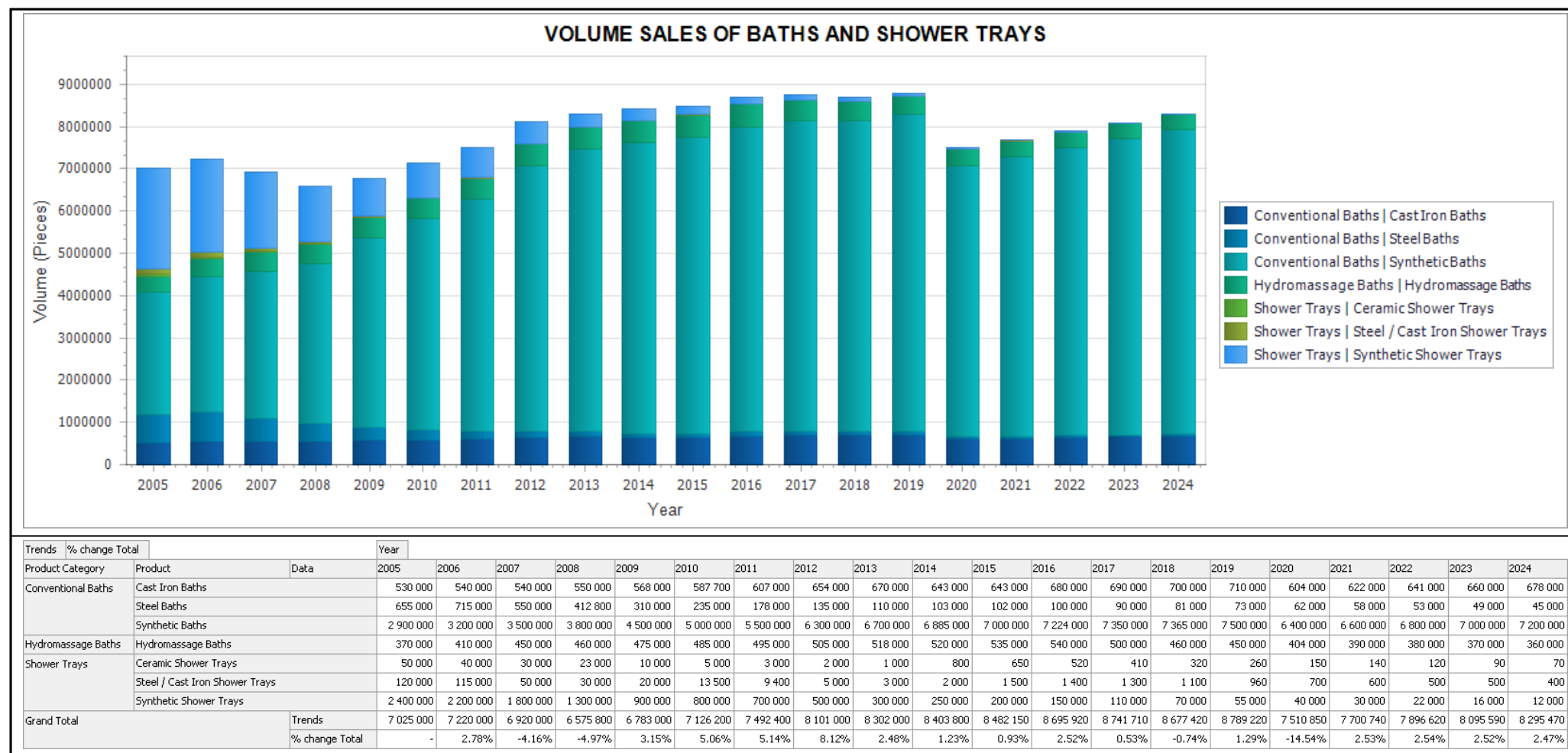
The installation of baths depends on the size of bathroom and the size of bathrooms is unlikely to change much in new dwellings in China over the next few years. The average size of a bathroom is about 5-6 square meters, with a smaller size at about 3-4 m² and a few larger bathrooms at about 9-10 m² in villas. This situation should not change in next few years due to increased construction of high-rise apartments instead of houses. Therefore, baths will continue to face competition from shower enclosures when the consumers make the purchasing decisions.

There are still some growth opportunities for bath sales, especially for those products suitable for young consumers looking for comfort, functionality and smart design. Therefore, the sales of smaller sized, colorful, stylish and smart baths, such as cast iron and synthetic baths, could grow amongst the young and rich.

Post pandemic, the sales of baths to hotels and other non-housing segments should pick up again with the construction and renovation of new facilities driven by the tourist industry and increased leisure time for the consumer. Sales of obsolete products such as steel enamel baths and large products such as hydromassage baths will continue to decline due to the changes of consumer habits.

1.2 Historical Trends and Forecasts

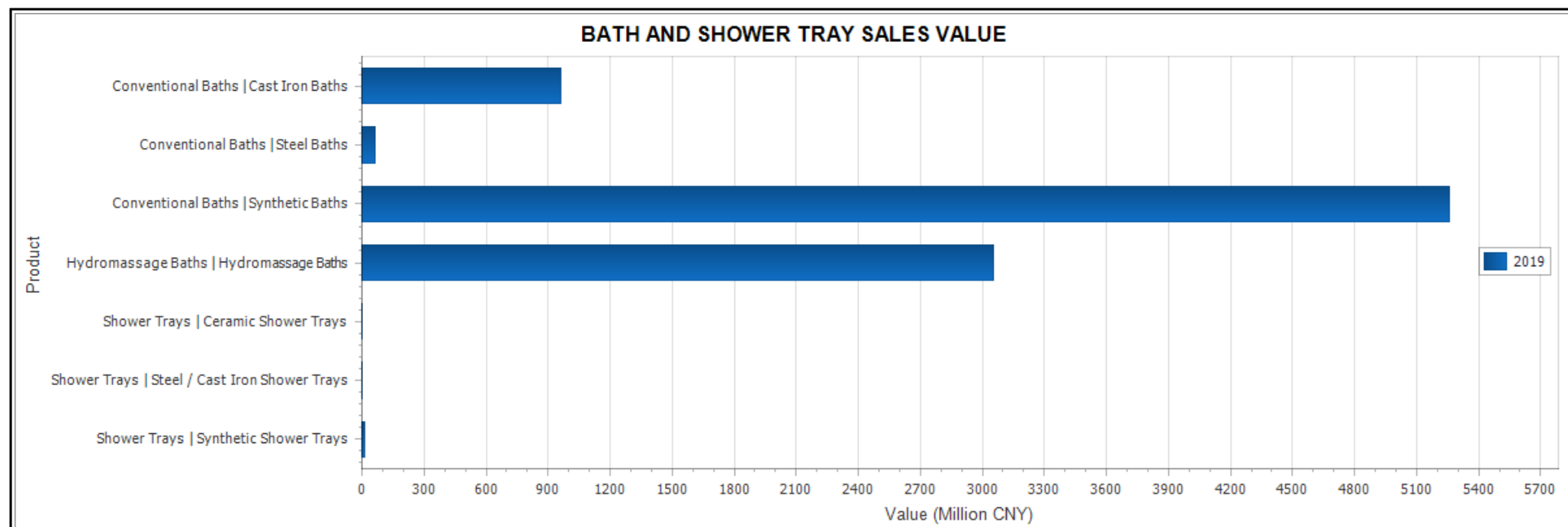
CN - SUMMARY OF TRENDS IN THE BATH AND SHOWER TRAY MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

1.3.1 Prices and Market Values : Baths and Shower Trays

CN - ESTIMATED VALUES OF THE BATH AND SHOWER TRAY MARKET

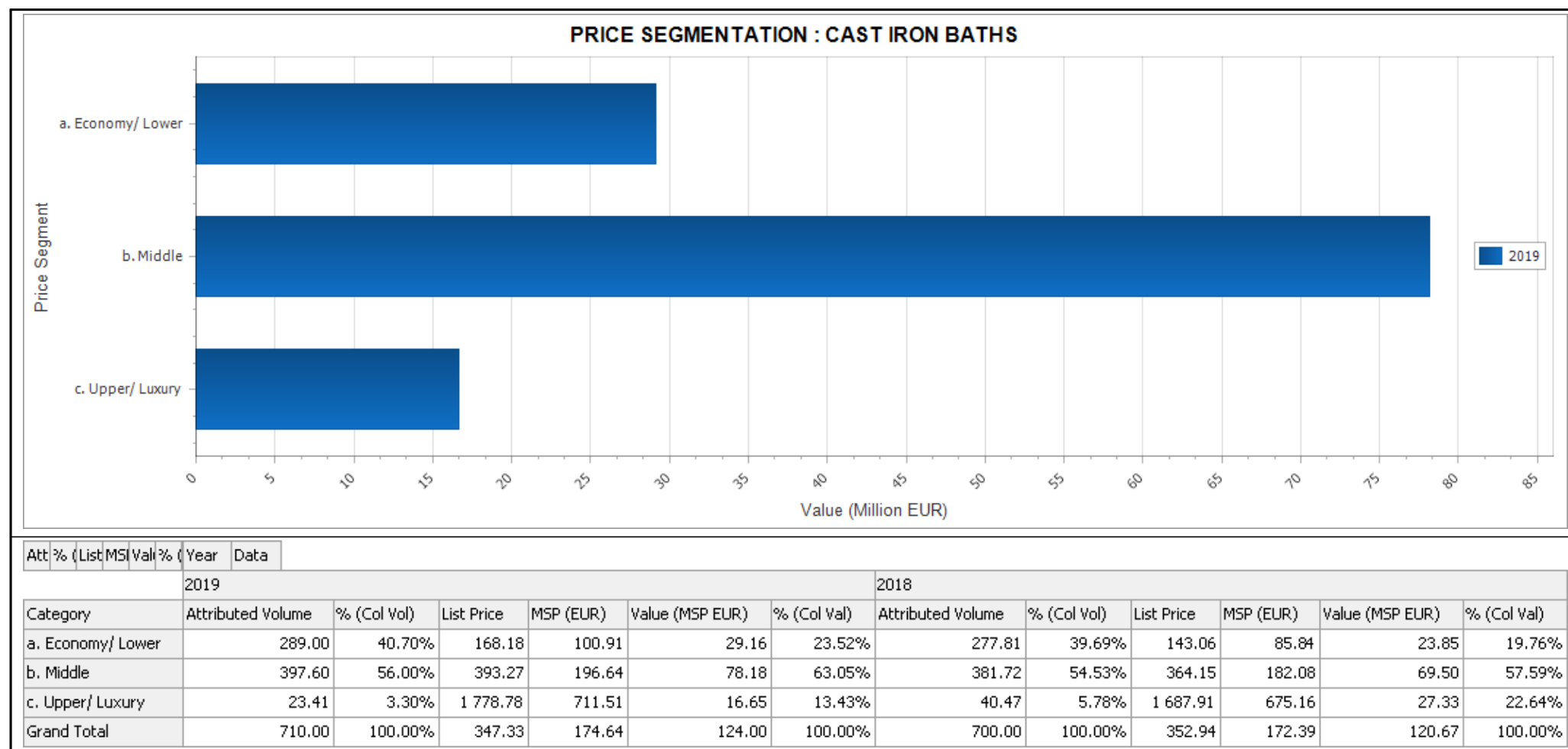


Volumes	MSP (Local)	Value (MSP)	MSP (EUR)	Value (MSP)	Year	Data						
2019						2018						
Product Category	Product	Volumes	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	Volumes	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	
Conventional Baths	Cast Iron Baths	710 000	1,349.97	958.48	174.64	123.99	700 000	1,345.59	941.91	172.39	120.67	
	Steel Baths	73 000	885.47	64.64	114.55	8.36	81 000	875.54	70.92	112.17	9.09	
	Synthetic Baths	7 500 000	701.27	5 259.53	90.72	680.40	7 365 000	699.29	5 150.27	89.59	659.83	
Hydromassage Baths	Hydromassage Baths	450 000	6,780.37	3 051.17	877.15	394.72	460 000	6,770.72	3 114.53	867.43	399.02	
Shower Trays	Ceramic Shower Trays	260	153.44	0.04	19.85	0.01	320	152.99	0.05	19.60	0.01	
	Steel / Cast Iron Shower Trays	960	215.59	0.21	27.89	0.03	1 100	213.17	0.23	27.31	0.03	
	Synthetic Shower Trays	55 000	242.03	13.31	31.31	1.72	70 000	239.78	16.78	30.72	2.15	
Grand Total		8 789 220	1,063.50	9 347.37	137.58	1 209.23	8 677 420	1,071.14	9 294.70	137.23	1 190.79	

Source: BRG based on industry interviews. Local Prices Shown in CNY. Values Shown in Millions (at MSP = Manufacturers' Selling Price).

1.3.2 Price Segmentation : Cast Iron Baths

CN - ESTIMATED PRICE SEGMENTATION FOR CAST IRON BATHS

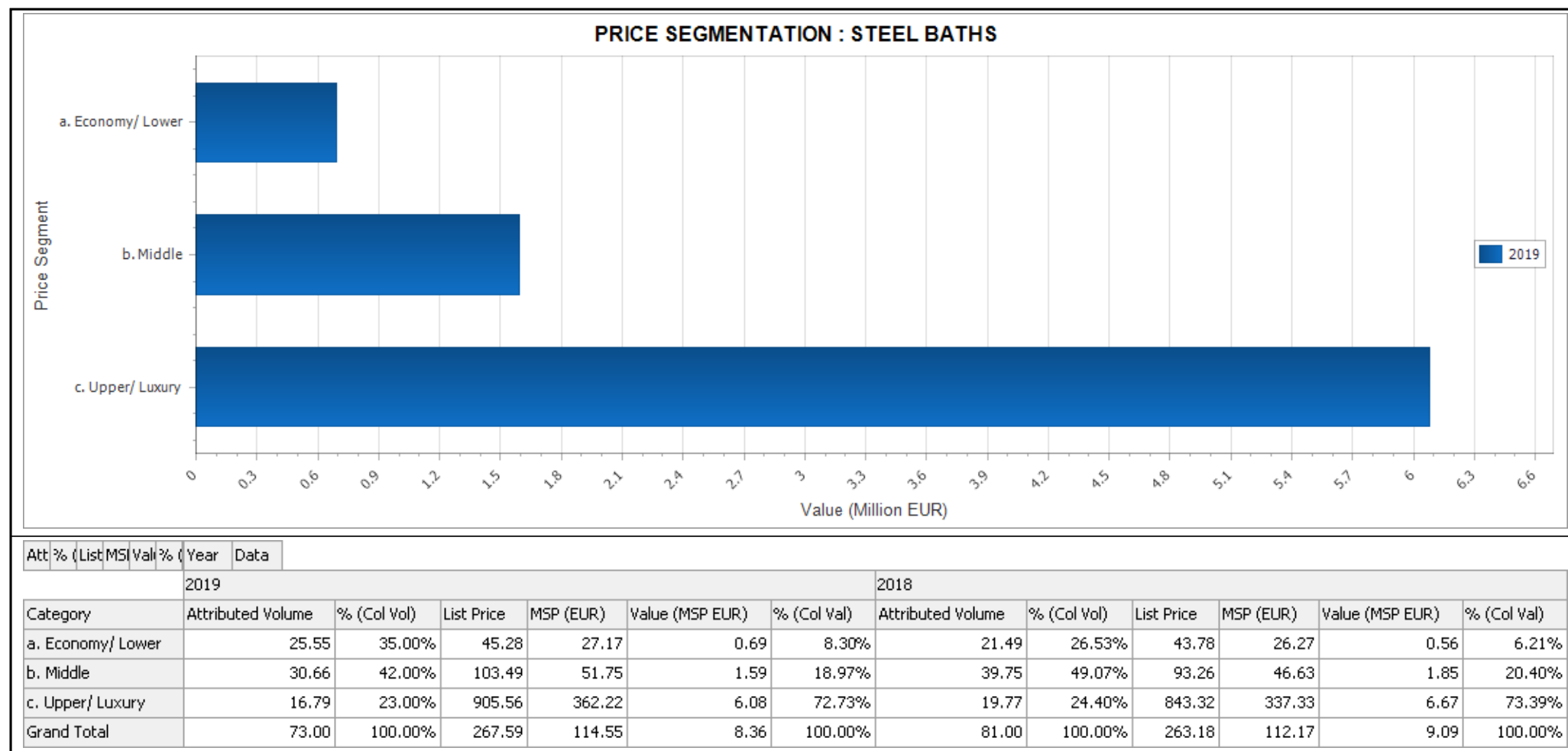


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€150, Middle €150-€300, Upper/ Luxury >€300

1.3.3 Price Segmentation : Steel Baths

CN - ESTIMATED PRICE SEGMENTATION FOR STEEL BATHS

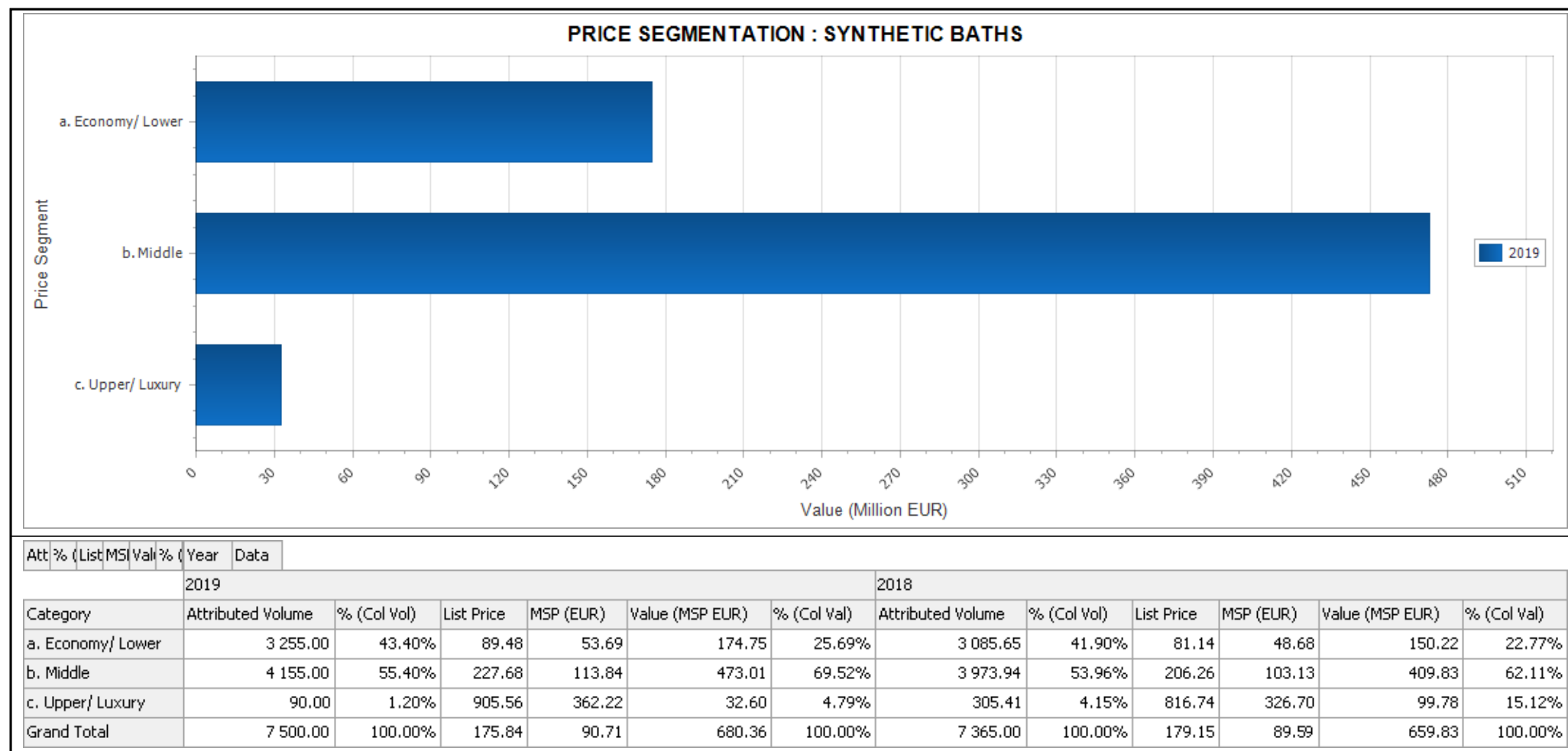


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€30, Middle €30-€150, Upper/ Luxury >€150

1.3.4 Price Segmentation : Synthetic Baths

CN - ESTIMATED PRICE SEGMENTATION FOR SYNTHETIC BATHS

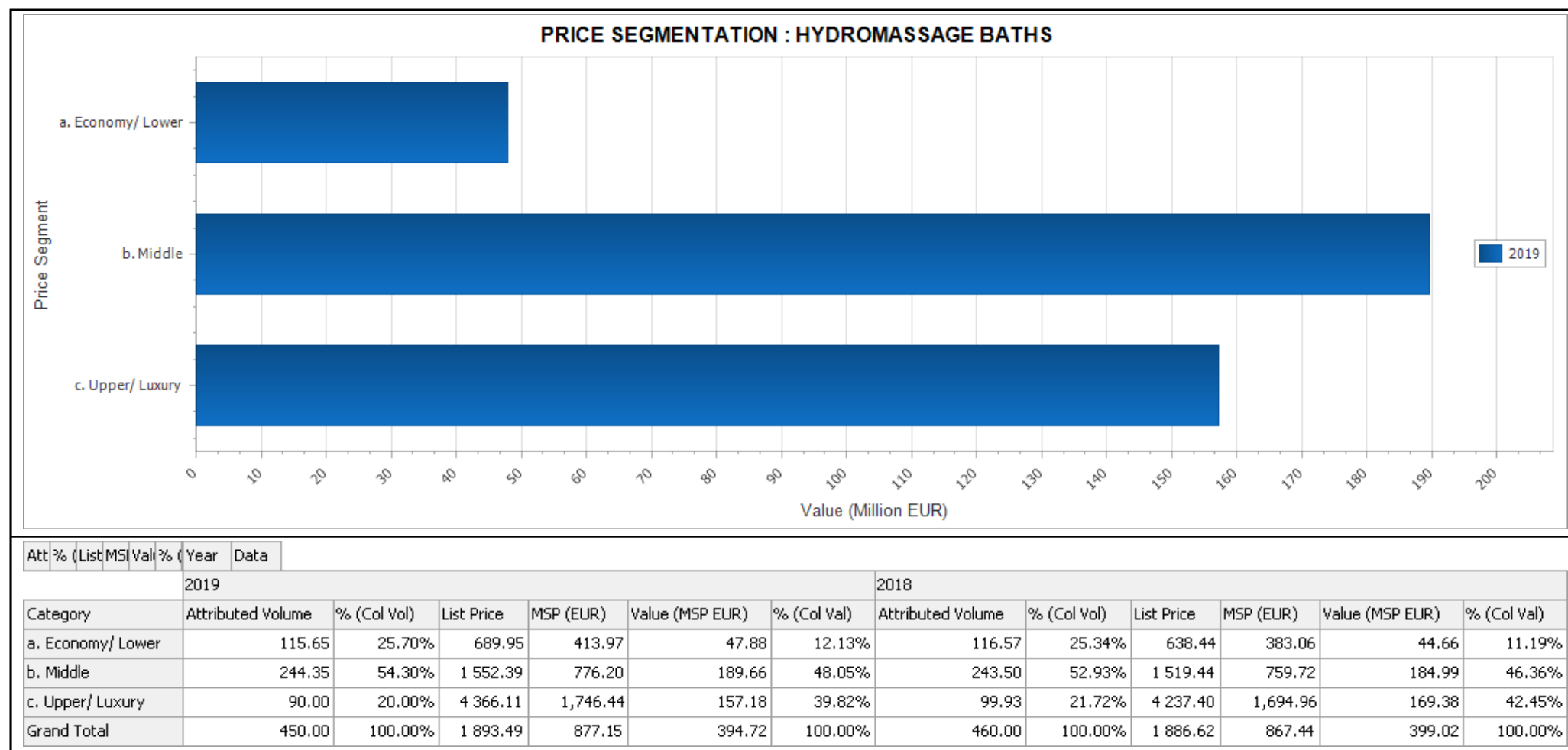


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€100, Middle €100-€300, Upper/ Luxury >€300

1.3.5 Price Segmentation : Hydromassage Baths

CN - ESTIMATED PRICE SEGMENTATION FOR HYDROMASSAGE BATHS

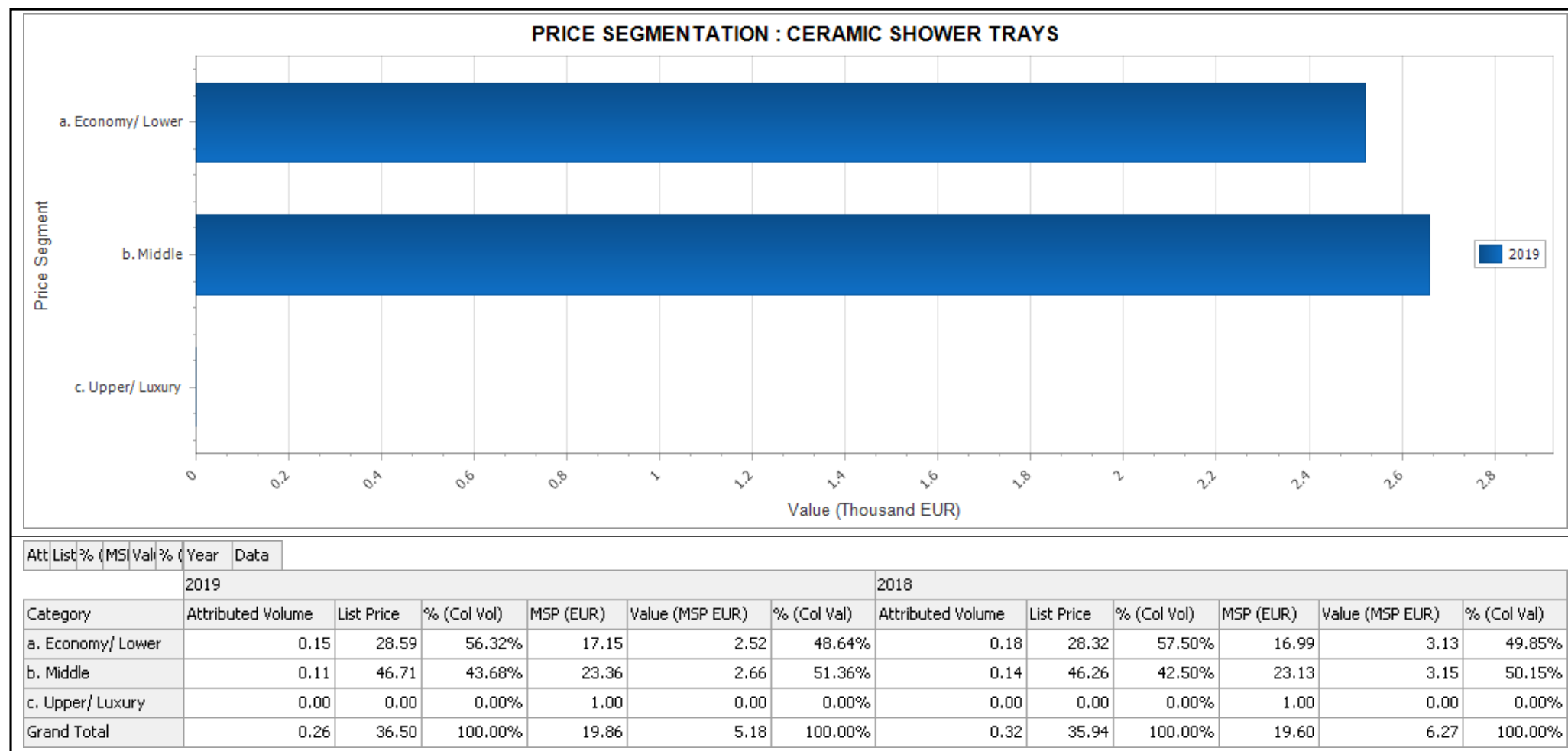


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€500, Middle €500-€1,000, Upper/ Luxury >€1,000

1.3.6 Price Segmentation : Ceramic Shower Trays

CN - ESTIMATED PRICE SEGMENTATION FOR CERAMIC SHOWER TRAYS

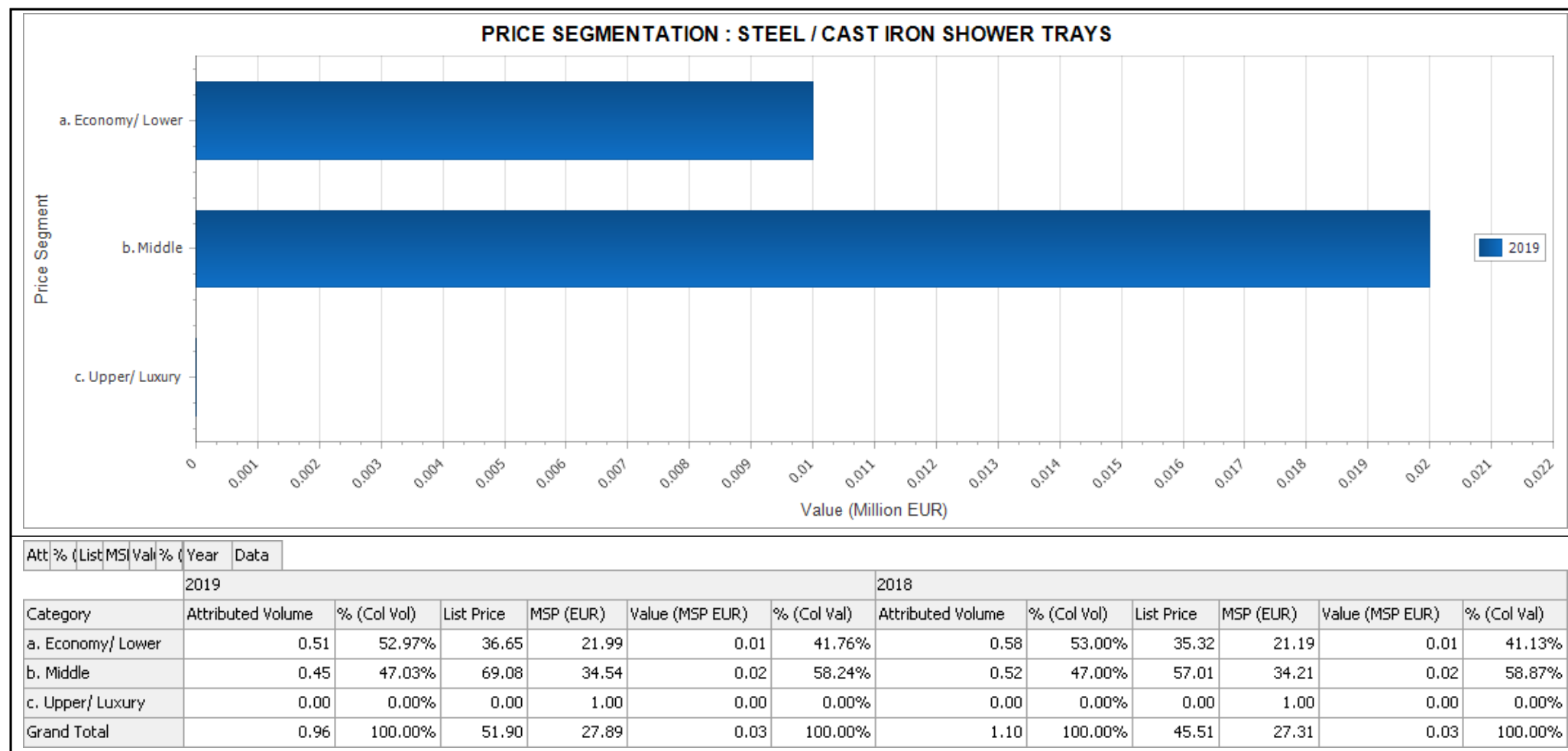


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€20, Middle €20-€50, Upper/ Luxury >€50

1.3.7 Price Segmentation : Steel/ Cast Iron Shower Trays

CN - ESTIMATED PRICE SEGMENTATION FOR STEEL / CAST IRON SHOWER TRAYS

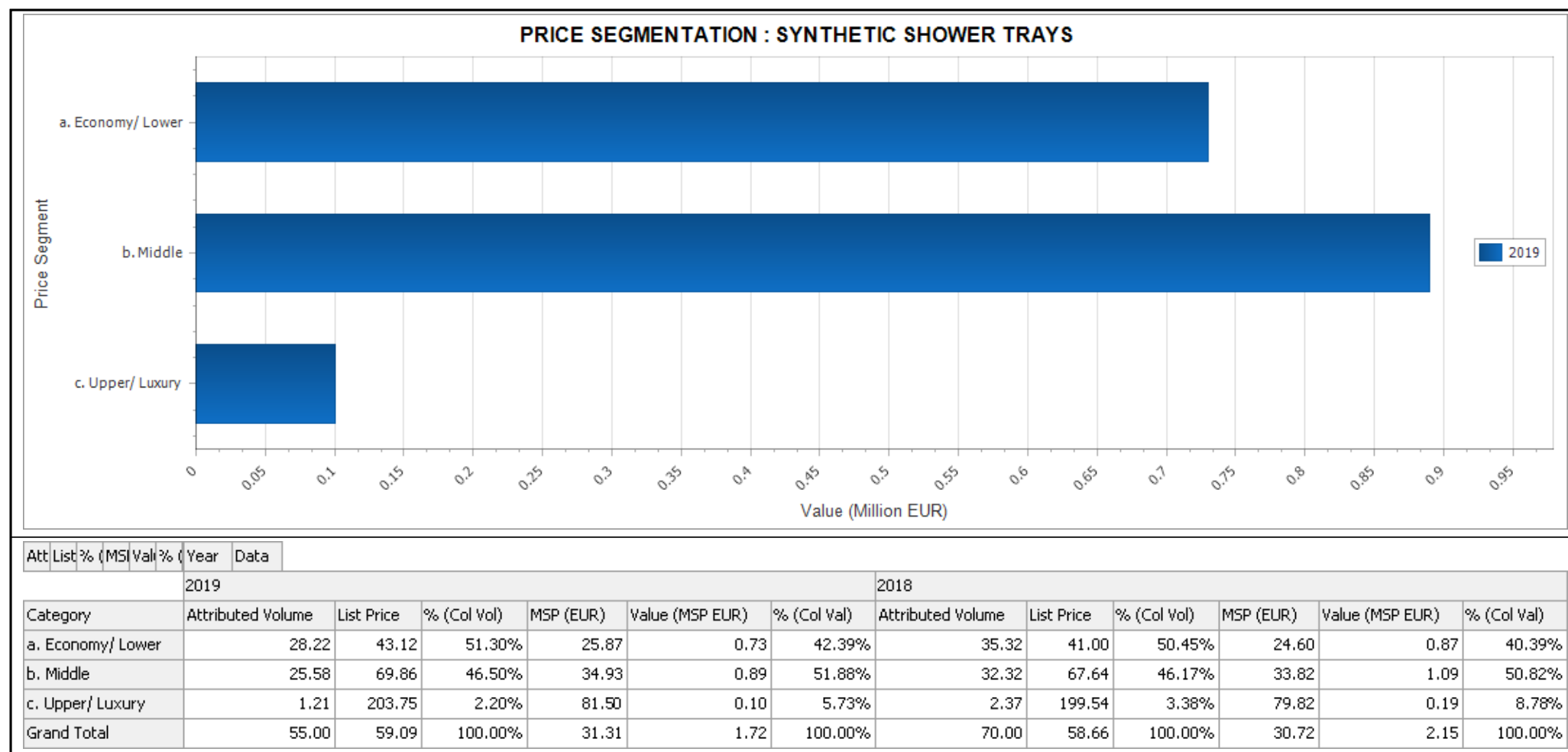


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€25, Middle €25-€50, Upper/ Luxury >€50

1.3.8 Price Segmentation : Synthetic Shower Trays

CN - ESTIMATED PRICE SEGMENTATION FOR SYNTHETIC SHOWER TRAYS

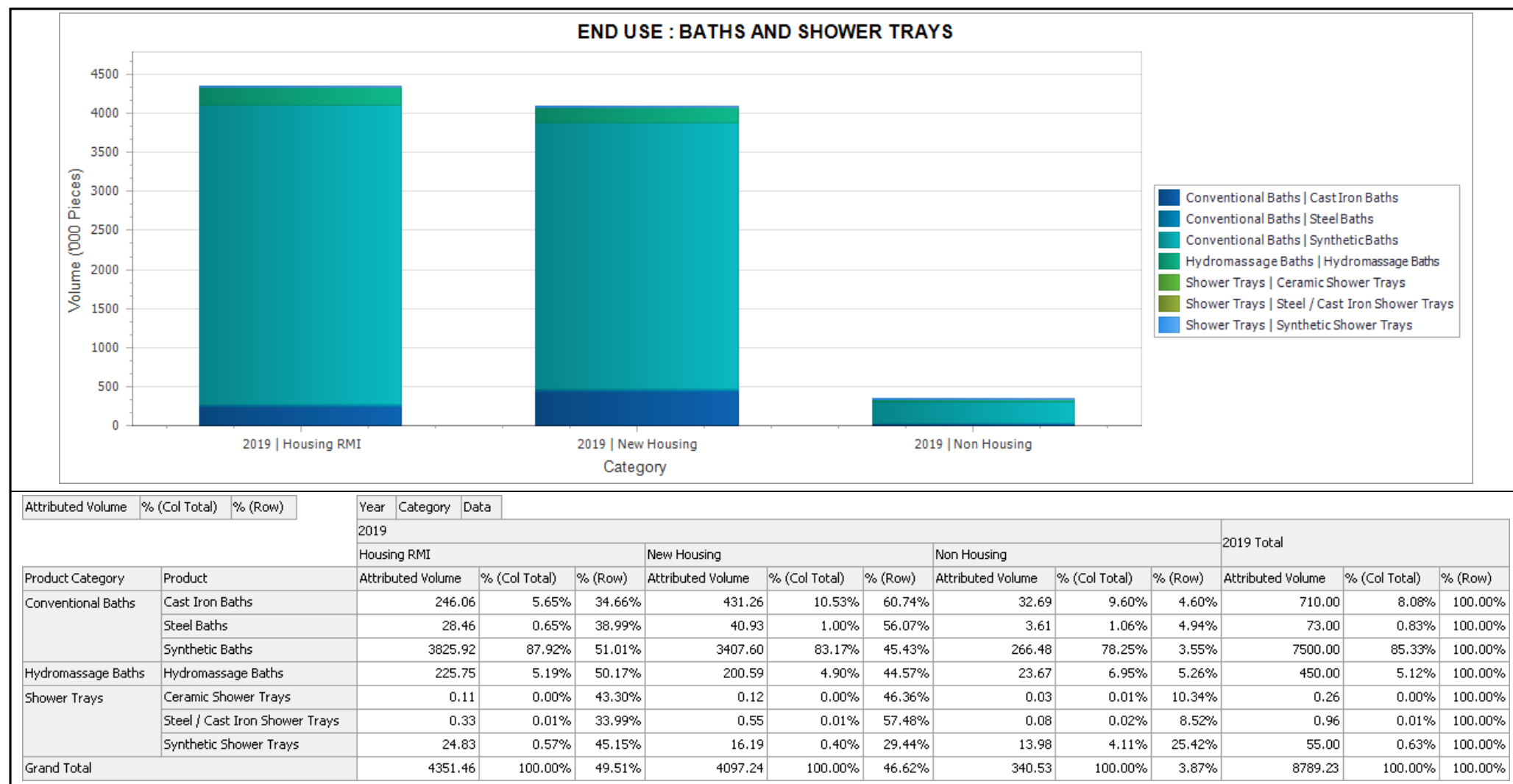


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€30, Middle €30-€60, Upper/ Luxury >€60

1.4.1 End Use : Base Year

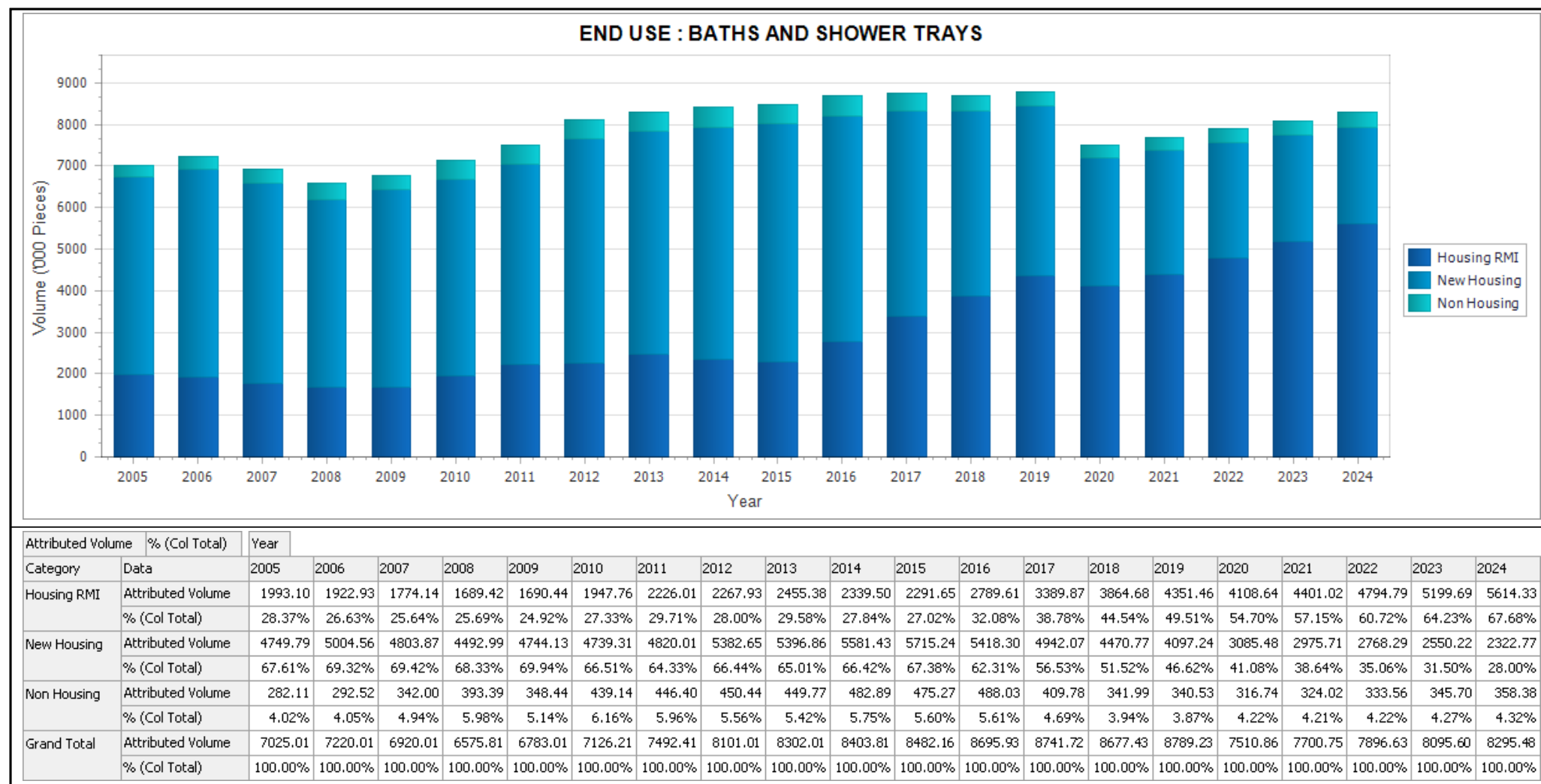
CN - SALES OF BATHS AND SHOWER TRAYS BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.2 End Use : Historical and Forecasts

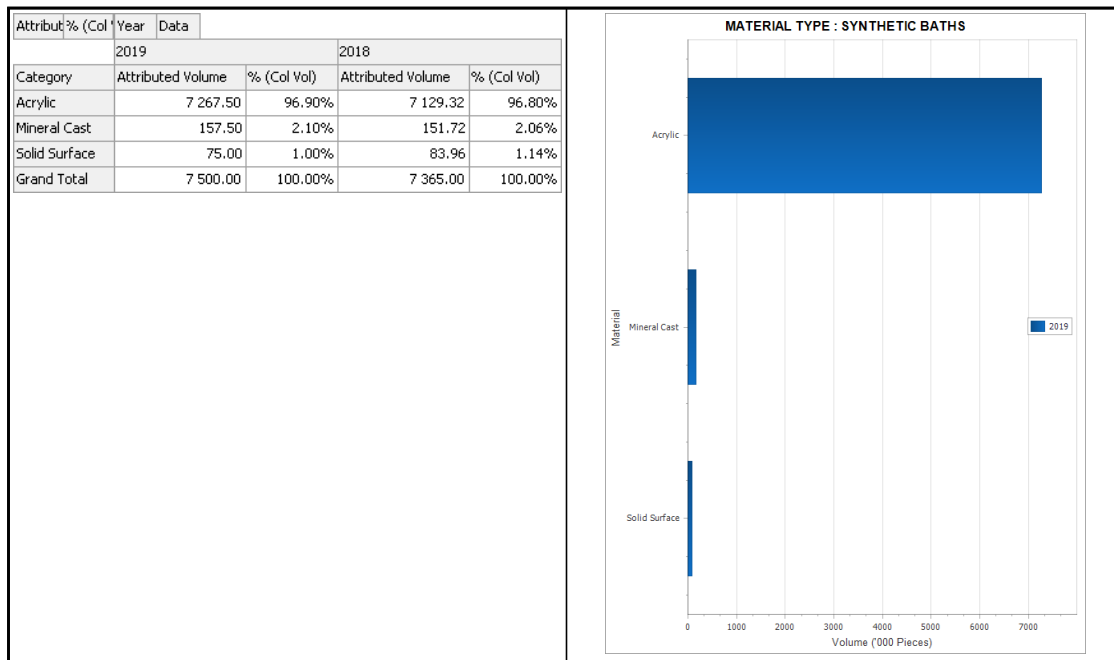
CN - SALES OF BATHS AND SHOWERS TRAYS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.3 Sales by Material : Synthetic Baths

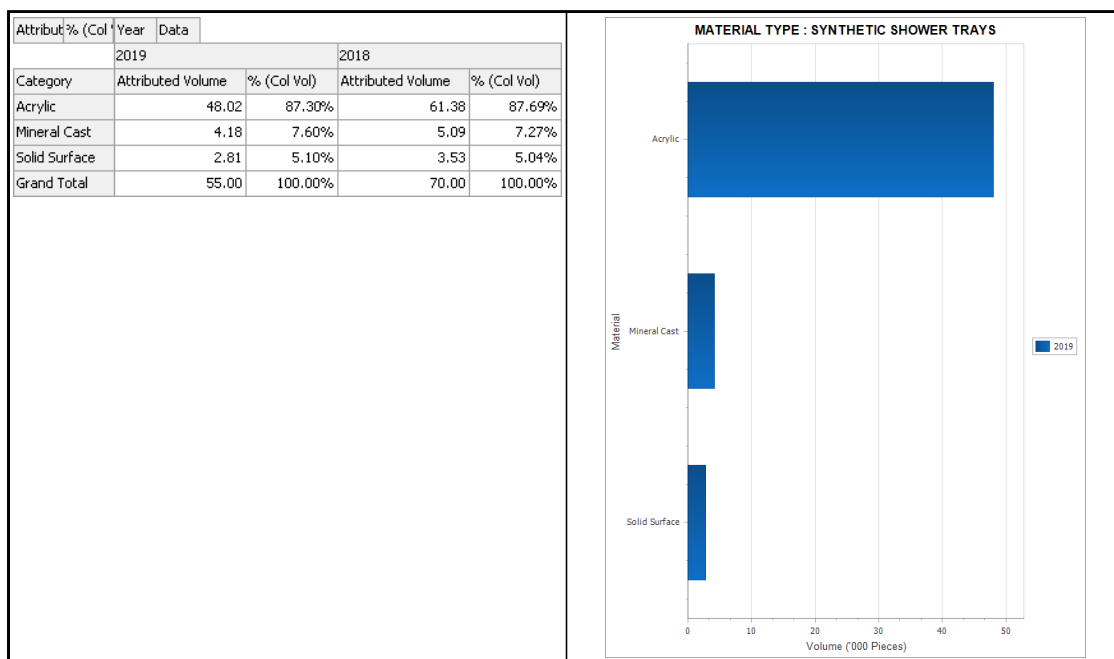
CN - SYNTHETIC BATH SALES BY MATERIAL TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.4.4 Sales by Material : Synthetic Shower Trays

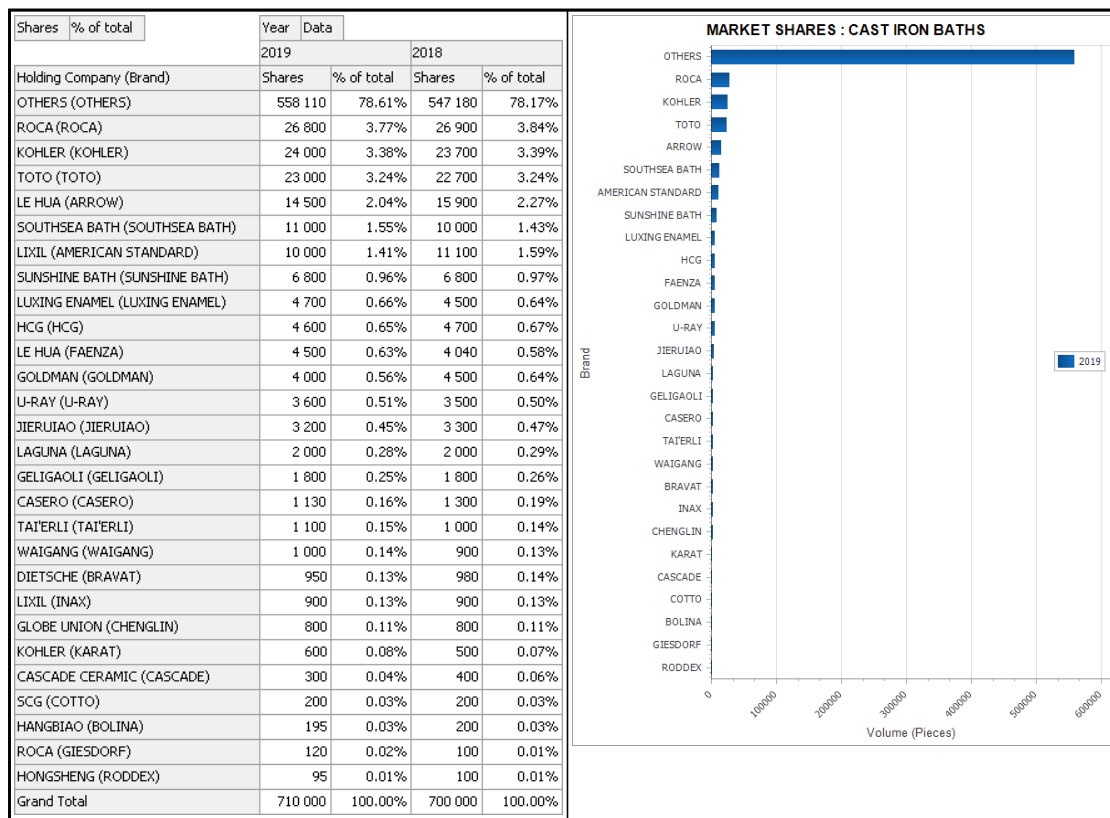
CN - SYNTHETIC SHOWER TRAY SALES BY MATERIAL TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

1.5.1 Market Shares : Cast Iron Baths

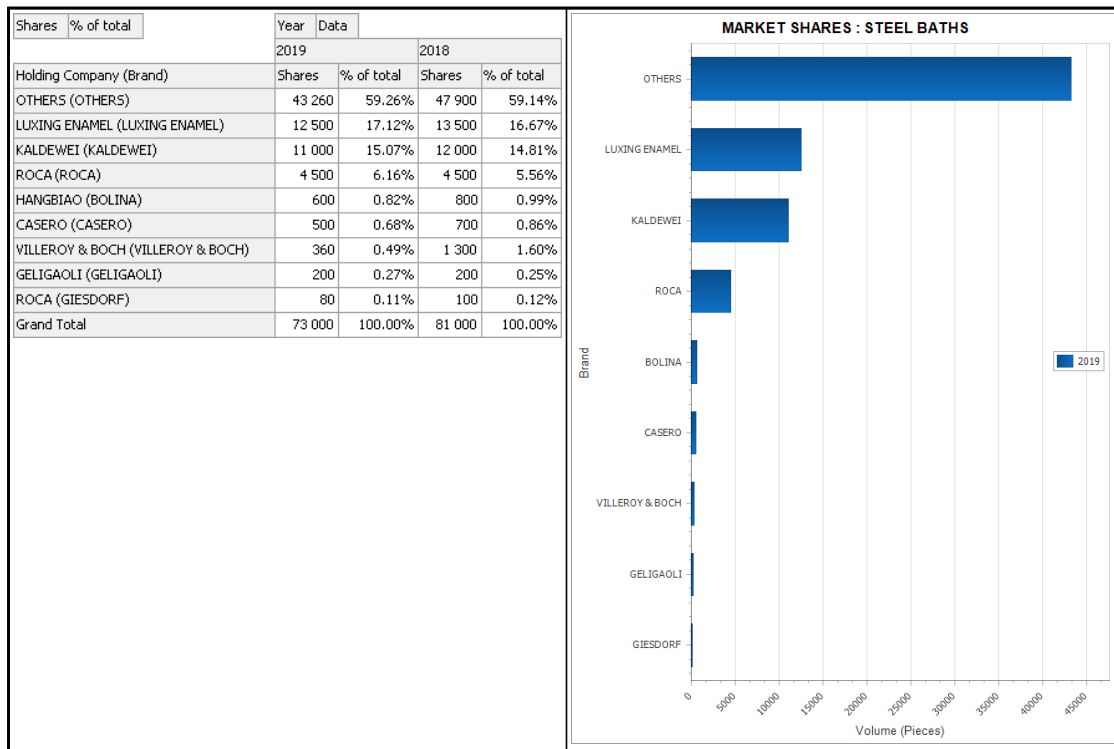
CN - ESTIMATED MARKET SHARES FOR CAST IRON BATHS



Source: BRG based on industry interviews.

1.5.2 Market Shares : Steel Baths

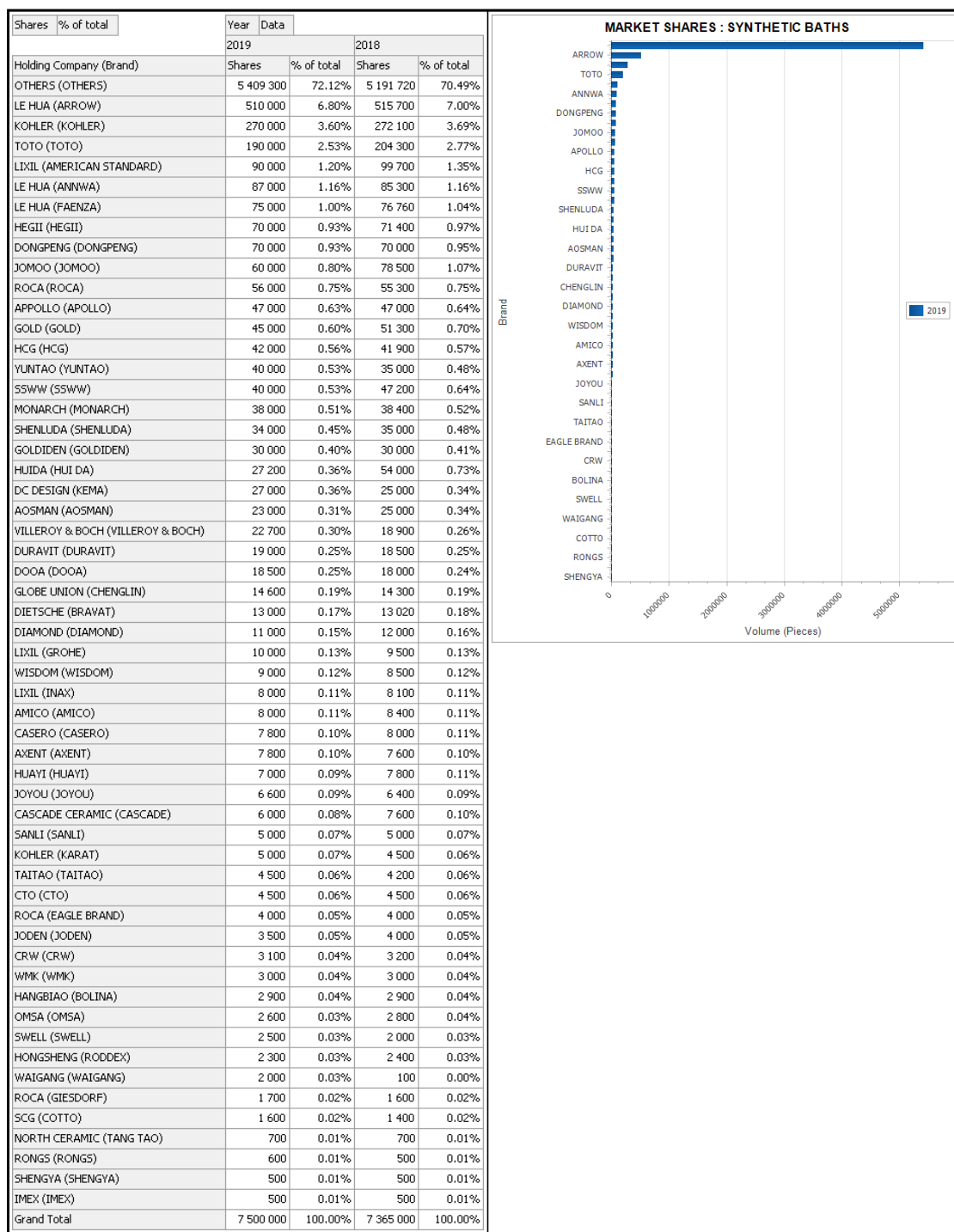
CN - ESTIMATED MARKET SHARES FOR STEEL BATHS



Source: BRG based on industry interviews.

1.5.3 Market Shares : Synthetic Baths

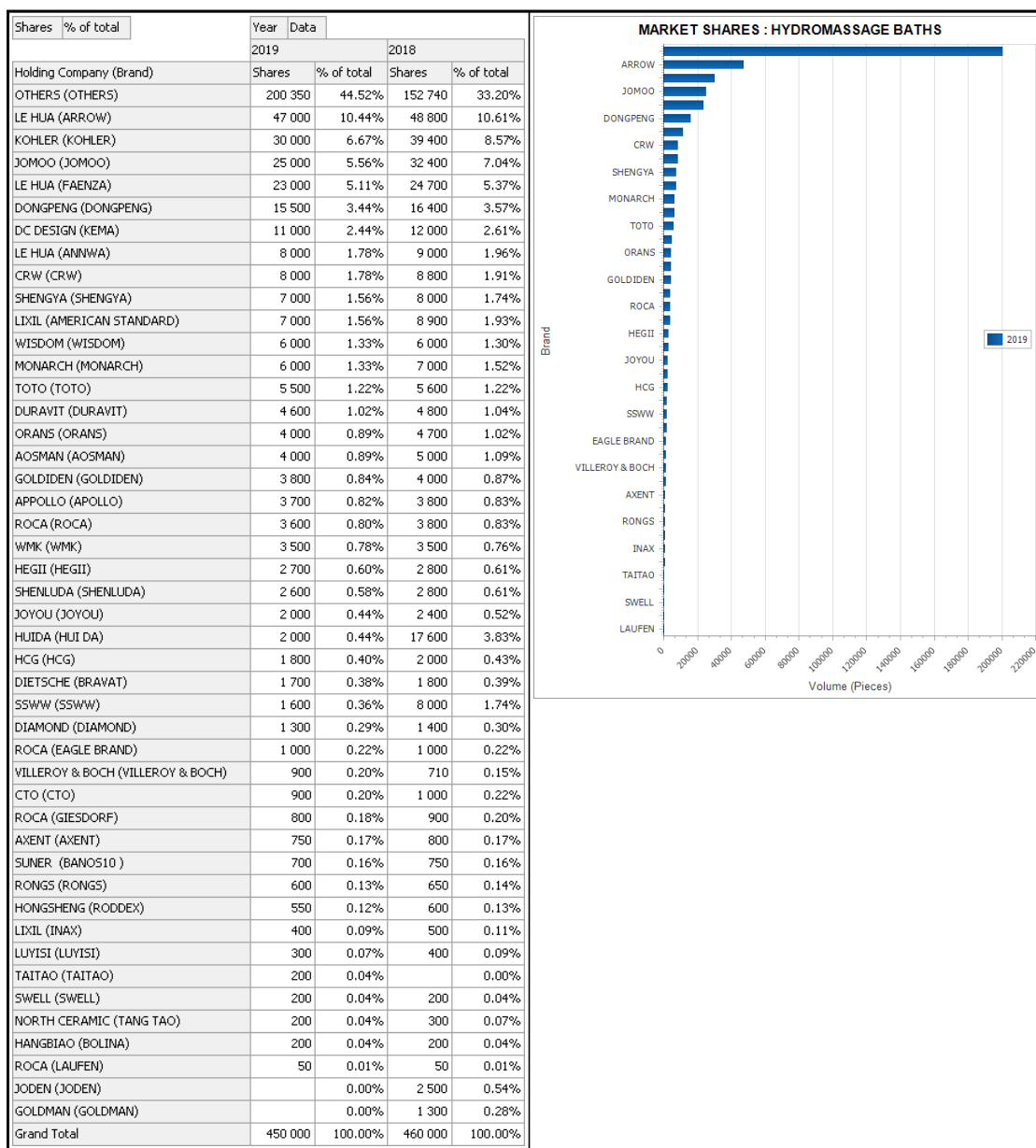
CN - ESTIMATED MARKET SHARES FOR SYNTHETIC BATHS



Source: BRG based on industry interviews.

1.5.4 Market Shares : Hydromassage Baths

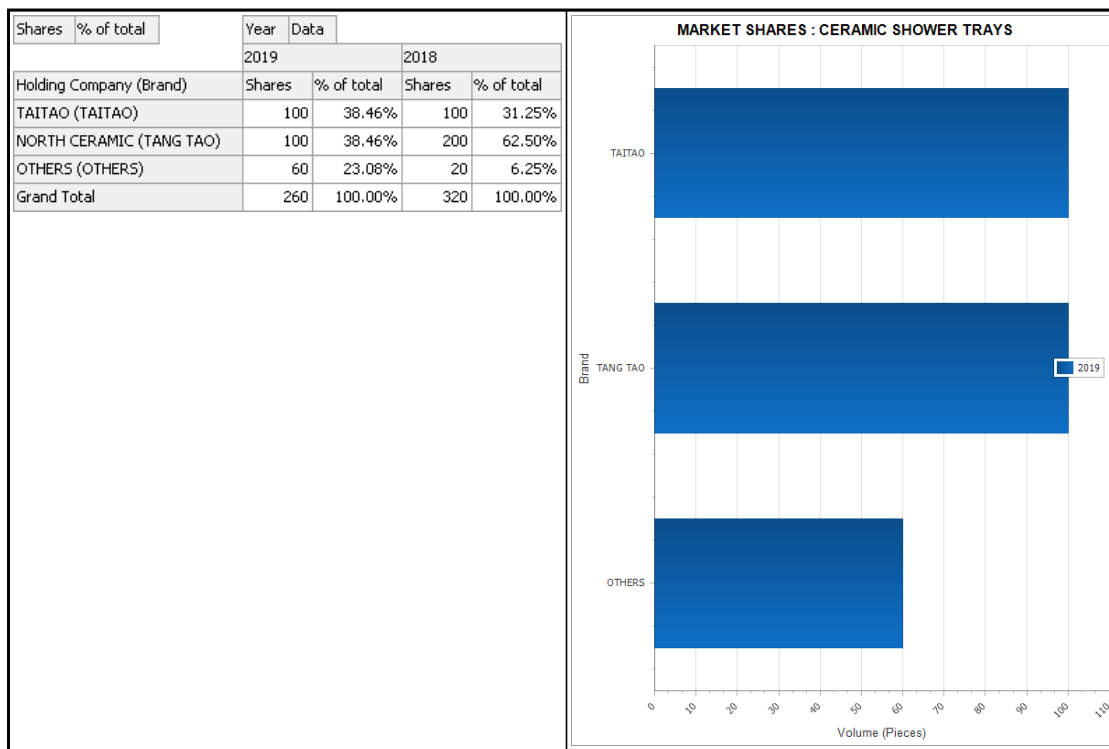
CN - ESTIMATED MARKET SHARES FOR HYDROMASSAGE BATHS



Source: BRG based on industry interviews.

1.5.5 Market Shares : Ceramic Shower Trays

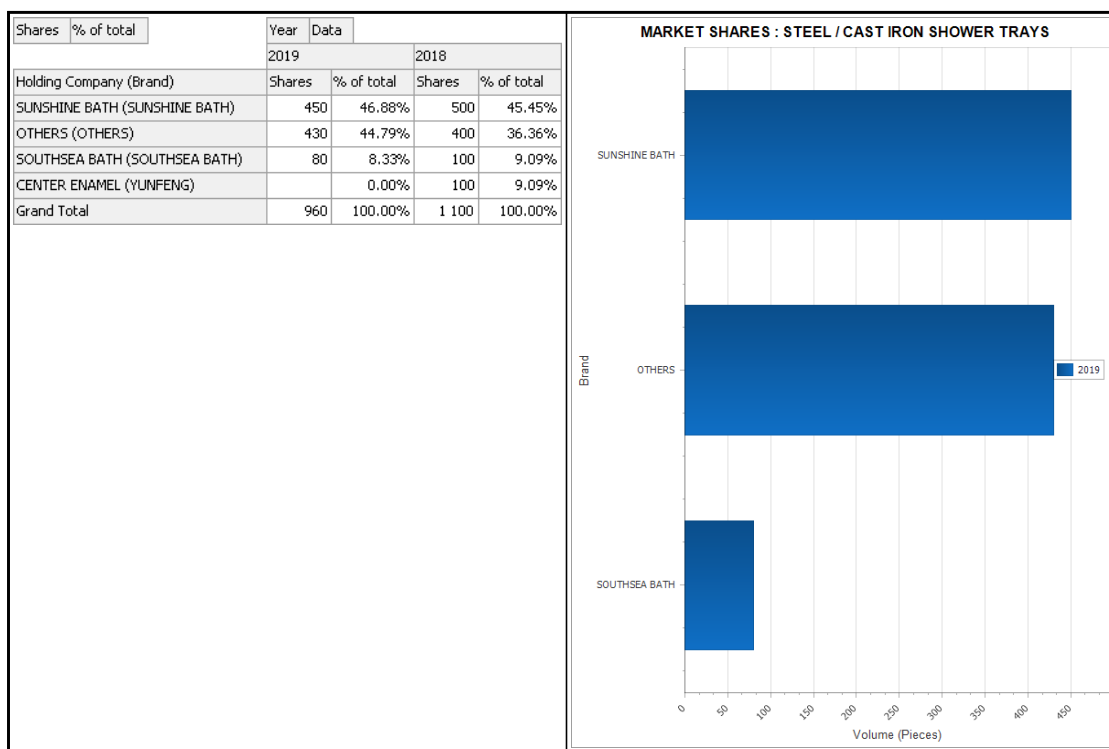
CN - ESTIMATED MARKET SHARES FOR CERAMIC SHOWER TRAYS



Source: BRG based on industry interviews

1.5.6 Market Shares : Steel / Cast Iron Shower Trays

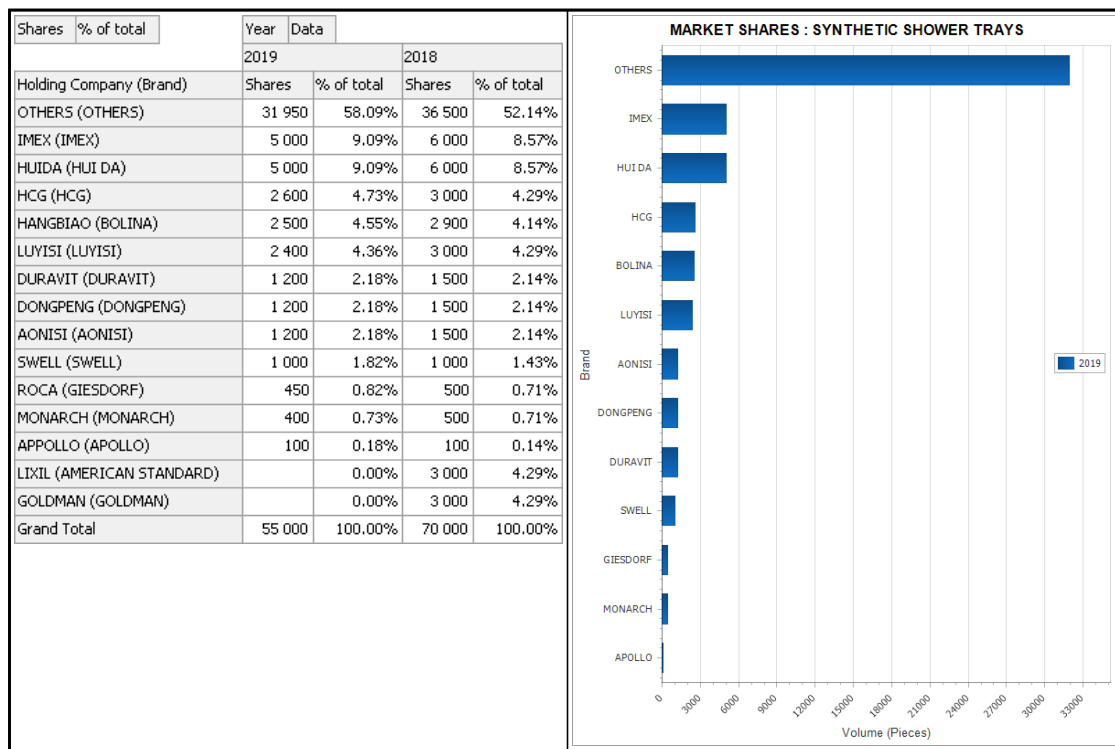
CN - ESTIMATED MARKET SHARES FOR STEEL / CAST IRON SHOWER TRAYS



Source: BRG based on industry interviews.

1.5.7 Market Shares : Synthetic Shower Trays

CN - ESTIMATED MARKET SHARES FOR SYNTHETIC SHOWER TRAYS



Source: BRG based on industry interviews.

1.6.1 CN - IMPORTS OF BATHS AND SHOWER TRAYS - CAST IRON 2019

Source Country	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Germany	64	77.5%	272	69.0%	69	10.5%	280	18.8%
Spain	12	14.5%	52	13.2%	507	76.6%	616	41.3%
Italy	3	3.1%	43	10.9%	5	0.7%	64	4.3%
Portugal	2	2.1%	13	3.3%	77	11.7%	484	32.5%
United States of America	1	1.4%	8	2.0%	1	0.1%	10	0.7%
China	1	1.1%	2	0.5%	1	0.1%	2	0.1%
OTHERS	0	0.2%	4	1.0%	2	0.2%	34	2.3%
Total	83	100.0%	394	100.0%	661	100.0%	1 490	100.0%

Source : Trademap

1.6.2 CN - EXPORTS OF BATHS AND SHOWER TRAYS - CAST IRON 2019

Destination	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Russian Federation	4 595	35.8%	6 631	24.1%	3 386	19.9%	5 026	15.7%
United States of America	3 988	31.1%	11 416	41.5%	7 930	46.5%	16 884	52.6%
Tunisia	631	4.9%	389	1.4%	470	2.8%	268	0.8%
Uzbekistan	448	3.5%	527	1.9%	417	2.4%	673	2.1%
Hong Kong, China	380	3.0%	849	3.1%	888	5.2%	2 177	6.8%
United Kingdom	362	2.8%	1 309	4.8%	328	1.9%	865	2.7%
France	321	2.5%	759	2.8%	375	2.2%	826	2.6%
Portugal	236	1.8%	187	0.7%	432	2.5%	348	1.1%
Mongolia	189	1.5%	55	0.2%	288	1.7%	77	0.2%
Lithuania	158	1.2%	210	0.8%	536	3.1%	625	1.9%
Philippines	135	1.0%	516	1.9%	65	0.4%	260	0.8%
OTHERS	1 398	10.9%	4 662	16.9%	1 924	11.3%	4 059	12.6%
Total	12 841	100.0%	27 510	100.0%	17 039	100.0%	32 088	100.0%

Source : Trademap

1.6.3 CN - IMPORTS OF BATHS AND SHOWER TRAYS - STEEL 2019

Source Country	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Germany	1 442	90.6%	6 717	94.0%	1 420	89.8%	6 120	90.3%
Spain	134	8.4%	346	4.8%	140	8.9%	378	5.6%
OTHERS	15	0.9%	83	1.2%	21	1.3%	277	4.1%
Total	1 591	100.0%	7 146	100.0%	1 581	100.0%	6 775	100.0%

Source : Trademap

1.6.4 CN - EXPORTS OF BATHS AND SHOWER TRAYS - STEEL 2019

Destination	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Chile	1 193	21.8%	1 916	20.0%	1 522	19.8%	2 246	18.2%
Russian Federation	1 156	21.1%	1 748	18.3%	1 830	23.8%	2 637	21.4%
United Kingdom	953	17.4%	1 435	15.0%	956	12.4%	1 318	10.7%
Algeria	680	12.4%	1 172	12.3%	1 188	15.5%	1 902	15.4%
Sweden	269	4.9%	745	7.8%	273	3.6%	678	5.5%
United Arab Emirates	193	3.5%	272	2.8%	156	2.0%	165	1.3%
Argentina	137	2.5%	208	2.2%	339	4.4%	506	4.1%
United States of America	91	1.7%	403	4.2%	50	0.7%	198	1.6%
Ukraine	88	1.6%	127	1.3%	107	1.4%	149	1.2%
Botswana	86	1.6%	141	1.5%	148	1.9%	233	1.9%
Australia	65	1.2%	141	1.5%	72	0.9%	150	1.2%
OTHERS	559	10.2%	1 259	13.2%	1 036	13.5%	2 165	17.5%
Total	5 469	100.0%	9 567	100.0%	7 677	100.0%	12 347	100.0%

Source : Trademap

1.6.5 CN - IMPORTS OF BATHS AND SHOWER TRAYS - SYNTHETIC 2019

Source Country	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Egypt	398	23.3%	2 412	17.3%	310	20.2%	1 937	14.2%
Netherlands	334	19.6%	3 590	25.8%	221	14.3%	2 526	18.5%
Belgium	272	15.9%	1 914	13.8%	246	16.0%	2 418	17.7%
South Africa	115	6.7%	443	3.2%	42	2.7%	168	1.2%
Poland	96	5.6%	465	3.3%	31	2.0%	543	4.0%
Japan	90	5.2%	1 216	8.7%	97	6.3%	1 256	9.2%
Germany	82	4.8%	1 041	7.5%	129	8.4%	1 598	11.7%
Italy	78	4.6%	719	5.2%	147	9.5%	898	6.6%
Spain	68	4.0%	239	1.7%	112	7.3%	335	2.5%
Hungary	44	2.6%	192	1.4%	49	3.2%	192	1.4%
Korea, Republic of	39	2.3%	577	4.1%	51	3.3%	551	4.0%
OTHERS	94	5.5%	1 107	8.0%	105	6.8%	1 239	9.1%
Total	1 708	100.0%	13 915	100.0%	1 539	100.0%	13 661	100.0%

Source : Trademap

1.6.6 CN - EXPORTS OF BATHS AND SHOWER TRAYS - SYNTHETIC 2019

Destination	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
United States of America	20 170	20.1%	85 810	20.7%	19 485	20.0%	74 406	21.6%
United Kingdom	9 625	9.6%	40 142	9.7%	9 797	10.0%	32 990	9.6%
Australia	8 180	8.2%	37 085	9.0%	8 888	9.1%	35 712	10.4%
Canada	6 549	6.5%	27 122	6.6%	6 881	7.1%	25 807	7.5%
France	4 830	4.8%	14 139	3.4%	4 884	5.0%	12 645	3.7%
Netherlands	3 216	3.2%	16 633	4.0%	2 706	2.8%	10 327	3.0%
Germany	2 673	2.7%	12 235	3.0%	1 836	1.9%	8 876	2.6%
Italy	2 514	2.5%	7 450	1.8%	2 064	2.1%	5 389	1.6%
New Zealand	2 362	2.4%	9 369	2.3%	2 565	2.6%	8 694	2.5%
Belgium	2 197	2.2%	10 003	2.4%	2 418	2.5%	8 506	2.5%
South Africa	1 910	1.9%	4 529	1.1%	1 885	1.9%	3 865	1.1%
OTHERS	35 898	35.9%	149 036	36.0%	34 122	35.0%	116 655	33.9%
Total	100 124	100.0%	413 553	100.0%	97 533	100.0%	343 872	100.0%

Source : Trademap

1.7.1

Route to Market: Baths
CN - ESTIMATED DISTRIBUTION FOR BATHS: 2019

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	87.1%
	Wholesalers / Merchants
	87.1%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	0.3%	0.5%	4.9%	5.5%	3.6%	3.7%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	0.8%		10.4%		7.3%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
2.2%	3.5%	0.2%	3.0%	5.0%
Installers				
13.9%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	1.6%	72.4%	0.5%	7.2%	2.0%	12.6%	0.3%	1.5%	0.1%	0.2%	0.3%	1.3%
	Residential						Non-Residential					
	96.3%						3.7%					
New Build			Residential RMI			New Build			Non-Residential RMI			
46.7%			49.5%			2.0%			1.7%			
Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project		
37.0%	9.7%		9.0%	40.5%		1.6%	0.4%		1.1%	0.6%		

1.7.2

Route to Market: Baths
CN - ESTIMATED DISTRIBUTION FOR BATHS: 2024

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	88.0%
	Wholesalers / Merchants
	88.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	0.4%	0.7%	4.5%	6.0%	3.3%	3.5%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	1.1%		10.5%		6.8%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
2.0%	3.7%	0.3%	3.1%	4.5%
Installers				
13.6%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	1.6%	72.4%	0.6%	7.2%	2.0%	11.9%	0.2%	1.7%	0.2%	0.2%	0.3%	1.7%
	Residential						Non-Residential					
	95.7%						4.3%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	28.0%			67.7%			2.5%			1.8%		
	Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project	
	20.1%	7.9%		14.8%	52.9%		2.1%	0.4%		1.3%	0.5%	

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

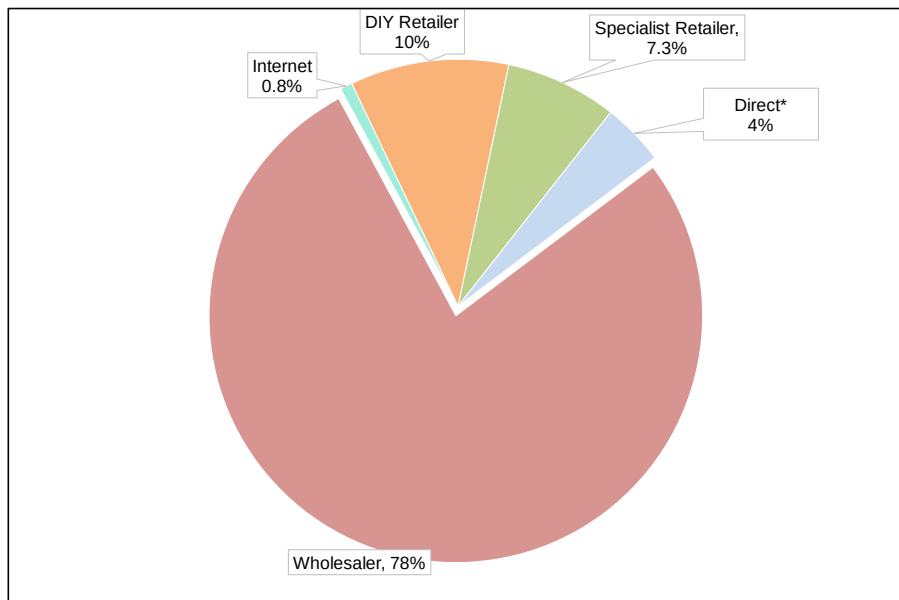
1.7.3 Route to Market - Consolidated View : Baths
CN - ESTIMATED DISTRIBUTION FOR BATHS : 2019

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
77.4%	0.8%	10.4%	7.3%	4.1%	46.0%	2.7%	50.3%	1.0%
100%					48.7%		51.3%	

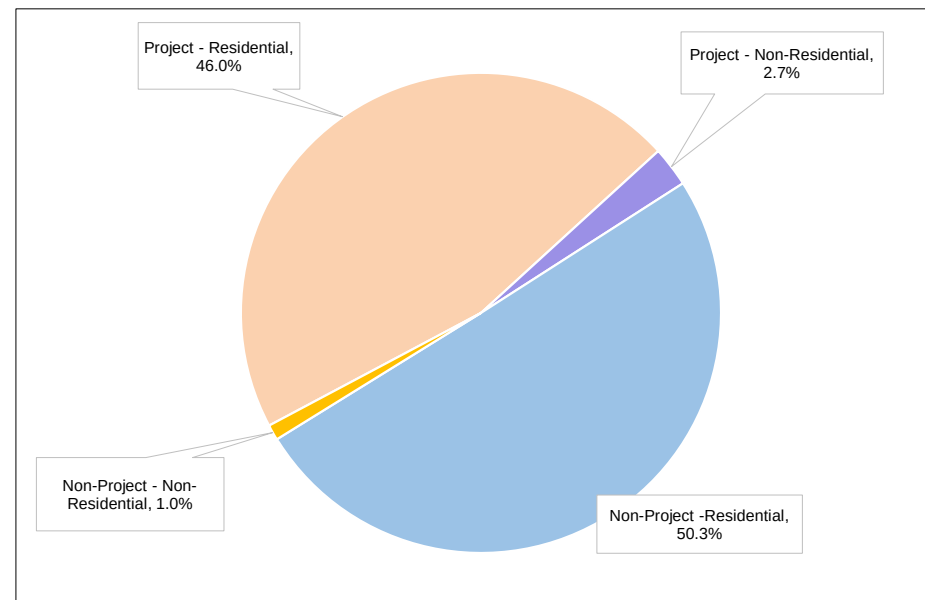
1.7.4 Route to Market - Consolidated View : Baths
CN - ESTIMATED DISTRIBUTION FOR BATHS : 2024

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
77.8%	1.1%	10.5%	6.8%	3.8%	34.9%	3.4%	60.8%	0.9%
100%					38.3%		61.7%	

CN - SALES OF BATHS BY DISTRIBUTION CHANNEL: 2019



CN - SALES OF BATHS BY POINT OF DECISION AND END-USE SECTOR: 2019



Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

1.7.5

Route to Market: Shower Trays
CN - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS: 2019

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	84.6%
	Wholesalers / Merchants
	84.6%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	0.2%	1.1%	3.8%	4.8%	4.2%	6.9%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	1.3%		8.6%		11.1%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
3.9%	3.5%	0.5%	2.5%	8.2%
Installers				
18.6%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer			
	1.1%	56.3%	0.7%	4.9%	2.0%	10.0%	2.3%	12.0%	0.1%	1.2%	0.9%	8.6%			
	Residential						Non-Residential								
	74.9%						25.1%								
New Build				Residential RMI				New Build				Non-Residential RMI			
30.0%				44.9%				17.3%				7.8%			
Project		Non-Project		Project		Non-Project		Project		Non-Project		Project		Non-Project	
21.8%		8.2%		31.9%		13.0%		16.1%		1.2%		6.2%		1.6%	

1.7.6

Route to Market: Shower Trays
CN - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS: 2024

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	85.0%
	Wholesalers / Merchants
	85.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	0.3%	1.6%	3.3%	5.0%	4.1%	6.9%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	1.9%		8.3%		11.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
4.4%	5.0%	0.8%	2.1%	8.1%
Installers				
20.4%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.7%	49.5%	0.6%	4.2%	1.8%	6.6%	2.2%	17.0%	0.5%	2.0%	1.1%	13.8%
	Residential						Non-Residential					
	63.4%						36.6%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	13.6%			49.7%			25.1%			11.5%		
	Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project	
	10.0%	3.6%		33.1%	16.6%		22.0%	3.1%		8.5%	3.0%	

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

1.7.7 Route to Market - Consolidated View : Shower Trays

CN - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS : 2019

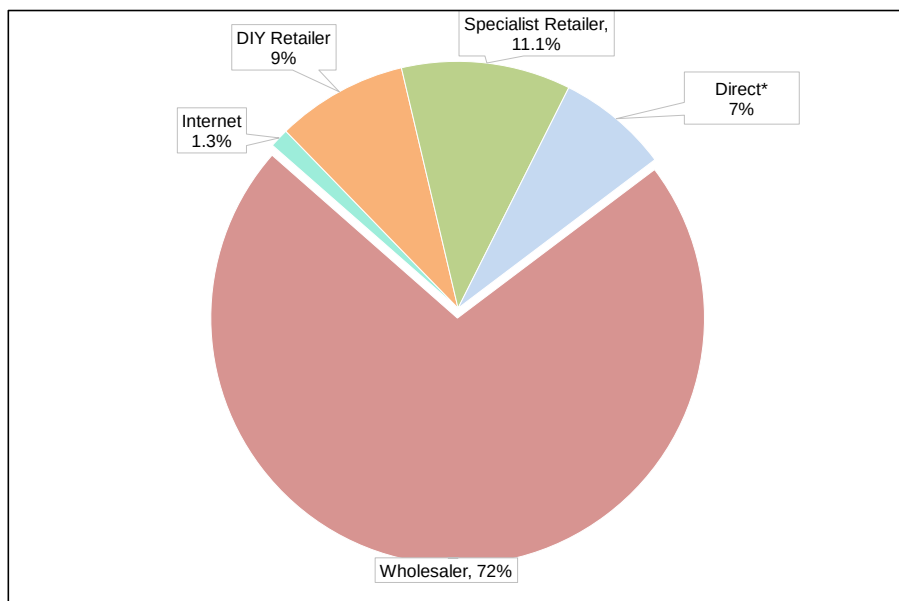
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
71.8%	1.3%	8.6%	11.1%	7.3%	53.7%	22.3%	21.2%	2.8%
100%					76.0%		24.0%	

1.7.8 Route to Market - Consolidated View : Shower Trays

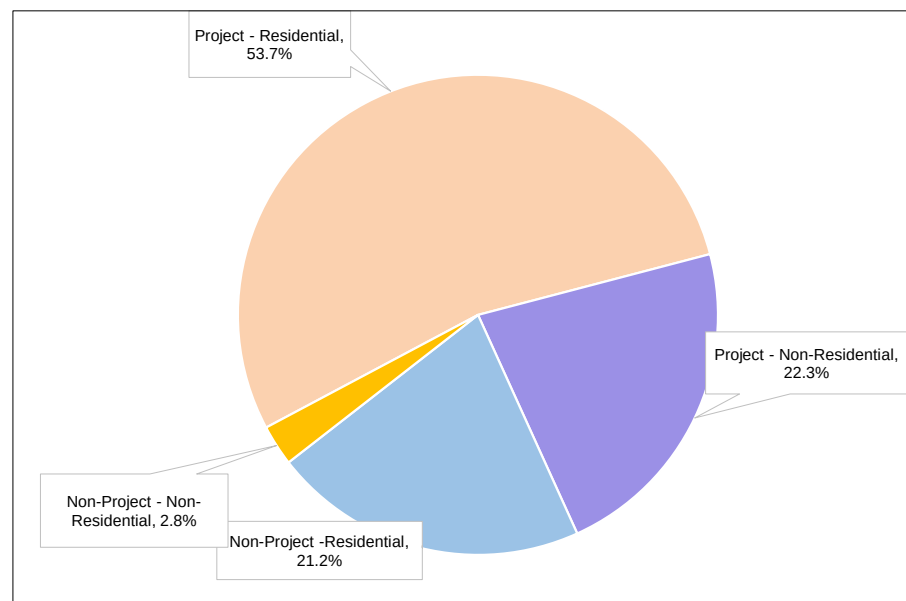
CN - ESTIMATED DISTRIBUTION FOR SHOWER TRAYS : 2024

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
71.5%	1.9%	8.3%	11.0%	7.3%	43.1%	30.5%	20.3%	6.1%
100%					73.6%		26.4%	

CN - SALES OF SHOWER TRAYS BY DISTRIBUTION CHANNEL: 2019



CN - SALES OF SHOWER TRAYS BY POINT OF DECISION AND END-USE SECTOR: 2019



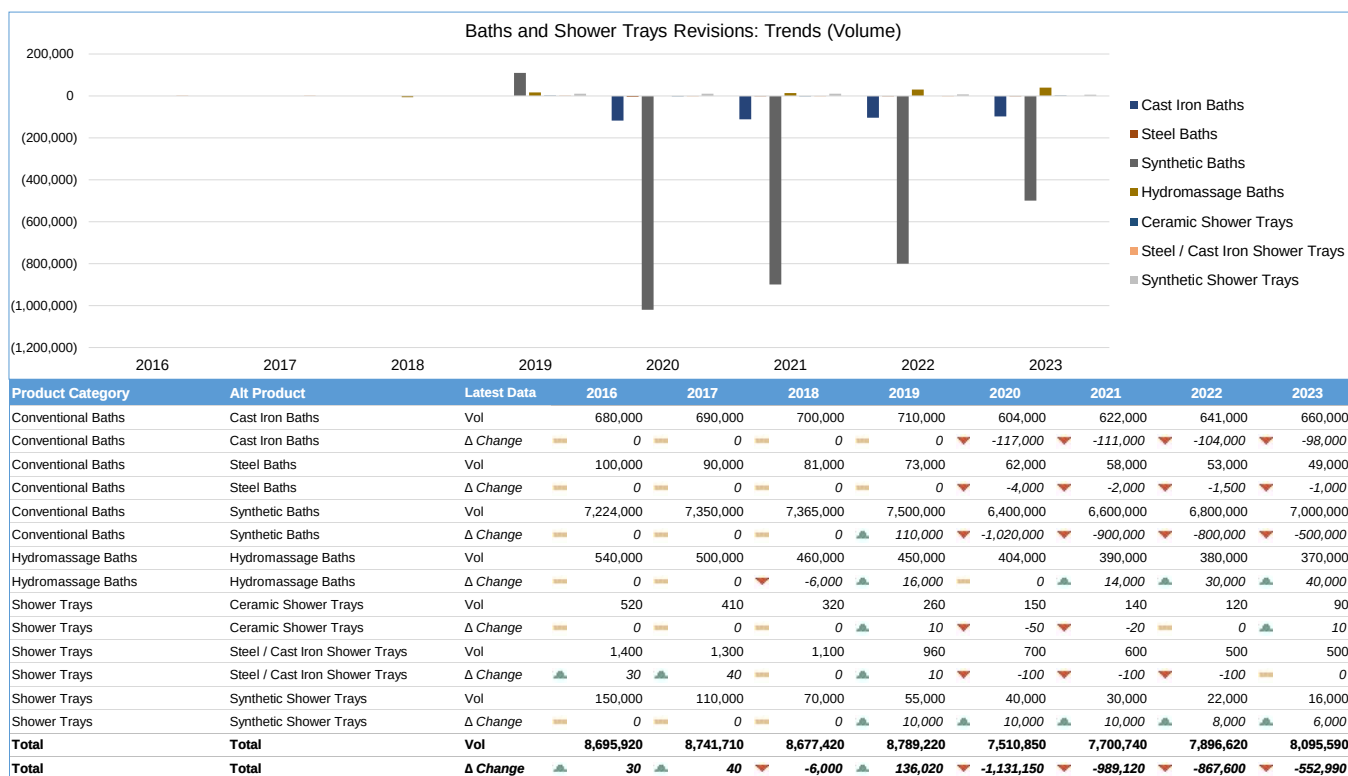
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

1.8 Revisions in the Chinese Baths and Shower Trays Market (July 2020)

CN - REVISIONS IN THE CHINESE BATHS AND SHOWER TRAYS MARKET

Data Comparison: July 2019



Only minor changes were made to the historical and base year figures for baths and shower trays.

CHINA

SANITARY WARE

THE EAST AND SOUTH EAST ASIAN BATHROOM PRODUCT MARKETS



Market Characteristics and Product Mix

The Chinese ceramic sanitary ware market has grown considerably over the last few decades, driven by the country's rapid economic growth, rising disposable incomes, the rapid progress of urbanization, and the development of a thriving construction and real estate market. Historically, the market has evolved following fast expansion throughout the 1990s and 2000s, which saw the influx of foreign investment in local production and the growth of domestic private manufacturers. This was succeeded by a period of more moderate growth from 2011, with a market decline seen in 2014 due to China's reduced economic growth rate and the government's control policies on the real estate market. The market has since, however, returned to growth.

New housing construction is still the major driver of sales, as the market is not yet mature and not all Chinese dwellings are equipped with sanitary ware. However, the demand for sanitary ware in the RMI segment has grown in recent years, with the improvement of disposable income and growth of interest in home improvement work. The sanitation levels in rural and public sectors have also been improving due to government measures to increase urbanization.

In the WC/toilet segment, floor standing models account for the majority of sales as they are cheaper and more easily installed than wall hung models. Sales of wall hung WCs are normally limited to the high end commercial and residential market for design and space saving purposes. Most recently, however, sales of wall hung products have grown due to the increase in small apartment construction and the upgrading of upscale office buildings.

Sales of intelligent WCs and electronic bidet seats have been increasing as consumer incomes and aspirations rise. However, sales of 2-piece WCs are declining, losing out to 1-piece WCs. Squat WCs have been widely installed in low end public and commercial buildings, as well as a large number of rural bathrooms in China, however, with the upgrade of these facilities, and improving bathroom facilities in rural areas, sales of squat WCs/toilets are declining.

Sales of plastic cisterns are much smaller than the sales of WCs in China as most of the WCs in China are one piece. Exposed cisterns are only installed with squat WCs. Sales of concealed plastic cisterns remain relatively small in relation to the smaller size of the wall hung WCs market; however, this market is now growing due to the growth of small bathrooms and changes of WCs drainages from horizontal to vertical pipe in some regions in China.

Water efficiency and anti-bacterial has become the major focuses for WCs technology. The low flush design such as 3/6 liters per flush for WCs have been the main products in the market since 2011 when the government launched regulations on sanitary ware, while the lower flush designs with less than 3/5 liters per flush WCs emerged after 2015 with the implementation of new mandatory regulations

GB31436-2015. As the price of water has been growing since 2014, many large cities have adopted the time-of-use water pricing system, which has led to the replacement of existing WCs with high efficient products with smaller flush volumes.

The design of WCs tends to be sophisticated, including intelligent design with cleaning and drying functions, digital design with audio systems, artistic design integrated with bathroom interiors, and temperature controls on the toilet seats. Nano self-cleaning glazes and antibacterial glazes are two major glaze materials for WCs in the market. The WCs flushing technology has been evolving from wash down to siphon flushing with the majority of the WCs now being siphon flush, such as spiral siphon, jet siphon and super swirling siphon. Jet siphon WCs are the most popular in the market currently.

In the washbasin segment, countertop basins account for most washbasin sales in China and increasingly replace pedestal/half pedestal models as the installation of bathroom furniture in homes rises. The design of countertops tends to be aesthetic, practical and multifunctional to appeal to various tastes and personal needs. Sales of hand washbasins are small in China and only limited in commercial and public segments.

Although new dwelling completions have slowed down in recent years, there has been an increase in en-suite installations and second to third bathrooms, as well as expansion of average bathroom size per dwellings.

Sales of urinals continued to grow quickly in China with the new build and renovation in the non-housing sector. The sales of ceramic sinks, shower trays and other ceramic sanitary ware, such as mop basins have remained small. Sales of bidets have notably decreased to a niche level since Chinese households have switched to intelligent toilets and separate bidet seats.

Recent Market Trends

The Chinese sanitary ware market was effectively flat 2019 with some increases in remodeling in residential dwellings and non-housing construction, rapid growth of fully decorated dwelling completions, and the expansion in size of tier 2 cities to tier 4 cities and rural markets. Larger manufacturers did see growth, at the expense of smaller ones particularly due to some environmental directives which pushed smaller companies out of business.

The construction and real estate markets continued to be sluggish in 2019, as government restrictions on purchases affected the economy and slowed economic growth. The decline in engineered sales was greater than that of retail sales but the overall market demand was basically balanced. The demand from new housing was stable, the demand for replacement is growing rapidly, but sales in non-housing sector declined.

At the same time, consumers focus on personalization which led to demand for diversified sanitary ware products increase and sales of high-end products grew whereas the mid-range and low-end market declined.

With industry integration and the changes in product lines, some low-end product suppliers and product lines have been eliminated, but in general the quality of product increased over the year.

Within the product mix, the share of WCs and washbasins continued to grow as demand increased at the expense of ceramic cisterns, bidets and pedestals. In 2019, the growth in sales of wall hung WCs drove growth of concealed plastic cisterns in both residential and commercial sectors due to increased installation in small apartment projects and commercial and public refurbishment projects. Regional government policies in Zhejiang and Jiangsu no require horizontal drainage in apartment blocks. The sales of exposed plastic cisterns remained stable due to slowing down of squat WCs market in general despite of some regional growth in Sichuan, Guizhou, Yunnan, and other rural markets.

Prices and Market Values

Average MSPs, in local currency, for washbasins, toilets and other sanitary ware increased in 2019 due to the rising costs of main raw material such as ceramic glaze, utility power, environmental costs due to conversion from coal to gas or electricity, labor, packaging and logistics. The sales of middle and upper end products continued to grow which also pushed the average MSPs up overall. Another factor affecting prices is the rise in rents for manufacturers.

The average price of plastic cisterns also grew in 2019 due to the rising costs of plastic raw materials and components. The price of concealed products grew more quickly than exposed cisterns due to the sudden rapid growth of the demands in the non-housings and small apartment projects under some regional governments' mandatory policies on horizontal WC drainage design in buildings.

New Housing

New housing is the largest segment for sanitary ware sales in China due to an expansive housing market and low levels of product saturation as parts of China remain very rural.

The large land developers such as WANKE, HENGDA, BIGUIYUAN, etc. have invested heavily in construction of fully decorated housing and stopped the construction of rough case shell housings which has seen new housing sales grow although in 2019, sales declined into the segment due to the onset of the COVID pandemic. The construction of fully decorated housings has not been limited to the large cities but also expanded into tier 3 to tier 4 cities. According to the Alliance of Fully Decorated Housing Construction Industry of China, China is aiming to target the penetration of fully decorated housings at 80% by 2020 (pre COVID expectation).

Housing RMI

Sales of sanitary ware to housing RMI grew in 2019.

The sales of WCs to housing RMI also increased in 2019 in line with increased renovation and home improvement activity, and the replacement of old WCs with water efficient products as the result of government directives. There was also increased installation of intelligent WC and bidet seats.

In the tier 1 and tier 2 cities, the sales of medium to upper end products grew, while the sales of medium to economy end products increased well in tier 3 to tier 4 cities and rural areas. Sales associated with housing RMI grew more in developed areas, such as the north, the east and the south, than the west.

The replacement of other ceramic sanitary wares remained stable overall and sales of washbasins into housing RMI grew in line with the WC segment.

Non-Housing

Non-housing construction is important for sanitary ware sales in China due to the construction of new commercial and public buildings with the urbanization, and large park of renovation of non-housing buildings such as offices, hotels, hospitals, schools, and transportation hubs. In 2019, the sales of ceramic sanitary ware to non-housing declined even though there is some traction from the construction of public and commercial infrastructures such as Beijing Daxing Airports, Beijing Sub-city centre, Beijing Olympic sports venues in Zhangjiakou, and some hotels and exhibition centres.

Sales of WCs to hotels declined slightly due to the slow construction of new hotels, but the sales of medium and high-end products to luxury hotels grew as a result of increased RMI. Sales of WCs to other non-housing segments such as schools, hospitals, entertainment and recreational facilities continued to grow well.

Sales of washbasins were also affected by the slow growth of new construction of hotels, and this segment decreased in 2019. For other ceramic sanitary wares, sales of urinals grew in offices, schools, hospitals, stations, hotels and others, while the sales of other products declined. For plastic cisterns, the sales of concealed products grew dramatically due to the rapid growth of demand from offices, hotels and schools, etc.

Supply Structure

The Chinese sanitary ware market is highly fragmented with thousands of local suppliers, mostly medium to small players serving the local market, hence, there is a large share of “Others” in the market. The top 50 suppliers only accounted for less than one quarter of the market, and the top 10 large players have less than 20% market share.

The market is predominately in hands of Chinese players, which accounted for about 95% of the market led by four large Chinese sanitary ware groups including HUIDA, JOMOO, HEGII, and ARROW. The international companies have the remaining 5%, and mainly focus on the medium to high end of the market with TOTO, KOHLER and the ROCA GROUP being the three largest.

More and more small suppliers were affected by competition, and pressure from environmental regulations, resulting in declining sales and in some cases business closure.

The washbasins market is highly competitive since many bathroom vanity suppliers, taps and mixers suppliers and even other non-bathroom suppliers have entered the market. HUIDA, JOMOO, HEGII, ARROW, KOHLER, TOTO, ROCA, FAENZA, DONGPENG, and GOLD are the ten large suppliers of washbasins.

The toilet segment is the largest segment in sanitary ware and the toilet is the most profitable product. Some of the large to medium suppliers grew market share in 2019 due to their ability to provide better quality and innovative products. The top manufacturers are HUIDA, JOMOO, HEGII, ARROW, TOTO, KOHLER, FAENZA, DONGPENG, AMERICAN STANDARD and GOLD.

The major activities of sanitary ware suppliers in 2019 include:

HUIDA started the production of its ceramic sanitary ware in its factory in Fengnan, Tangshan in January 2019. The factory started construction in 2017 with an investment of RMB 160 million and has an annual capacity of 2.8 million pieces. HUIDA is also proceeding with the construction of its intelligent sanitary ware plant in Chongqing to produce 800,000 ceramic sanitary ware per year and the production is expected to start in September 2020. In May 2019, HUIDA also announced to start the production of full bathroom projects in Tangshan with an investment of RMB 300 million. The full bathroom factory in Tangshan will produce 150,000 sets of full bathrooms per year by the end of 2021.

JOMOO set up a new ceramic sanitary ware factory in Nan'an, Fujian province, which is to start operation from the second half of 2021.

DONGPENG invested RMB 980 million to set up an intelligent full bathroom production base in Jiangmen, Guangdong province. The factory is to produce 2.6 million water saving sanitary ware and 1 million faucets. DONGPENG started the production of its intelligent home creative industrial park in Yongchuan, Chongqing in November 2019. The industrial park has 10 tile lines, 3 lines of ceramic sanitary wares (including intelligent WCs), 1 line of bathroom cabinets, 1 line of shower enclosures, 1 line of faucets, a R&D center and a logistic center.

KOHLER started the trial operation of its sanitary ware production line Phase II in Zibo, Shandong province, with an investment of RMB 560 million. The production line occupied a land space of 100,000 square meters and has production capacity

of 1.74 million pieces per year. On January 7th, 2019, KOHLER's third ceramic sanitary ware manufacturing base, SHENYANG KOHLER SANITARY BATHROOM CO. LTD. (formerly LIAONING SUPOR CERAMIC INDUSTRY CO. LTD.) started operation in Faku, Liaoning Province. This company completed the registration on December 20th, 2018, which is to produce 1.2 million pieces of ceramic sanitary ware per year. It is reported that SHENYANG KOHLER will also invest RMB 1 billion on building a ceramic sanitary ware industry park and making the capacity up to 6 million pieces per year in the future. In February 2019, KOHLER claimed to set up a sanitary ware factory in Changzhou to produce water purifying equipment, bathroom master suites, concealed cisterns, and composite full bathrooms. The total investment on the Changzhou factory is RMB 475 million and the planned turnover will be RMB 3 billion per year in the future. KOHLER is reported to have invested RMB 360 million to expand its sanitary ware factory in Sanshui, Foshan, Guangdong province, and the new facility is to be completed by 2021.

TOTO invested RMB 1 billion on a sanitary ware factory Phase II with a designated annual capacity of 1.2 million pieces in Zhangzhou, Fujian province. The construction has started in April 2019 and the production is to start in April 2021. In addition, it claimed to close its ceramic sanitary ware factory in Beijing.

HEGII built a production line for intelligent sanitary ware with an annual capacity of a million pieces in Sanshui, Foshan, Guangdong province, and the production is to start in June 2020.

MONARCH started to build a factory to produce intelligent bathroom products in Chongqing in July 2019, with an investment of RMB 800 million. The factory is to produce 5 million intelligent bathroom products.

ANNWA invested RMB 550 million to set up a production facility in Gaoming, Foshan to produce ceramic sanitary ware, shower enclosures and bathroom furniture, which is expected to be completed by 2021.

MILIN invested RMB 200 million in February 2019 on setting up a sanitary ware factory in Weifang with an annual capacity of 3.6 million pieces.

RUIZHU YOUKA started the production of its first automatic composite full bathroom production in Gaoming, Foshan, Guangdong.

YIHE BATHROOM completed its intelligent WCs factory with an annual capacity of 1.5 million pieces per year in May 2019.

BIGUIYUAN, a large real estate developer in China, invested RMB 5 billion to set up an intelligent bathroom industrial park, Guangdong ZHUZHI Ceramic Technology Co. Ltd. in Chaozhou, Guangdong province, with an annual capacity of 700,000 intelligent WCs and 700,000 washbasins.

DYNASTY closed its factory in Zhongshan and used the ceramic sanitary ware factory in Chaozhou to do OEM production for them.

From January to July 2019, there were over 19 bathroom companies starting an expansion plan in Nan'an, Fujian province. The overall investment was about RMB 600 million, and most of these factories are new and small plastic cistern components and faucet manufacturers.

In 2019, many intelligent WCs production projects started, including WANJIE and SHUERMO's intelligent WC factories and BAILI's intelligent WC component factory in Taizhou; VIVI's new factory with an investment of RMB 2 billion; SIMA's intelligent automatic factory Phase I completed by the end of 2019; JITIAN started the operation of its 500,000 intelligent WCs per year factory at the end of 2019; XIERKANG's 1 million intelligent WCs per year factory; and JIAYU's 1 million intelligent WCs started in July 2019.

China is also the largest producer and exporter of sanitary ware in the world. USA still remained the largest destination of Chinese ceramic sanitary ware exports in 2019 despite the raised tariff from 10% to 25% from May 2019, accounting for over 24.2% in terms of export volume comparing with 26% in 2018. Other destinations include Singapore, Malaysia, Saudi Arab, India, Vietnam, South Korea, the UAE, Spain, and Nigeria. Affected by the trade war, the exports to Vietnam, Indonesia, Greece and Netherlands increased rapidly for the recent two years. The imports are relatively small with the major sourcing countries include Thailand and Germany, which are believed to be imported from COTTO, VILLEROY & BOCH and others.

The supplies and production of plastic cisterns are mainly in Fujian (Xiamen), Guangdong (Zhongshan), and Hebei (Renqiu) in China. The exposed plastic cistern market is almost completely supplied by Chinese manufacturers, such as TOYO, KONIG, WDI, ANBOLY, BESTTER, MEITU, WEAL WELL, RUIERTE and HTD. AMASHI and NISON focus on exports. TOYO is the largest plastic cistern manufacturer in China, with nearly half of the exposed plastic cistern market in China.

There are a few suppliers of concealed plastic cisterns in China and the majority of the market is in the hands of GEBERIT. In 2019, RUIERTE and WDI turned to focus on domestic market and sales of their own branded products from the previous exports and OEM production, which help them to improve their domestic market share greatly.

Forecasts

Given the exceptional circumstances related to the spread of the COVID-19 virus and measures in place for its containment, forecasts for 2020 and beyond are based on the latest information available at the time of publication. BRG has endeavored to factor in the latest developments – see Macroeconomic data in Table A.3 and supporting summary table at the end of the Background section.

The COVID-19 pandemic has widely affected the sales of all sanitary ware in China due to the cessation of both production and construction in the first quarter. With the resumption of production and construction/RMI activities, it is estimated that the market will decline by over 14% in 2020 as there are still pockets of reoccurring virus infections particularly in the bigger cities curtailing building starts.

The market is expected to recover in 2021 as building resumes and the country goes back to implementing the “13th Five Year Plan” which looks towards building totally finished and decorated new homes.

The demands for sanitary wares in the housing RMI segment will continue to grow more quickly than the new housing market with the increase of consumer income and growing needs for improving life quality in urban areas in China. The improvement of housing conditions such as increased number of en-suite bathrooms and second to third bathrooms will continue to boost the demands for more sanitary wares in the next few years. In addition, the availability of innovative products such as smart WCs and increased penetration of bathroom furniture, as well as popularity of full bathroom products, will further stimulated the demands for more sanitary wares in Chinese dwellings in next few years.

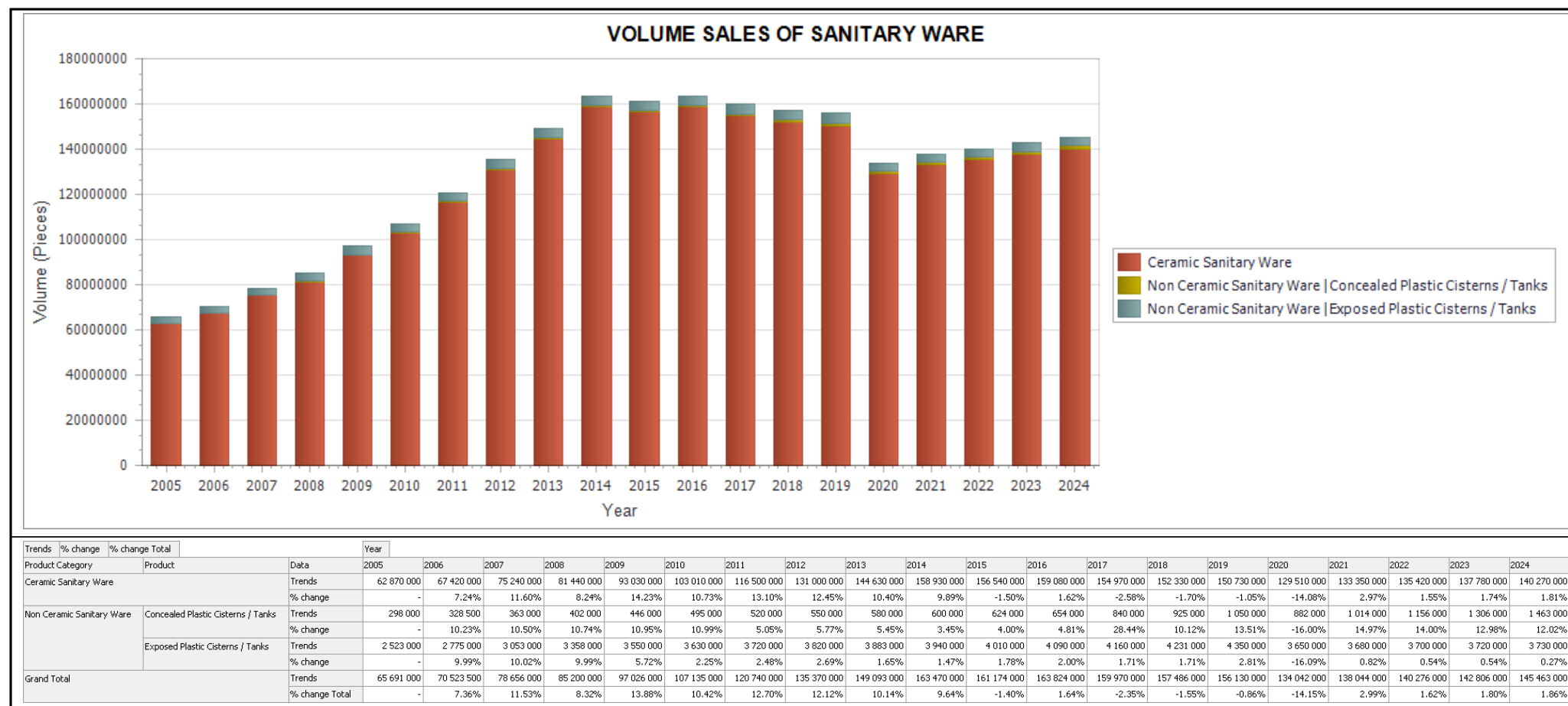
The demands for sanitary wares in non-housing segments, especially in schools, hospitals, shopping malls, care homes, and hotel renovation will continue to grow with the urbanization, construction and upgrading of urban infrastructure.

The demands for water efficient sanitary wares should continue to grow with the implementation of new legislations on water efficiency implemented in December 2015. Product innovation and design should further enhance the demands for ceramic sanitary ware, such as intelligent and water saving WCs.

Some large construction projects such as Beijing New Administration Centre, Xiong'an New Administration Zone, Asian Game 2022 in Hangzhou, etc. will further enhance the demands for more sanitary wares in these regions in the next few years.

2.2.1 Historical Trends and Forecasts : Sanitary Ware

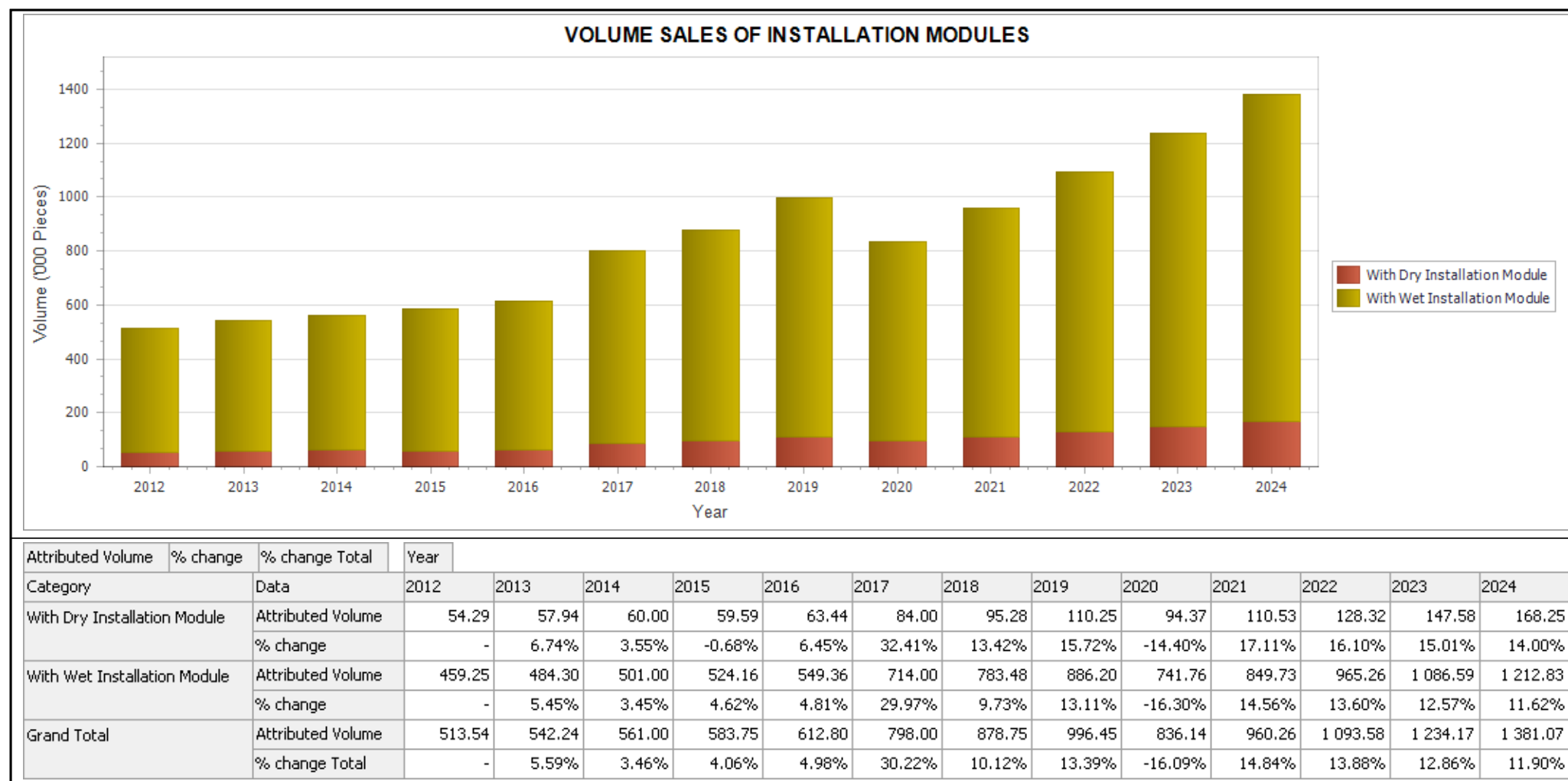
CN - SUMMARY OF TRENDS IN THE SANITARY WARE MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

2.2.2 Historical Trend and Forecasts: Installation Mod.

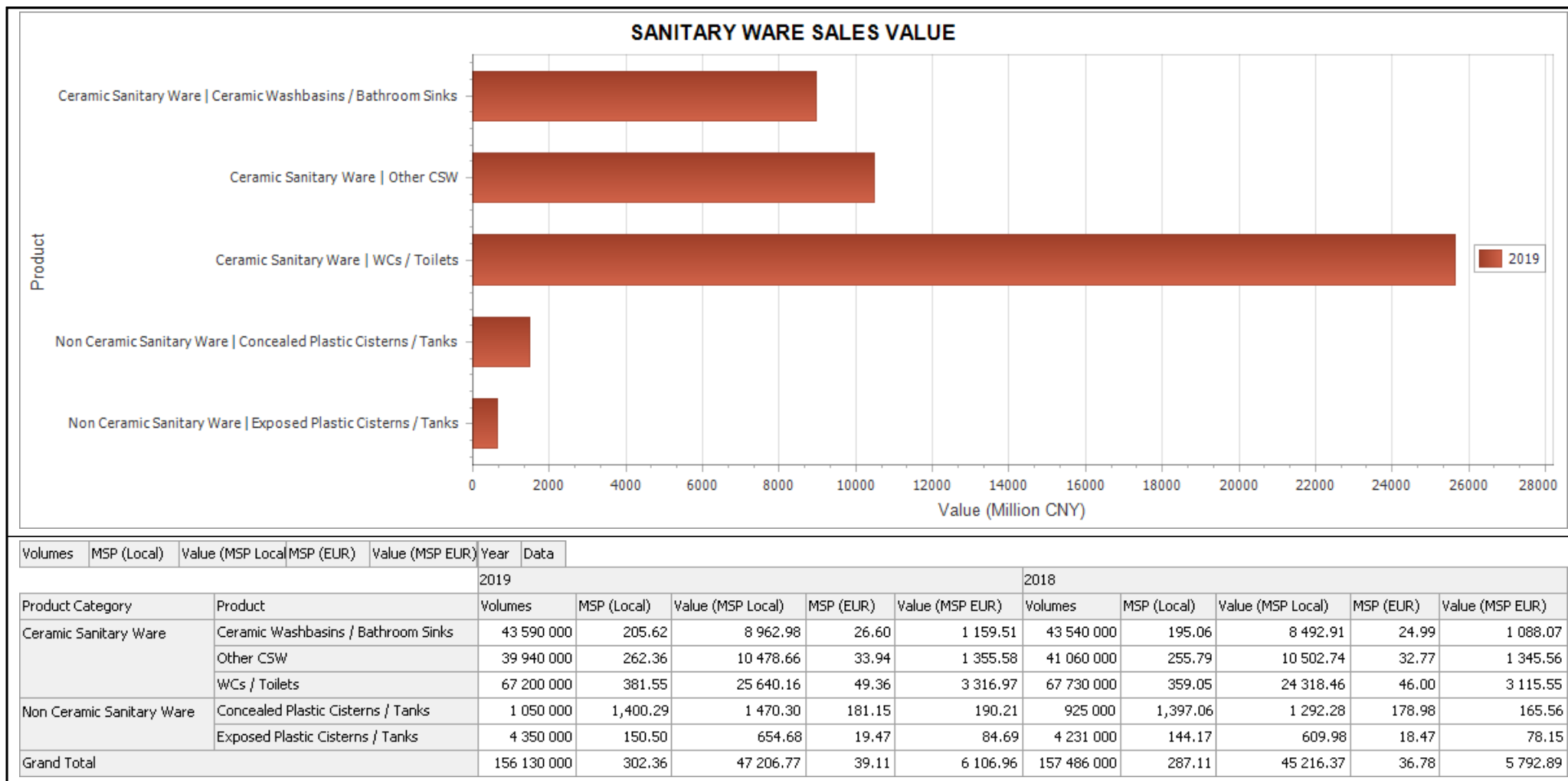
CN - SUMMARY OF TRENDS IN THE INSTALLATION MODULE MARKET : HISTORICAL AND FORECASTS ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

2.3.1 Prices and Market Values : Sanitary Ware

CN - ESTIMATED VALUES OF THE SANITARY WARE MARKET



Source: BRG based on industry interviews. Local Prices Shown in CNY. Values Shown in Millions (at MSP = Manufacturers' Selling Price).

2.3.2 Prices and Market Values by Type : CSW

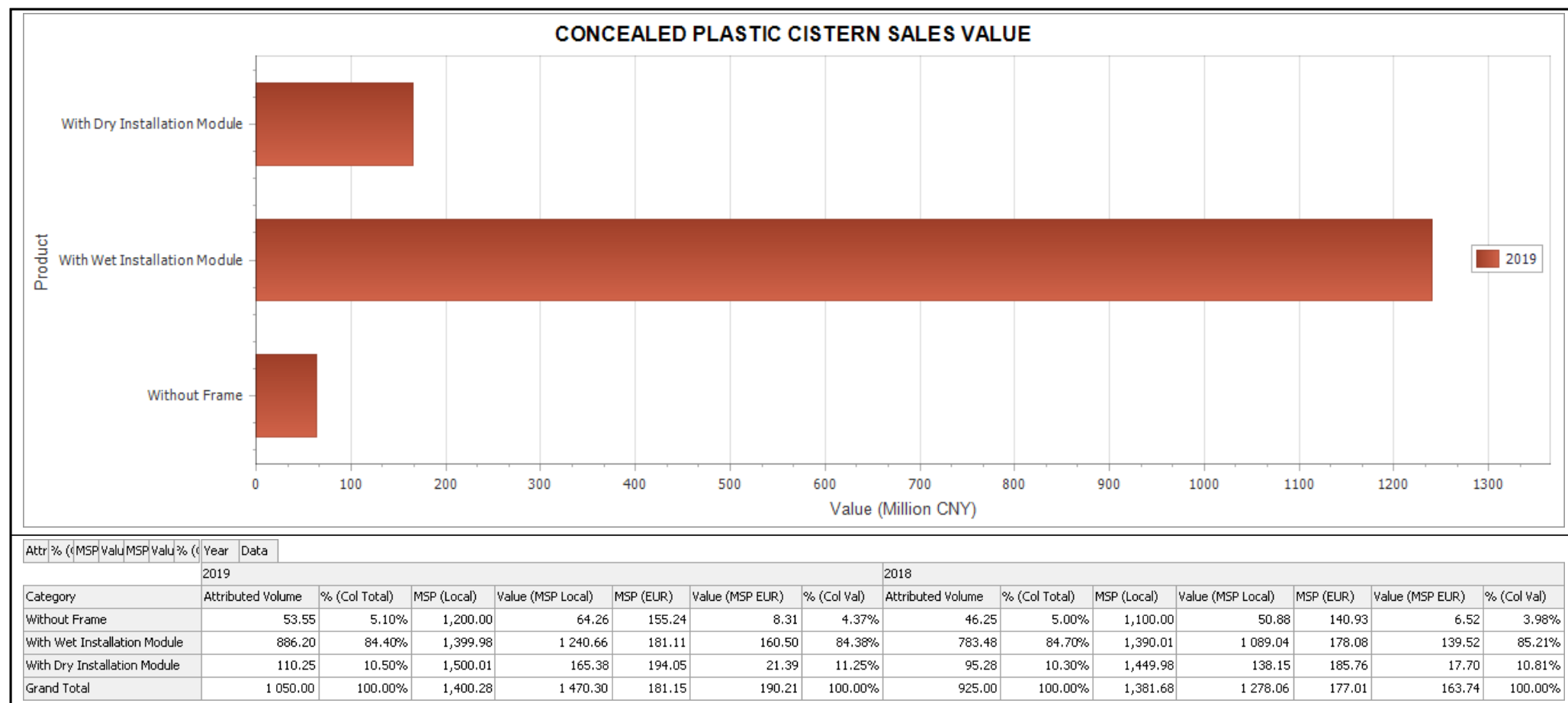
CN - ESTIMATED VALUES OF THE CERAMIC SANITARY WARE MARKET BY TYPE OF PRODUCT

Attributed	% (Col Vol)	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	% (Col Val)	Year	Data																	
								2019									2018								
Alt Product	Category		Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	% (Col Val)	Attributed Volume	% (Col Vol)	MSP (Local)	Value (MSP Local)	MSP (EUR)	Value (MSP EUR)	% (Col Val)									
Ceramic Washbasins / Bathroom Sinks	Countertop Washbasins		5 884.65	13.50%	350.66	2 063.53	45.36	266.95	23.03%	5 660.20	13.00%	333.95	1 890.25	42.78	242.17	22.26%									
	Hand Washbasins		13 425.72	30.80%	168.02	2 255.78	21.74	291.82	25.17%	13 236.16	30.40%	160.00	2 117.80	20.50	271.32	24.94%									
	Pedestal Washbasins		15 256.50	35.00%	186.60	2 846.84	24.14	368.28	31.77%	15 674.40	36.00%	177.69	2 785.16	22.76	356.82	32.79%									
	Wall Hung Washbasins		9 023.13	20.70%	199.01	1 795.72	25.75	232.30	20.04%	8 969.24	20.60%	189.54	1 700.00	24.28	217.80	20.02%									
Other CSW	Bidets		62.91	0.16%	227.01	14.28	29.37	1.85	0.13%	63.64	0.15%	219.99	14.00	28.18	1.79	0.13%									
	Kitchen Sinks		459.99	1.15%	124.46	57.25	16.10	7.41	0.54%	479.99	1.17%	119.13	57.18	15.26	7.33	0.53%									
	Others		1 725.41	4.32%	158.02	272.65	20.44	35.27	2.56%	1 728.63	4.21%	149.97	259.25	19.21	33.21	2.42%									
	Pedestals & Half Pedestals		15 556.63	38.95%	165.21	2 570.04	21.37	332.48	24.17%	16 054.46	39.10%	162.00	2 600.81	20.75	333.20	24.32%									
	Shower Trays		0.28	0.00%	153.45	0.04	19.85	0.01	0.00%	0.33	0.00%	152.97	0.05	19.60	0.01	0.00%									
	Two-Piece Toilet Tanks		16 103.85	40.32%	392.11	6 314.51	50.73	816.88	59.38%	16 775.15	40.86%	382.55	6 417.29	49.01	822.15	60.01%									
	Urinals		2 576.13	6.45%	283.98	731.58	36.74	94.64	6.88%	2 562.14	6.24%	270.02	691.82	34.59	88.63	6.47%									
	Washing Tanks		3 454.81	8.65%	195.00	673.69	25.23	87.15	6.34%	3 395.66	8.27%	192.59	653.98	24.67	83.78	6.12%									
WCs / Toilets	One-Piece Toilets		47 954.46	71.36%	452.02	21 676.35	58.48	2 804.18	84.54%	47 478.73	70.10%	429.99	20 415.35	55.09	2 615.51	83.94%									
	Squat Pans		2 620.60	3.90%	113.98	298.70	14.75	38.64	1.17%	3 048.39	4.50%	111.02	338.43	14.22	43.36	1.39%									
	Two-Piece Toilet Bowls		15 120.00	22.50%	205.01	3 099.71	26.52	401.00	12.09%	15 916.55	23.50%	194.99	3 103.56	24.98	397.61	12.76%									
	Wall Hung WC Bowls		1 504.94	2.24%	374.98	564.33	48.51	73.00	2.20%	1 286.33	1.90%	360.01	463.10	46.12	59.33	1.90%									
Grand Total			150 730.00	100.00%	300.11	45 235.01	38.82	5 851.88	100.00%	152 330.00	100.00%	285.62	43 508.02	36.59	5 574.02	100.00%									

Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in CNY. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

2.3.3 Prices and Market Values by Type : Conc. Cisterns

CN - ESTIMATED VALUE OF THE CONCEALED PLASTIC CISTERNS MARKET BY TYPE OF INSTALLATION

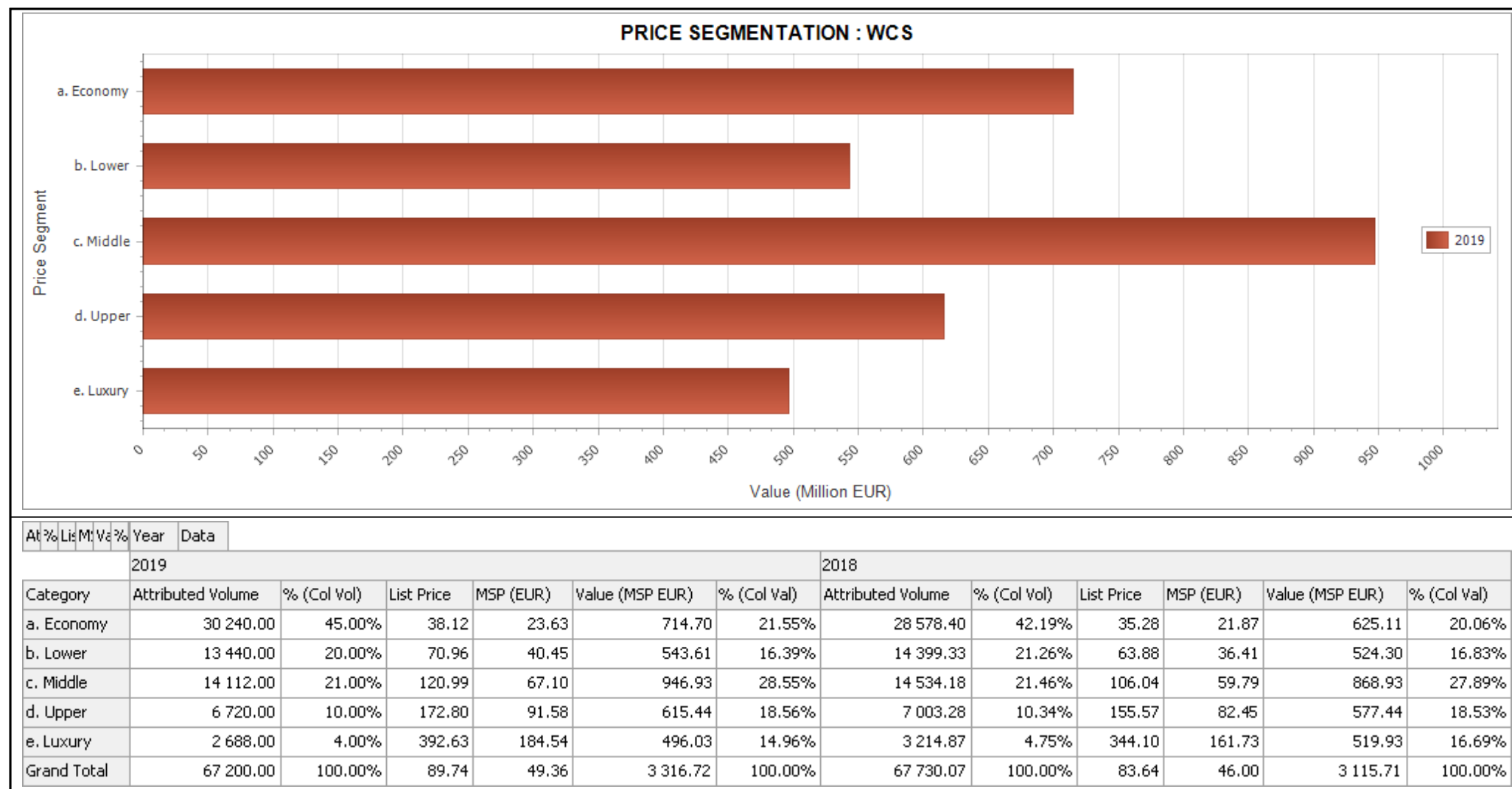


Totals may show slight variance compared to market data due to rounding. Local Prices Shown in CNY. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

Source: BRG based on industry interviews.

2.3.4 Price Segmentation : WCs

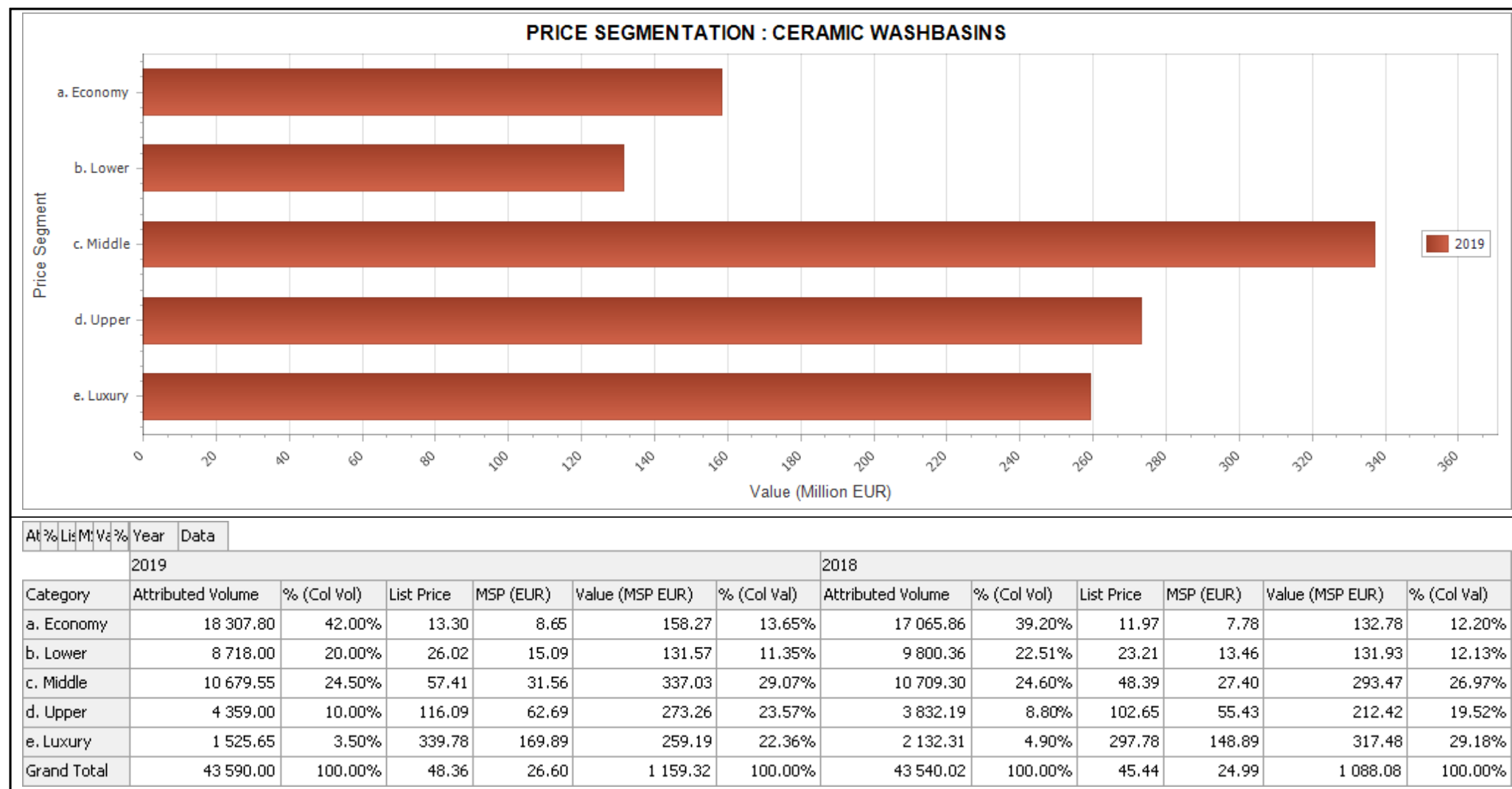
CN - ESTIMATED PRICE SEGMENTATION FOR WC



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes all WC Types. Volumes Shown in Thousands and Values Shown in Millions. Economy <€30, Lower €30-€45, Middle €45-€70, Upper >€70-€100, Luxury >€100

2.3.5 Price Segmentation : Ceramic Washbasins

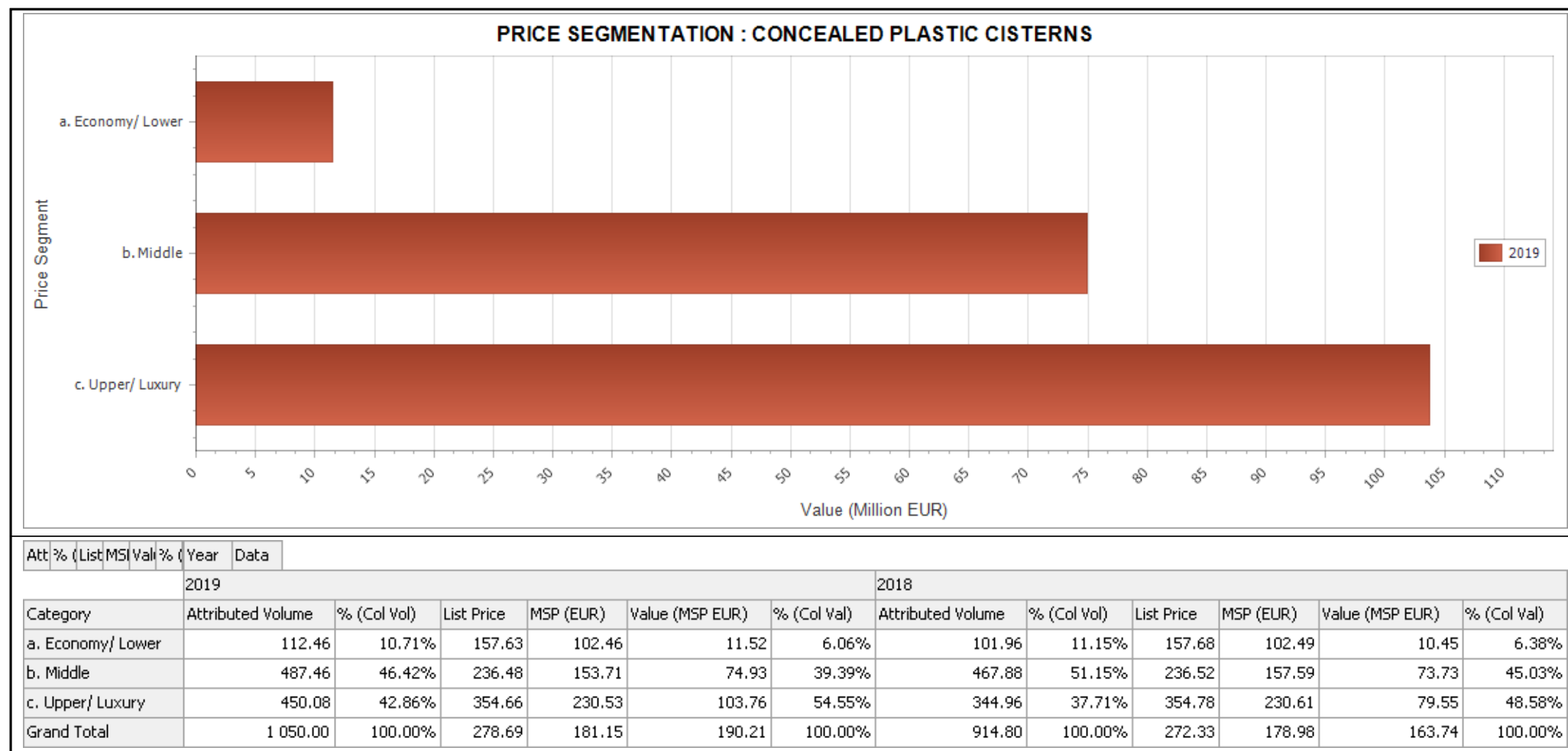
CN - ESTIMATED PRICE SEGMENTATION FOR CERAMIC WASHBASINS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Includes all Washbasin Types. Volumes Shown in Thousands and Values Shown in Millions. Economy <€12, Lower €12-€20, Middle €20-€45, Upper >€45-€70, Luxury >€70

2.3.6 Price Segmentation : Concealed Plastic Cisterns

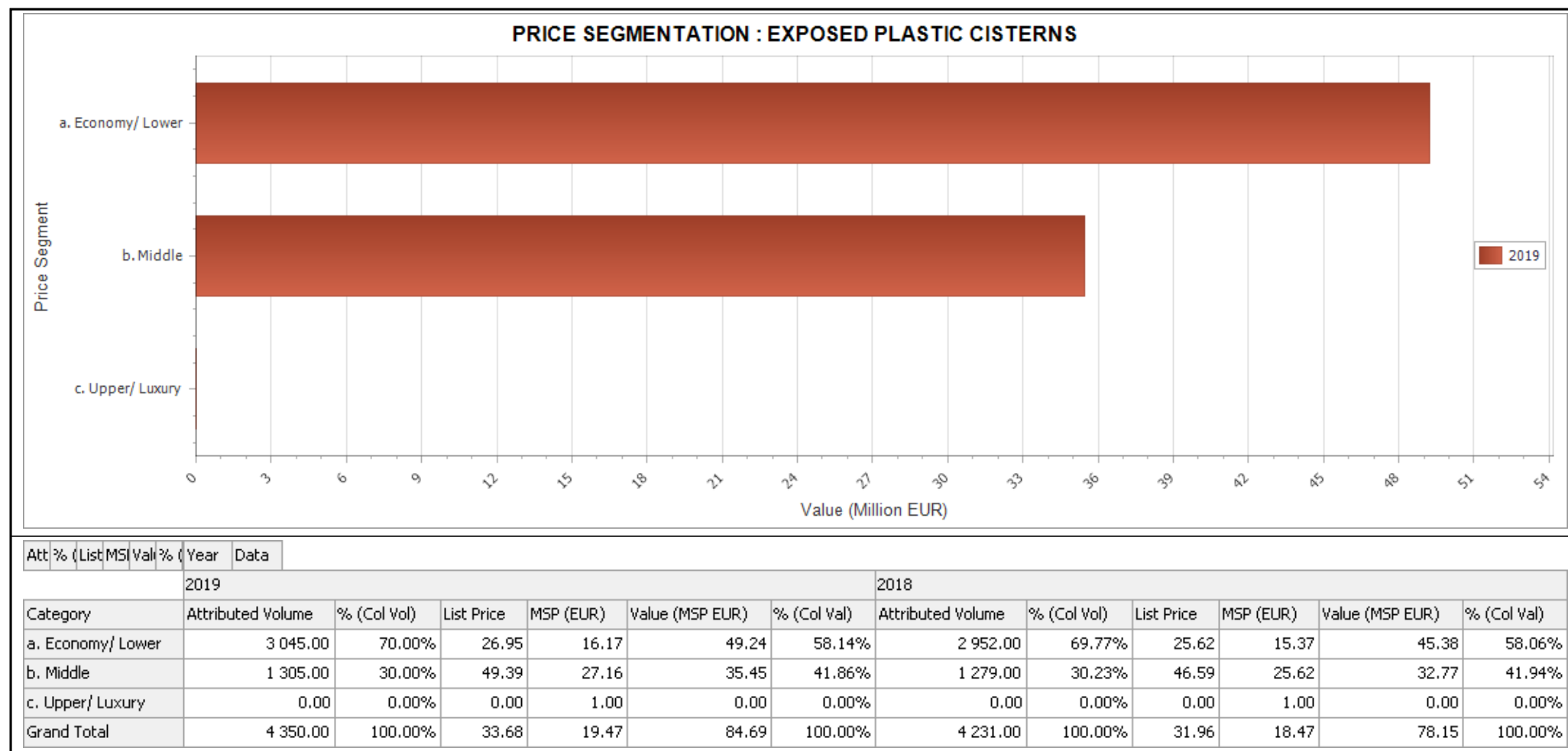
CN - ESTIMATED PRICE SEGMENTATION FOR CONCEALED PLASTIC CISTERNS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions. Economy/ Lower <€120, Middle €120-€200, Upper/ Luxury >€200

2.3.7 Price Segmentation : Exposed Plastic Cisterns

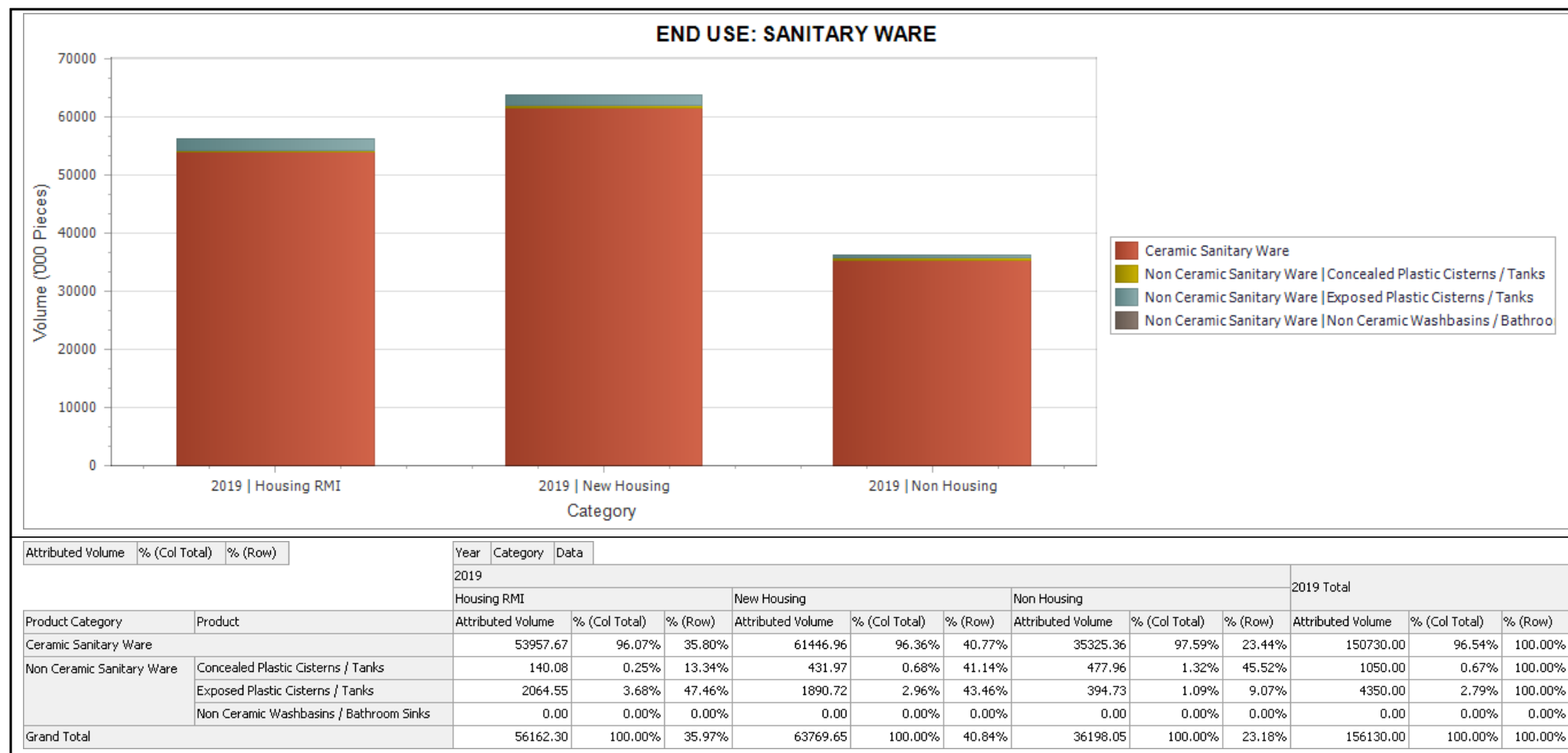
CN - ESTIMATED PRICE SEGMENTATION FOR EXPOSED PLASTIC CISTERNS



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions. Economy/ Lower <€20, Middle €20-€40, Upper/ Luxury >€40

2.4.1 End Use : Base Year

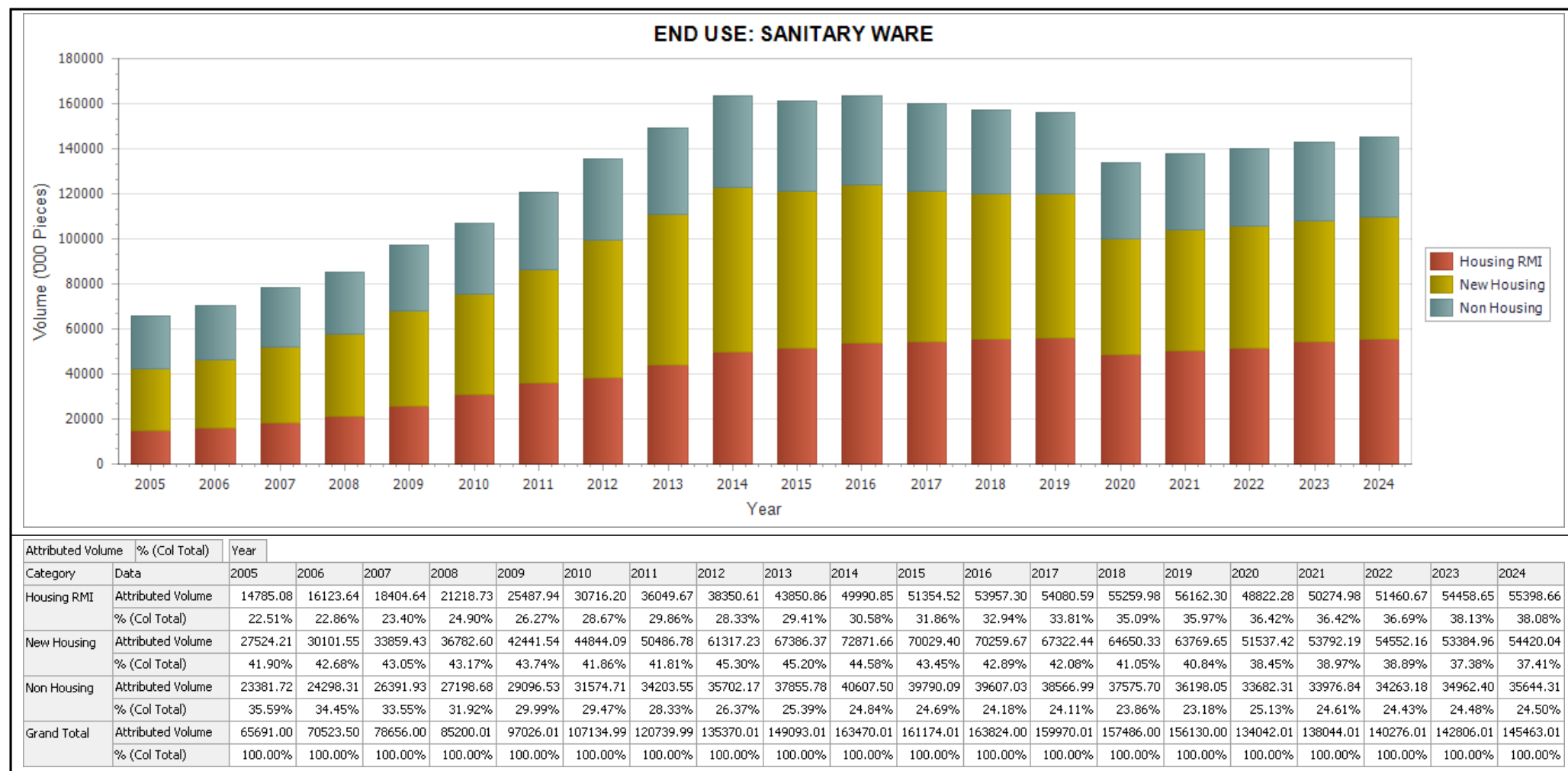
CN - SALES OF SANITARY WARE BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

2.4.2 End Use : Historical and Forecasts

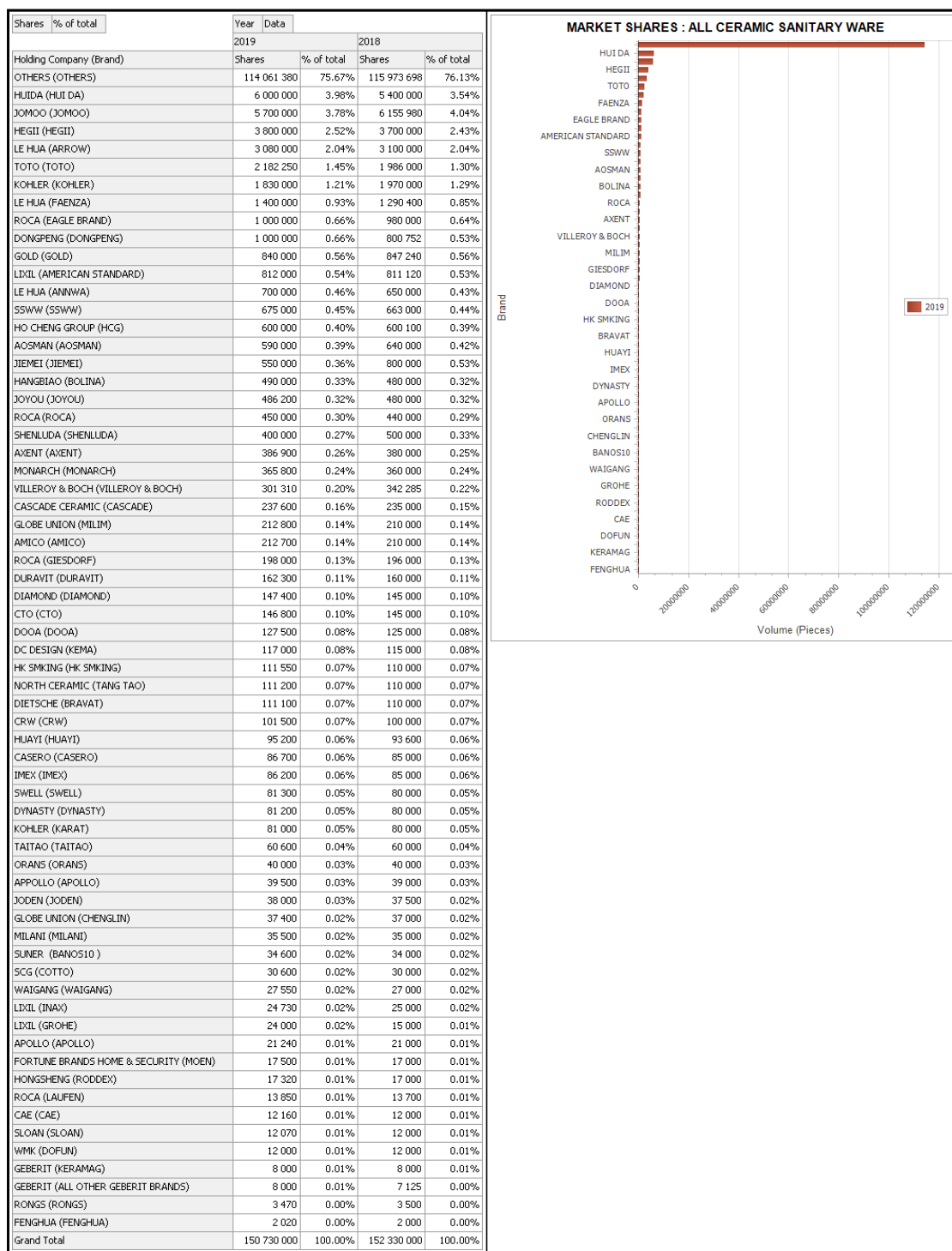
CN - SALES OF SANITARY WARE BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

2.5.1 Market Shares : All Ceramic Sanitary Ware

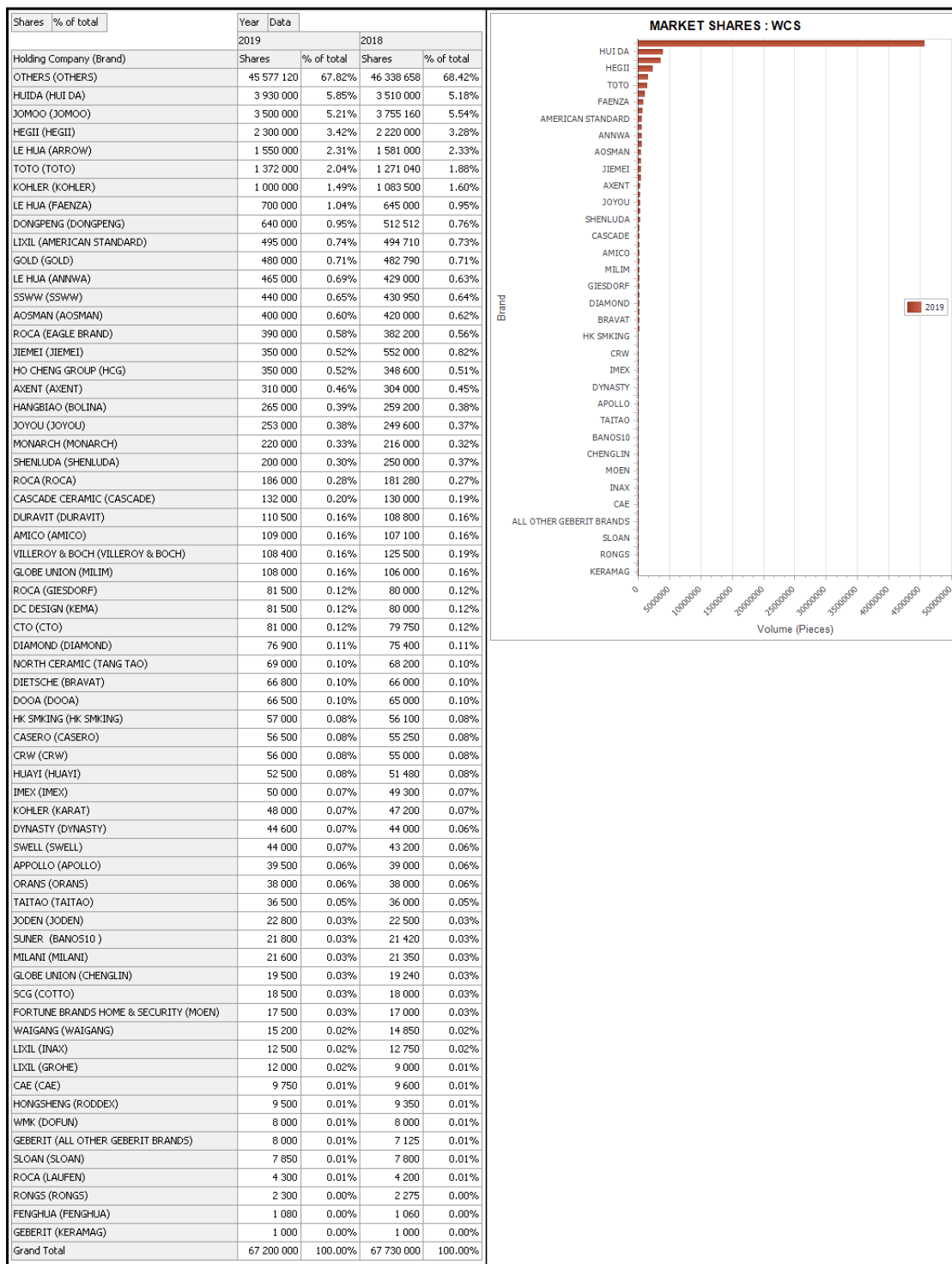
CN - ESTIMATED MARKET SHARES FOR ALL CERAMIC SANITARY WARE



Source: BRG based on industry interviews

2.5.2 Market Shares : WCs

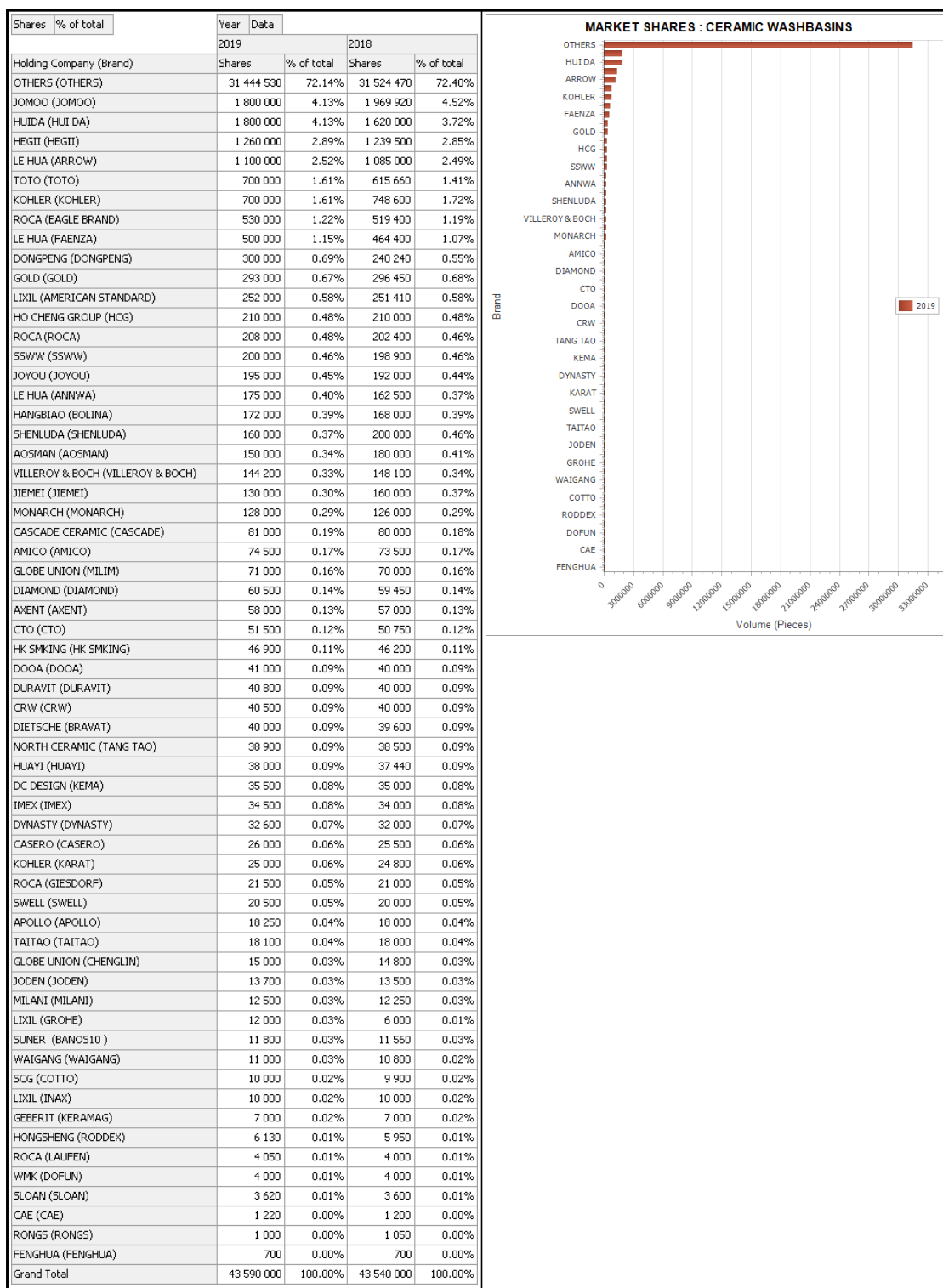
CN - ESTIMATED MARKET SHARES FOR WCS



Source: BRG based on industry interviews

2.5.3 Market Shares : Ceramic Washbasins

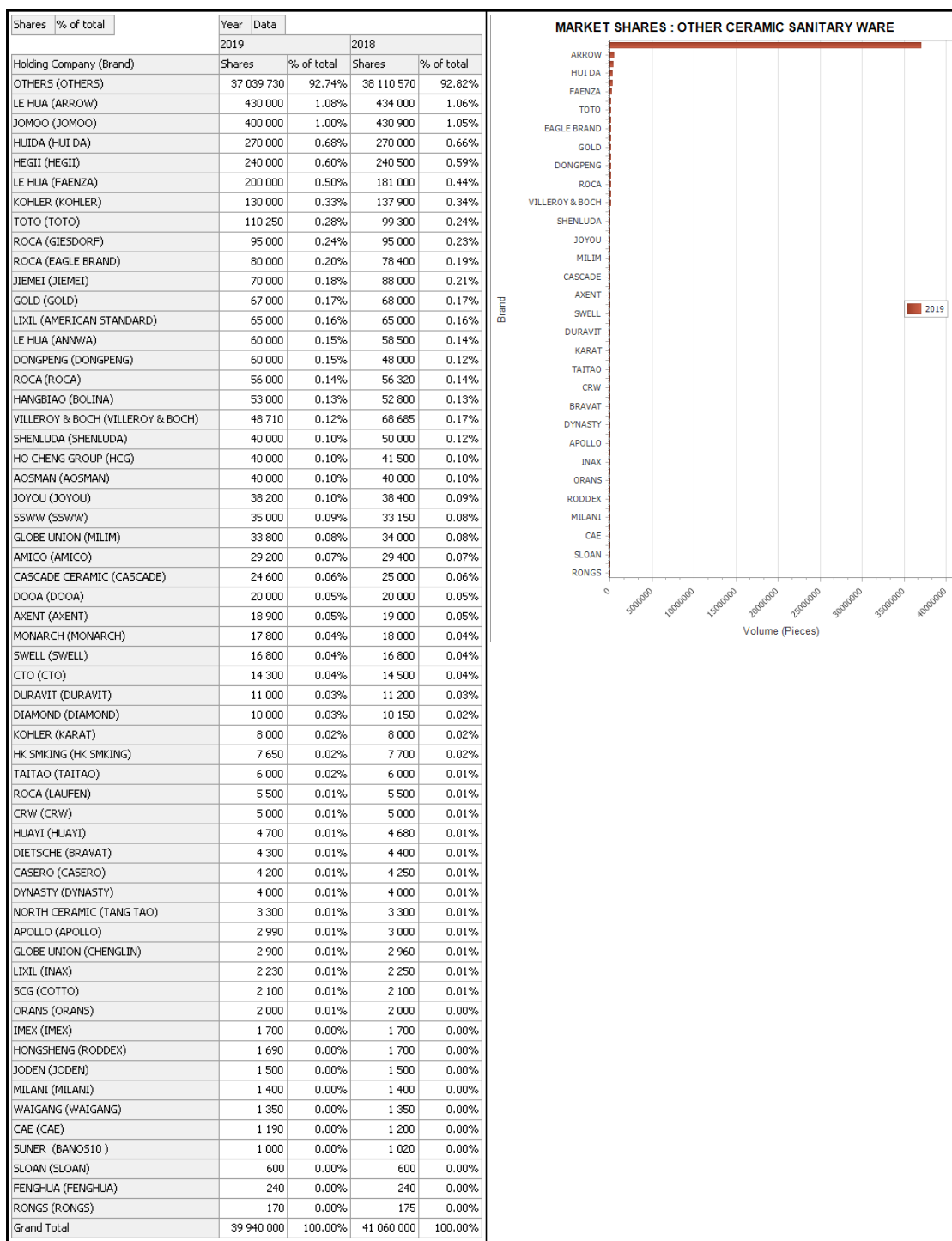
CN - ESTIMATED MARKET SHARES FOR CERAMIC WASHBASINS



Source: BRG based on industry interviews

2.5.4 Market Shares : Other Ceramic Sanitary Ware

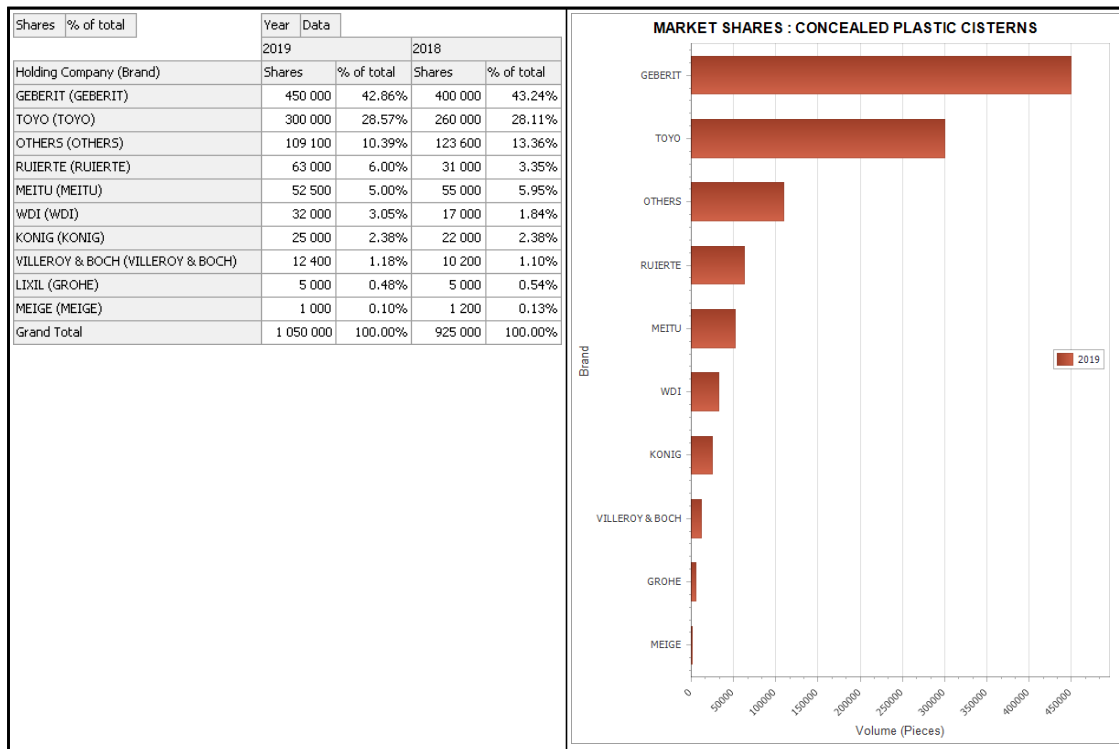
CN - ESTIMATED MARKET SHARES FOR OTHER CERAMIC SANITARY WARE



Source: BRG based on industry interviews

2.5.5 Market Shares : Concealed Plastic Cisterns

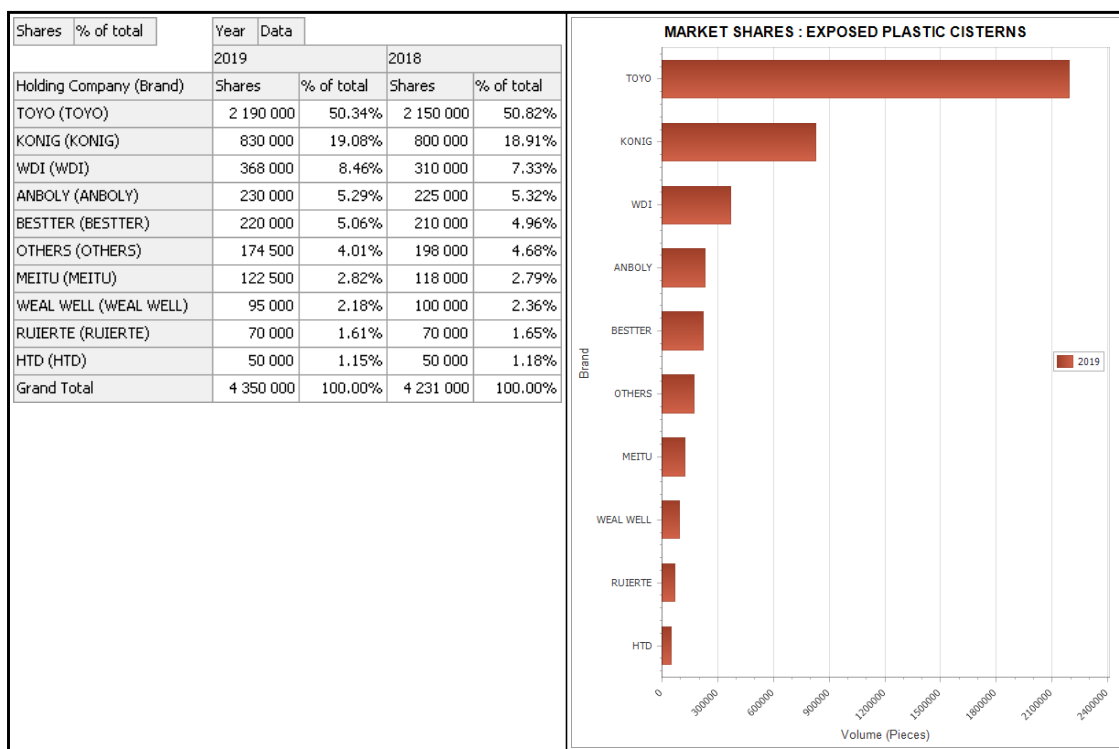
CN - ESTIMATED MARKET SHARES FOR CONCEALED PLASTIC CISTERNS



Source: BRG based on industry interviews

2.5.6 Market Shares : Exposed Plastic Cisterns

CN - ESTIMATED MARKET SHARES FOR EXPOSED PLASTIC CISTERNS



Source: BRG based on industry interviews.

2.6.1 CN - IMPORTS OF ALL CERAMIC SANITARY WARE 2019

Source Country	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Thailand	18 392	46.4%	38 690	35.8%	14 583	46.6%	34 111	38.4%
Viet Nam	9 235	23.3%	18 719	17.3%	7 484	23.9%	12 899	14.5%
Indonesia	3 456	8.7%	7 402	6.9%	1 397	4.5%	2 641	3.0%
Turkey	1 787	4.5%	4 970	4.6%	1 494	4.8%	3 802	4.3%
Germany	1 513	3.8%	11 433	10.6%	1 276	4.1%	10 645	12.0%
India	900	2.3%	1 345	1.2%	837	2.7%	1 423	1.6%
Japan	777	2.0%	9 436	8.7%	734	2.3%	7 367	8.3%
Czech Republic	594	1.5%	2 924	2.7%	594	1.9%	2 452	2.8%
Brazil	560	1.4%	1 653	1.5%	335	1.1%	905	1.0%
OTHERS	2 415	6.1%	11 475	10.6%	2 556	8.2%	12 648	14.2%
Total	39 629	100.0%	108 047	100.0%	31 289	100.0%	88 893	100.0%

Source : Trademap

2.6.2 CN - EXPORTS OF ALL CERAMIC SANITARY WARE 2019

Destination	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
United States of America	423 935	24.3%	817 406	11.5%	403 535	26.3%	668 075	13.4%
Korea, Republic of	139 147	8.0%	267 544	3.8%	141 292	9.2%	224 399	4.5%
Nigeria	76 792	4.4%	228 830	3.2%	52 307	3.4%	136 528	2.7%
Canada	62 405	3.6%	169 538	2.4%	57 317	3.7%	119 066	2.4%
United Kingdom	61 601	3.5%	210 426	3.0%	54 527	3.6%	135 355	2.7%
India	59 876	3.4%	358 860	5.0%	48 683	3.2%	222 427	4.5%
Viet Nam	51 946	3.0%	336 475	4.7%	27 781	1.8%	153 663	3.1%
Philippines	50 416	2.9%	211 814	3.0%	45 660	3.0%	160 292	3.2%
Saudi Arabia	48 416	2.8%	390 215	5.5%	29 798	1.9%	165 036	3.3%
Malaysia	47 365	2.7%	402 296	5.6%	46 900	3.1%	313 827	6.3%
Australia	45 933	2.6%	195 862	2.7%	50 812	3.3%	169 546	3.4%
OTHERS	679 754	38.9%	3 543 303	49.7%	575 328	37.5%	2 509 044	50.4%
Total	1 747 585	100.0%	7 132 569	100.0%	1 533 941	100.0%	4 977 258	100.0%

Source : Trademap

2.6.3 CN - IMPORTS OF PLASTIC CISTERNS 2019

Source Country	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Germany	683	53.8%	5 775	46.5%	647	53.4%	4 474	49.5%
Japan	79	6.2%	774	6.2%	59	4.9%	589	6.5%
Turkey	60	4.7%	301	2.4%	51	4.2%	233	2.6%
United States of America	60	4.7%	830	6.7%	69	5.7%	712	7.9%
Portugal	59	4.6%	254	2.0%	39	3.2%	210	2.3%
Netherlands	55	4.3%	876	7.1%	57	4.7%	763	8.4%
Sweden	55	4.3%	563	4.5%	9	0.7%	157	1.7%
China	45	3.6%	242	1.9%	166	13.7%	770	8.5%
Viet Nam	39	3.1%	917	7.4%	4	0.4%	84	0.9%
Korea, Republic of	33	2.6%	721	5.8%	49	4.1%	399	4.4%
Italy	30	2.4%	180	1.4%	17	1.4%	132	1.5%
OTHERS	71	5.6%	982	7.9%	44	3.6%	521	5.8%
Total	1 269	100.0%	12 415	100.0%	1 211	100.0%	9 044	100.0%

Source : Trademap

2.6.4 CN - EXPORTS OF PLASTIC CISTERNS 2019

Destination	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
United States of America	4 154	12.7%	28 865	16.6%	5 129	16.4%	30 154	20.2%
India	2 497	7.7%	10 346	6.0%	1 628	5.2%	6 332	4.2%
Japan	2 495	7.6%	8 570	4.9%	2 526	8.1%	8 161	5.5%
United Kingdom	2 252	6.9%	15 344	8.8%	2 178	6.9%	12 228	8.2%
Australia	2 182	6.7%	12 205	7.0%	2 576	8.2%	13 353	8.9%
Netherlands	1 346	4.1%	6 341	3.7%	1 341	4.3%	5 816	3.9%
Germany	1 301	4.0%	9 074	5.2%	1 106	3.5%	8 093	5.4%
United Arab Emirates	1 084	3.3%	4 657	2.7%	794	2.5%	3 096	2.1%
Spain	818	2.5%	3 603	2.1%	716	2.3%	2 897	1.9%
Indonesia	754	2.3%	1 649	1.0%	182	0.6%	743	0.5%
Korea, Republic of	683	2.1%	3 704	2.1%	482	1.5%	2 158	1.4%
OTHERS	13 071	40.1%	69 030	39.8%	12 692	40.5%	56 550	37.8%
Total	32 636	100.0%	173 388	100.0%	31 351	100.0%	149 581	100.0%

Source : Trademap

2.7.1 Route to Market: Ceramic Sanitary Ware
CN - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE: 2019

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	81.7%
	Wholesalers / Merchants
	81.7%

Retail Level	Manufacturer 3.4%	Wholesaler 2.8%	Manufacturer 3.1%	Wholesaler 27.0%	Manufacturer 1.6%	Wholesaler 5.8%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	6.2%		30.1%		7.4%	

Manufacturer 4.2%	Wholesaler 3.0%	Internet 0.1%	DIY 2.0%	Specialist 3.4%
Installers				
12.7%				

End Use Level	Manufacturer 0.8%	Wholesaler 31.6%	Internet 5.9%	DIY 25.1%	Specialist 3.0%	Installer 10.2%	Manufacturer 5.2%	Wholesaler 11.5%	Internet 0.2%	DIY 3.0%	Specialist 1.0%	Installer 2.6%
	Residential						Non-Residential					
	76.6%						23.4%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	40.8%			35.8%			15.2%			8.2%		
	Project		Non-Project	Project		Non-Project	Project		Non-Project	Project		Non-Project
	32.5%		8.3%	7.0%		28.8%	13.0%		2.2%	7.0%		1.2%

2.7.2 Route to Market: Ceramic Sanitary Ware
CN - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE: 2024

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	82.0%
	Wholesalers / Merchants
	82.0%

Retail Level	Manufacturer 3.7%	Wholesaler 3.3%	Manufacturer 2.2%	Wholesaler 26.0%	Manufacturer 1.5%	Wholesaler 5.7%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	7.0%		28.2%		7.2%	

Manufacturer 4.8%	Wholesaler 3.0%	Internet 0.3%	DIY 1.8%	Specialist 3.4%
Installers				
13.3%				

End Use Level	Manufacturer 1.3%	Wholesaler 32.5%	Internet 6.2%	DIY 23.5%	Specialist 2.9%	Installer 8.8%	Manufacturer 4.5%	Wholesaler 11.5%	Internet 0.5%	DIY 2.9%	Specialist 0.9%	Installer 4.5%
	Residential						Non-Residential					
	75.2%						24.8%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	37.3%			37.9%			14.8%			10.0%		
	Project		Non-Project	Project		Non-Project	Project		Non-Project	Project		Non-Project
	29.1%		8.2%	8.9%		29.0%	12.5%		2.3%	8.0%		2.0%

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

2.7.3 Route to Market - Consolidated View : Ceramic Sanitary Ware

CN - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE : 2019

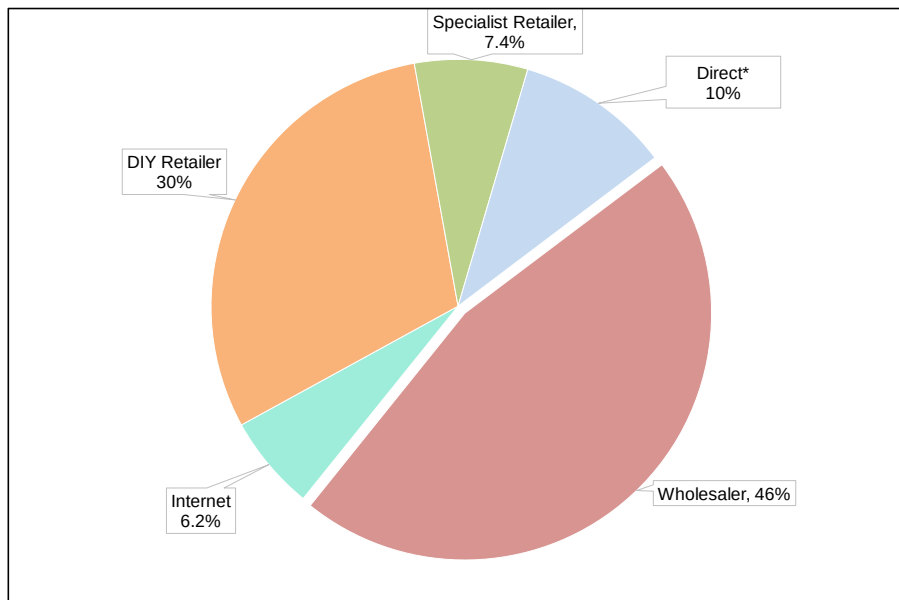
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
46.1%	6.2%	30.1%	7.4%	10.1%	39.5%	20.0%	37.1%	3.4%
100%					59.5%		40.5%	

2.7.4 Route to Market - Consolidated View : Ceramic Sanitary Ware

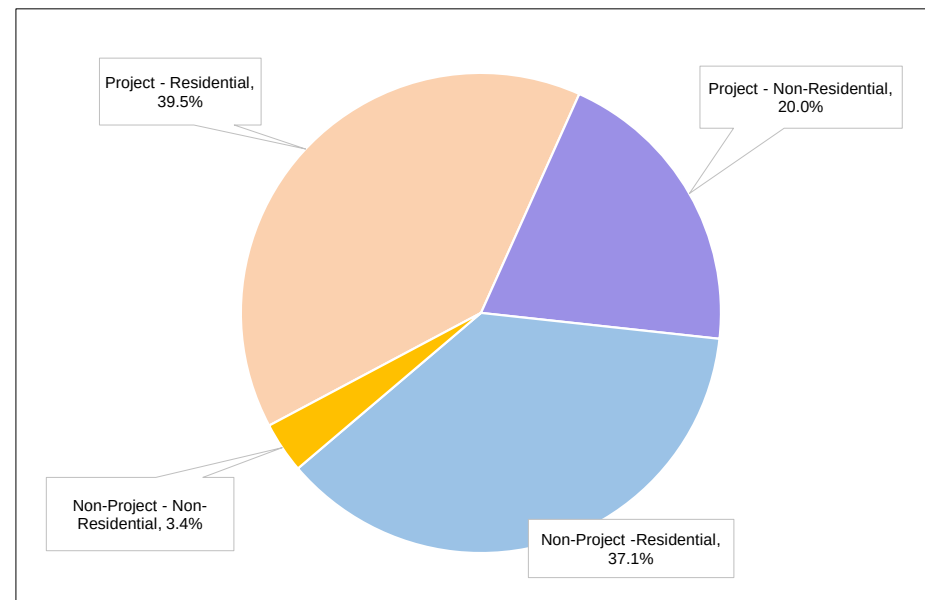
CN - ESTIMATED DISTRIBUTION FOR CERAMIC SANITARY WARE : 2024

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
47.0%	7.0%	28.2%	7.2%	10.6%	38.0%	20.5%	37.2%	4.3%
100%					58.5%		41.5%	

CN - SALES OF CERAMIC SANITARY WARE BY DISTRIBUTION CHANNEL: 2019



CN - SALES OF CERAMIC SANITARY WARE BY POINT OF DECISION AND END-USE SECTOR: 2019



Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

2.7.5

Route to Market: Plastic Cisterns
CN - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS: 2019

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	68.2%
	Wholesalers / Merchants
	68.2%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	2.0%	1.8%	5.1%	20.1%	4.3%	6.5%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	3.8%		25.2%		10.8%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
14.3%	8.1%	0.2%	3.9%	5.9%
Installers				
32.5%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	1.7%	29.2%	3.4%	16.3%	3.5%	29.8%	4.4%	2.5%	0.2%	5.0%	1.4%	2.7%
	Residential						Non-Residential					
	83.8%						16.2%					
New Build			Residential RMI			New Build			Non-Residential RMI			
43.0%			40.8%			11.0%			5.2%			
Project	Non-Project		Project	Non-Project		Project	Non-Project		Project	Non-Project		
38.0%	5.0%		7.5%	33.3%		10.5%	0.5%		3.8%	1.4%		

2.7.6

Route to Market: Plastic Cisterns
CN - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS: 2024

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	67.0%
	Wholesalers / Merchants
	67.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	2.2%	2.0%	5.0%	19.8%	4.4%	6.6%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	4.2%		24.8%		11.0%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
14.0%	8.0%	0.4%	4.0%	6.3%
Installers				
32.7%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	3.4%	28.2%	3.4%	16.0%	3.2%	28.3%	4.0%	2.4%	0.4%	4.8%	1.5%	4.4%
	Residential						Non-Residential					
	82.5%						17.5%					
	New Build			Residential RMI			New Build			Non-Residential RMI		
	40.4%			42.2%			11.5%			6.0%		
	Project		Non-Project	Project		Non-Project	Project		Non-Project	Project		Non-Project
	35.5%		4.9%	8.0%		34.2%	10.9%		0.6%	4.2%		1.8%

Source: BRG Based on Industry Interviews. N.B. Include both Exposed and Concealed Plastic Cisterns - Totals may vary due to roundings

2.7.7 Route to Market - Consolidated View : Plastic Cisterns

CN - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS : 2019

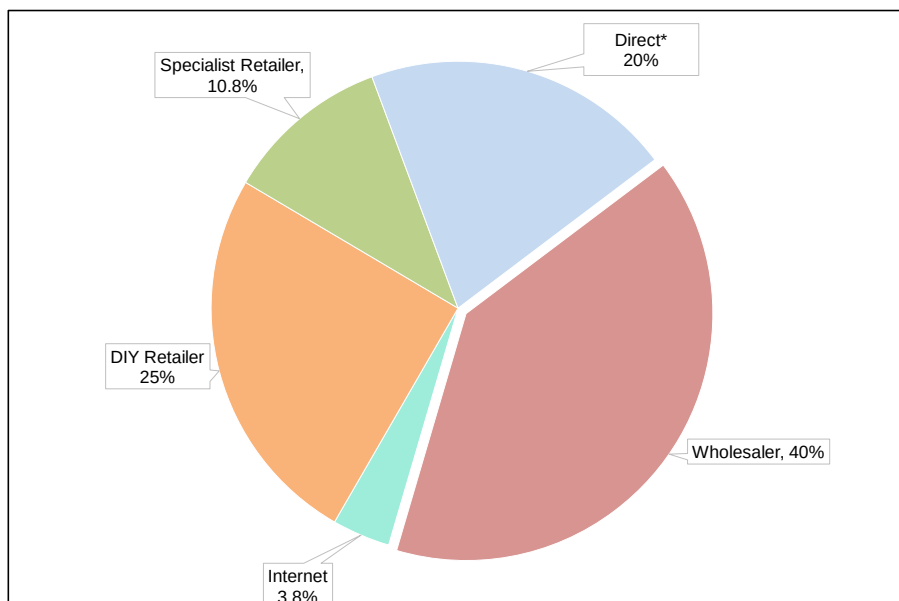
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
39.8%	3.8%	25.2%	10.8%	20.4%	45.5%	14.3%	38.3%	1.9%
100%					59.8%		40.2%	

2.7.8 Route to Market - Consolidated View : Plastic Cisterns

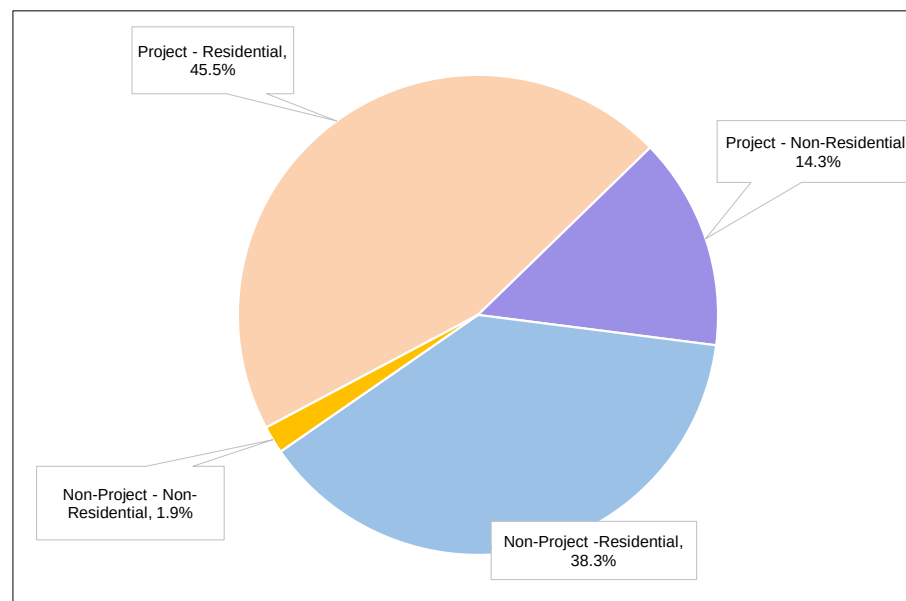
CN - ESTIMATED DISTRIBUTION FOR PLASTIC CISTERNS : 2024

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
38.6%	4.2%	24.8%	11.0%	21.4%	43.5%	15.1%	39.0%	2.4%
100%					58.6%		41.4%	

CN - SALES OF PLASTIC CISTERNS BY DISTRIBUTION CHANNEL: 2019



CN - SALES OF PLASTIC CISTERNS BY POINT OF DECISION AND END-USE SECTOR: 2019



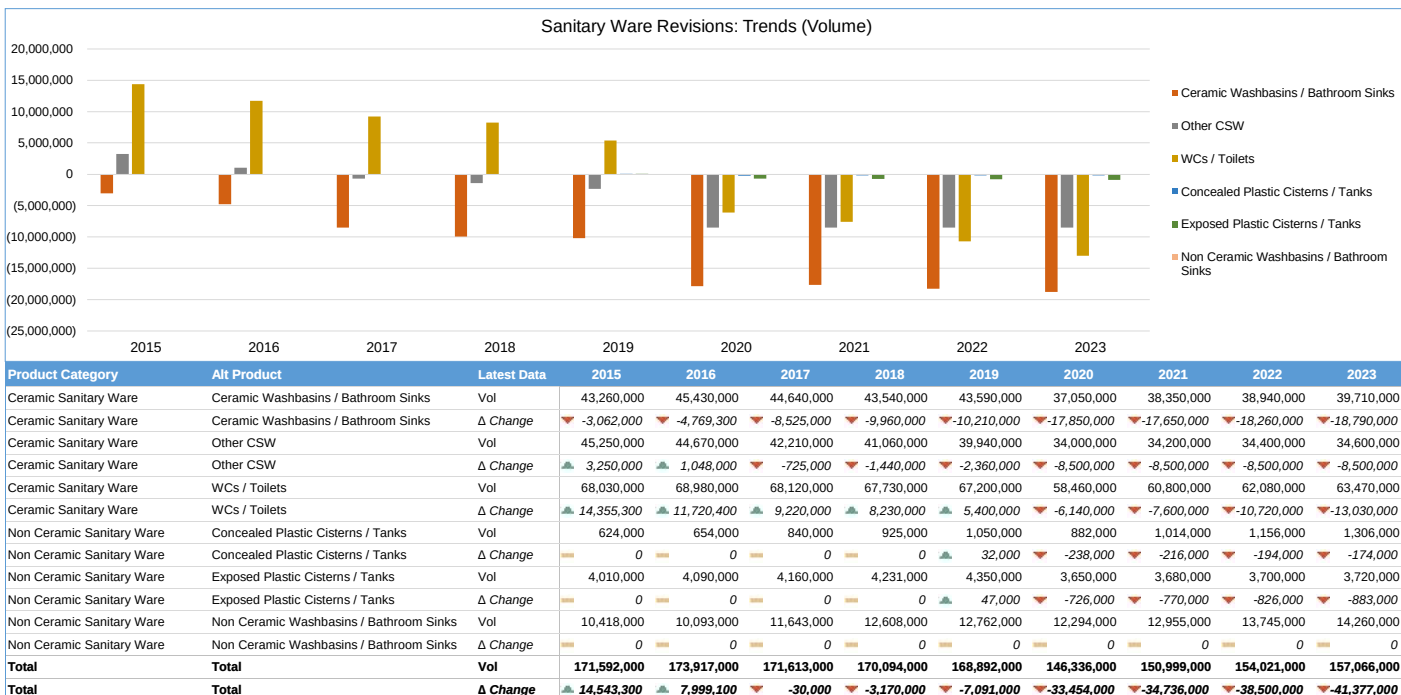
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

2.8 Revisions in the Chinese Sanitary Ware Market (July 2020)

CN - REVISIONS IN THE CHINESE SANITARY WARE MARKET

Data Comparison: July 2019



The ceramic sanitary ware as a whole has been revised historically due to further analysis of the commercial and residential markets in China. Official data on commercial building construction as for residential completions are difficult to confirm year on year and further analysis uncovered that the market had a higher percentage of toilets and hence our toilet number has been revised up at the expense of ceramic wash basins.

CHINA

FAUCETS

THE EAST AND SOUTH EAST ASIAN BATHROOM PRODUCT MARKETS



Market Characteristics and Product Mix

Much like the sanitary ware market, the Chinese faucets market has experienced rapid growth over the last three decades. This was largely driven by the construction boom, both residential and non-residential, as well as rising household incomes and investment in the improvement of sanitation facilities. Although the pace of market growth slowed during the economic crisis of 2008/2009, overall sales demand continued to grow, due a favorable property market combined with high residential construction output.

Single handle faucets have dominated the Chinese market for many years, but other innovations are also evident in the market place. Along with the use of lead-free and ceramic core materials, water-saving designs with extra water temperature setting handles have become popular on the middle and high-end markets. The recent product trend for single handle faucets is towards multifunction models, which have additional functions – for instance, for massage or hair-washing. In washbasins, mixers fitted with a flexible hose (similar to a small shower) for face and hair washing have gained popularity amongst young consumers. The valve and core materials used in the production of faucets tend to be copper and ceramics, materials that comply with new product standards and appeal to consumers concerned with environmental and health benefits. Apart from traditional chrome, colorful faucets with ornament materials, such as stones, fake crystals, plastics, and wood, are also available on the market. Two head mixers are more installed in non-housing segments such as hotels and offices rather than residential areas. They are considered inconvenience to adjust water temperature and taking extra space on washbasins for the residential bathrooms and kitchens.

Thermostatic faucets have gained share relatively quickly in recent years, due to increasing aesthetic and water saving concerns, as well as the prevalent intelligent bathroom and kitchen design. Growth of the middle class and the interest for more comfortable and privileged lifestyles have also stimulated sales of thermostatic faucets, in both housing and non-housing applications. Smart designs, such as touch operations for water temperature settings, water flow controls and anti-limescale, have become popular in the thermostatic faucet segment.

Thermostatic faucets were historically predominantly installed in baths and showers in the non-residential market, such as saunas, spas, swimming pools and hotels, but demand from the residential market has grown quickly over the last few years, often due to growing incomes, but also the general interest for thermostatic faucets and its benefits compared to traditional faucets. In addition, the installation of thermostatic faucets in washbasins have started to grow recently, mostly in commercial buildings.

Self-closing and electronic faucets are increasingly being fitted in public and commercial restrooms, largely due to increased hygiene concerns following the outbreak of SARS in Southern China in 2003 and also the more recent outbreak of

COVID-19. Self-closing faucets are considered as cheap, obsolete products, often installed in low-end commercial and public buildings. Electronic faucets have become increasingly popular and the market has developed rather quickly in recent years, due to high demand for these faucets from commercial and public buildings. On the residential market, there is clear demand for electronic kitchen faucets. Product designs have shifted from traditional and simple designs to more artistic designs with unique shapes, materials, colors, and functions, to fulfil consumer tastes for designer products. There are many Chinese suppliers of electronic faucets that have entered to the market in recent years, resulting a very fragmented supply structure.

Pillar faucets are typically installed in laundry ('mob') sinks, garages, rural bathrooms and kitchens, commercial kitchens, cleaning rooms, public toilets, schools and hospitals. Demand for pillar faucets is higher in the South than the North, due to the climate differences between these regions. However, it must be noted that there is a very high penetration of pillar faucets fitted in rural dwellings in China. Traditional pillar faucets were made of cast iron in China, with most of these now having reached the end of their lifespan, to be increasingly replaced with copper models.

The overcapacity of faucet supply, combined with the influx of poor quality, often defective, models and several incidents of lead poisoning, largely damaged consumers' confidence in the quality of faucets from 2010 to 2012. As a result, the Chinese government launched standards on the restriction of lead contents in faucets, which pushed many small manufacturers (that predominantly produced poor-quality products) out of the market and caused an influx of large manufacturers (who produced high-quality products). Consumers' confidence in the quality of domestically produced faucets resumed, and the market has returned to more healthy growth rates since 2013.

Foreign brands such as KOHLER, TOTO, AMERICAN STANDARD, and ROCA have been dominating the medium-to upper-end market, such as in Tier 1 cities, for many years, but have recently also started to move their focus to Tier 2 to Tier 3 and even Tier 4 cities in recent years by launching suitable products at reasonable prices. Chinese manufacturers, such as JOMOO and ARROW, have improved their production quality and services to expand their position on the market, especially in the medium-to-upper end. The consumer's perception on faucets have changed in recent years, in line with the improvement of household incomes, and affected by the pursuit for an improved lifestyle and cultural interests, resulting in increased demand for more high-end faucets. Strong growth in online sales and improving services offered have also led to growth.

Recent Market Trends

The Chinese faucet market remained practically static in 2019 as increased sales in housing RMI did not offset declines seen in new housing and commercial sector. In general, sales to the project market declined apart from certain areas such as Beijing and the luxury hotel market, but retail sales grew driven by the growth of

home redecoration and replacement for products with better function, design, materials and water efficiency. Sales of economy priced products in the tier 1 cities declined, but sales of medium- to upper-end products grew reflecting higher incomes and the pursuit for better quality products in large cities in China. Sales to hotel and office renovation and upgrading projects continued to grow steadily in the tier 1 cities, as well as sales to other non-housing segments such as schools, hospitals, and other public buildings due to accelerating urbanization.

In terms of product trends, the market continued to move towards safe, hygienic, water-efficient products, driven by more strict regulations and policies. The implementation of new standard Ceramic Cartridge Faucets (GB18145-2014) and Sanitary Fixture for Water Saving (GB/T31436-2015) at the end of 2014 and 2015 respectively forced the high lead and other 17 metal content faucets out of the market, and enhanced replacement demand for safe and environmentally friendly products amongst consumers. As a result, there are many medium and small firms whose products are unable to reach the new standard, which either closed down or were reformed.

Water efficiency is still the main R&D focus for many faucet manufacturers in China, largely driven by the government's supporting policies. For example, the Beijing government continues to subsidize the sale and installation of water saving faucets and showers. There is a water-efficient labelling scheme for bathroom and kitchen products in Beijing, and the government aim is to reduce the water flow per minute for WE Grade II faucets to 7.5 liters and 9 liters for ordinary faucets. The Beijing government is currently promoting the use of water efficient bathroom products and provide subsidies of up to 20% of the retail price of WE Grade II faucets to consumers and this resulted in sales of large faucets suppliers, such as TOTO, LEHUA, HEGII, DONGPENG, FAENZA, HUIDA, JOMOO and KOHLER. The Shanghai government continued to promote water efficient faucets and provide free WE products and installation to replace the old-fashioned cast iron faucets in old housing. There are various policies on promotion of water efficient faucets in other regions.

Prices and Market Values

Average prices for all types of faucets increased in 2019, largely as a result of increasing raw material costs, such as copper, steel and nickel grew, which led to increased production costs. In addition, wages, transportation and costs for environmental treatment also went up in 2019, resulting in average MSP increases. At the same time, there was also more demand for medium- to high-end priced products which contributed to the overall average MSP increase.

New Housing

Sales of faucets into new housing were stable for 2019 with some regional growth of sales to fully renovated new housing in Beijing, Jiangsu, Guangdong, Zhejiang, Shandong, Anhui and Sichuan. The construction of fully renovated new dwellings continued to grow moving from tier 1 and tier 2 cities to some tier 3 to tier 4.

Housing RMI

Sales of faucets to housing RMI were positive in 2019 due to increased replacement rates as a result of a growth in personal wealth which drives remodeling activity.

The majority of the sales of medium- to high-end faucets to the housing RMI segment are mainly in tier 1 and tier 2 cities, such as Beijing, Shanghai, Shenzhen, Tianjin and Guangdong due to the higher income of residents. There is also a considerable volume of sales for replacements in the vast rural areas with the improvement of incomes and transformation of bathroom and kitchens.

With the continuous medium growth rate of the economy, improvement of income, increasing concerns on water efficiency and health of products, and improving sanitary levels in small cities and rural areas, as well as the ending of the lifecycle of more installed faucets, it is expected that RMI rates will continue to grow in the next few years, after the affects of the COVID-19 pandemic have subsided.

Non-Housing

Sales of faucets into the non-housing segment were stable in 2019 due to the stagnation of demands in the major sub-segments of hotels and office buildings, offset by a growth of sales to schools, hospitals and other institutional buildings. Sales to both luxury hotel and office buildings and economy hotels and resorts increased, representing a polarized trend for faucets.

It is expected that the demand for faucets post-COVID in non-housing segment will continue to grow steadily for several years in China due to the expansion of urbanization and transformation of commercial and public infrastructures.

Supply Structure

The Chinese faucet market is highly fragmented with thousands of domestic manufacturers and a few international players. There are a large number of small players concentrated on sales to the economy and low-priced segments locally, giving the large identified Others category in the market share.

Chinese suppliers predominantly take the majority of the share of the market in China, led by JOMOO, SUNLOT, ARROW, HEGII, JOYOU, and FAENZA. The international companies such as MOEN, KOHLER, AMERICAN STANDARD, GROHE, TOTO, HANS GROHE, and ROCA, have only 7% share of the market. The international companies are mainly concentrated in high end market, while the Chinese players are in medium to low end segment.

MOEN, KOHLER, TOTO, and GROHE are the major four international suppliers. AMERICAN STANDARD, HANS GROHE, and the ROCA GROUP also have a reasonable presence on the market. In 2019, sales of faucets by most of

international companies grew healthily, due to the increasing demand for medium- to high-end products, which in turn was spurred by growth in housing RMI sales and increased demand for faucets from high grade offices, hotels, malls, and other commercial and public buildings.

The activities of major faucet suppliers in 2019 are:

DONGPENG started the production of its intelligent home creative industrial park in Yongchuan, Chongqing in November 2019. The industrial park has 10 tile lines, 3 lines of ceramic sanitary wares (including intelligent WCs), 1 line of bathroom cabinets, 1 line of shower enclosures, 1 line of faucets, a R&D center and a logistic center.

In 2019, SALLY stopped the supply of faucets and concentrated on the shower enclosure business.

Forecasts

Given the exceptional circumstances related to the spread of the COVID-19 virus and measures in place for its containment, forecasts for 2020 and beyond are based on the latest information available at the time of publication. BRG has endeavored to factor in the latest developments – see Macroeconomic data in Table A.3 and supporting summary table at the end of the Background section.

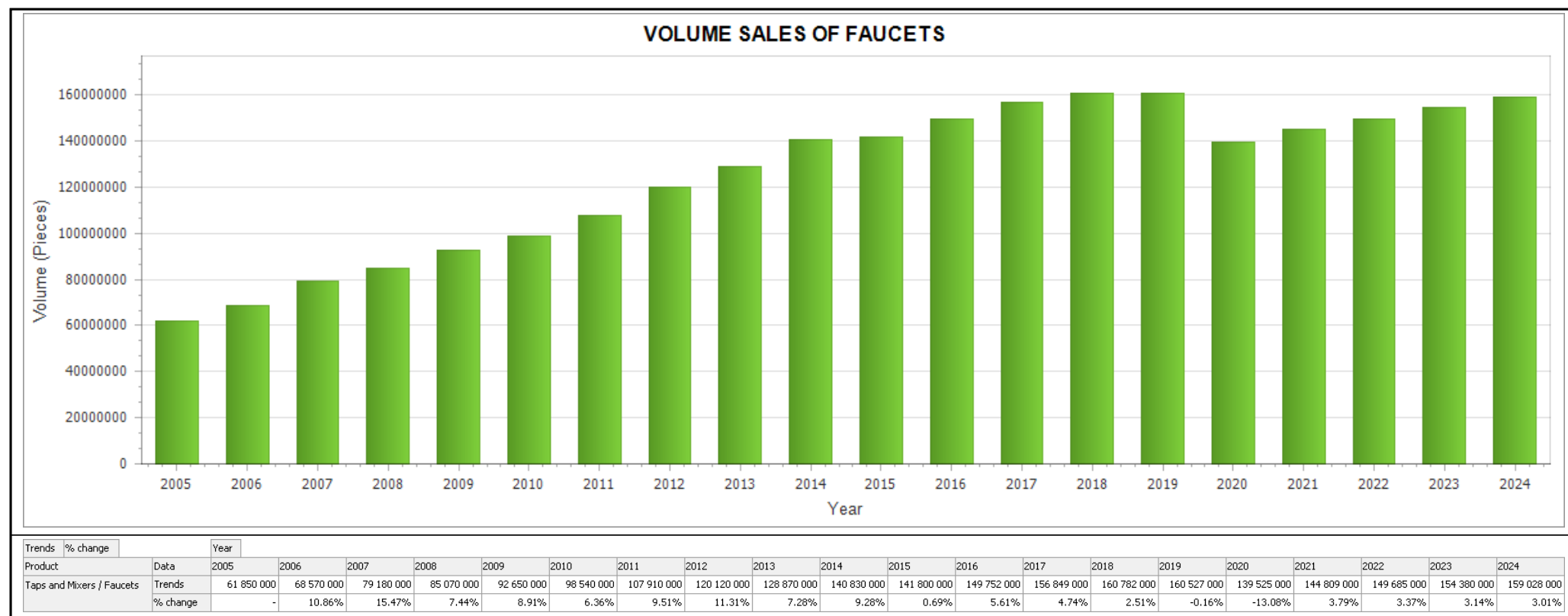
The COVID-19 pandemic has widely affected the sales of all faucets in China due to the cessation of both production and construction in the first quarter. With the resumption of production and construction/RMI activities, it is estimated that the market will decline by over 13% in 2020 as there are still pockets of reoccurring virus infections particularly in the bigger cities curtailing building starts.

The market is estimated to pick up however in 2021 due to the increased penetration of fully decorated new housings, the construction of government housing, and the expansion of the market from tier 1 and tier 2 cities to tier 3 to tier 4 cities and rural areas, which will continue to boost the sales through the reporting period. Housing RMI will continue to be an expanding segment with the continuous growth of home decoration, the ending of the lifespan of more and more products, growing concerns on water efficiency due to rising water prices, improvements in annual income, as well growth in the millennial population which prefer innovative products.

Post-COVID, the sales of taps and mixers to public and commercial segments will continue to grow steadily with the acceleration of urbanization, governments' efforts on infrastructural transformation, and booming tourism.

3.2 Historical Trends and Forecasts : Faucets

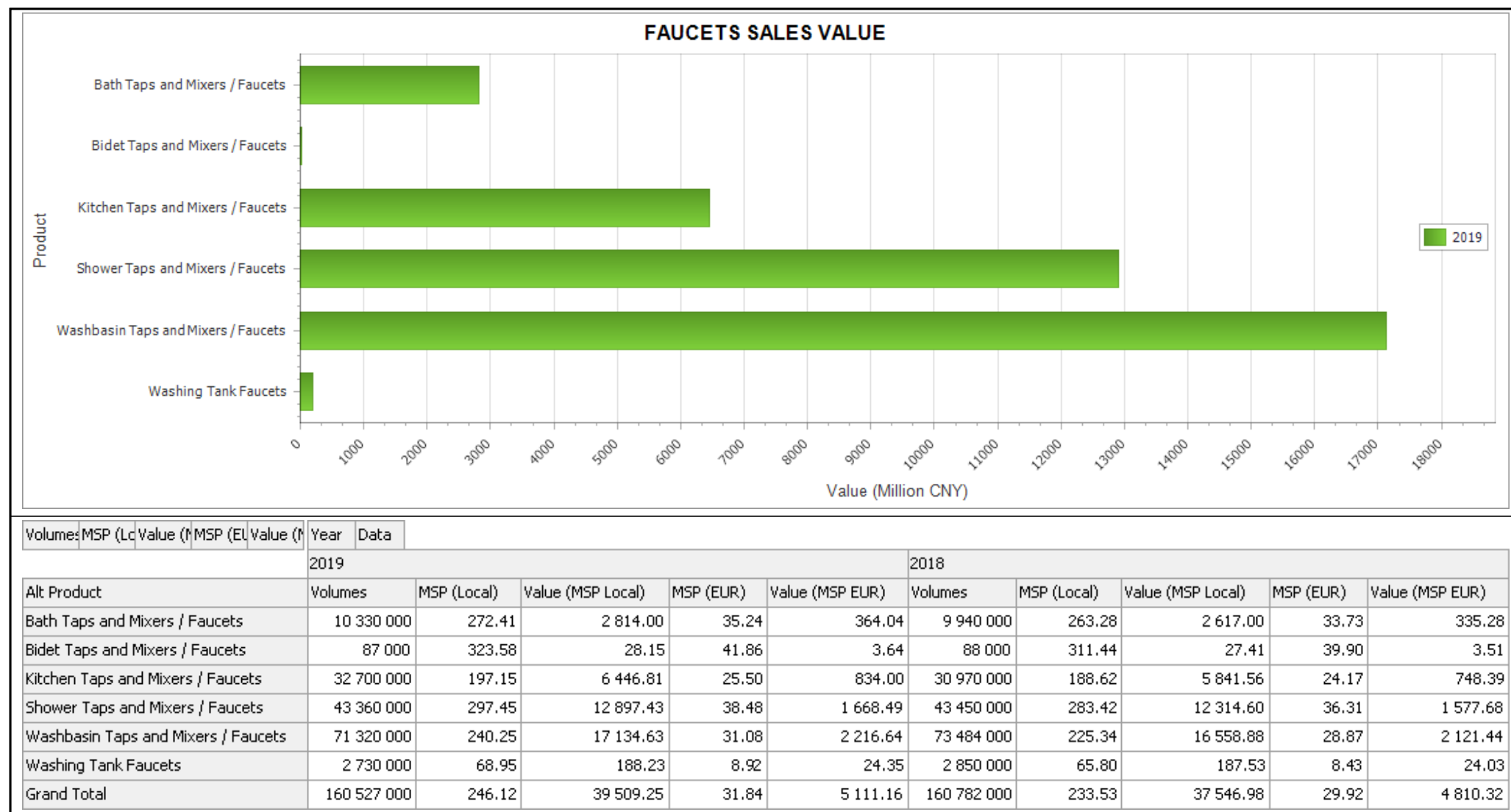
CN - SUMMARY OF TRENDS IN THE FAUCETS MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews. N.B. Includes Both Kitchen and Sanitary Taps & Mixers

3.3.1 Prices and Market Values : Faucets

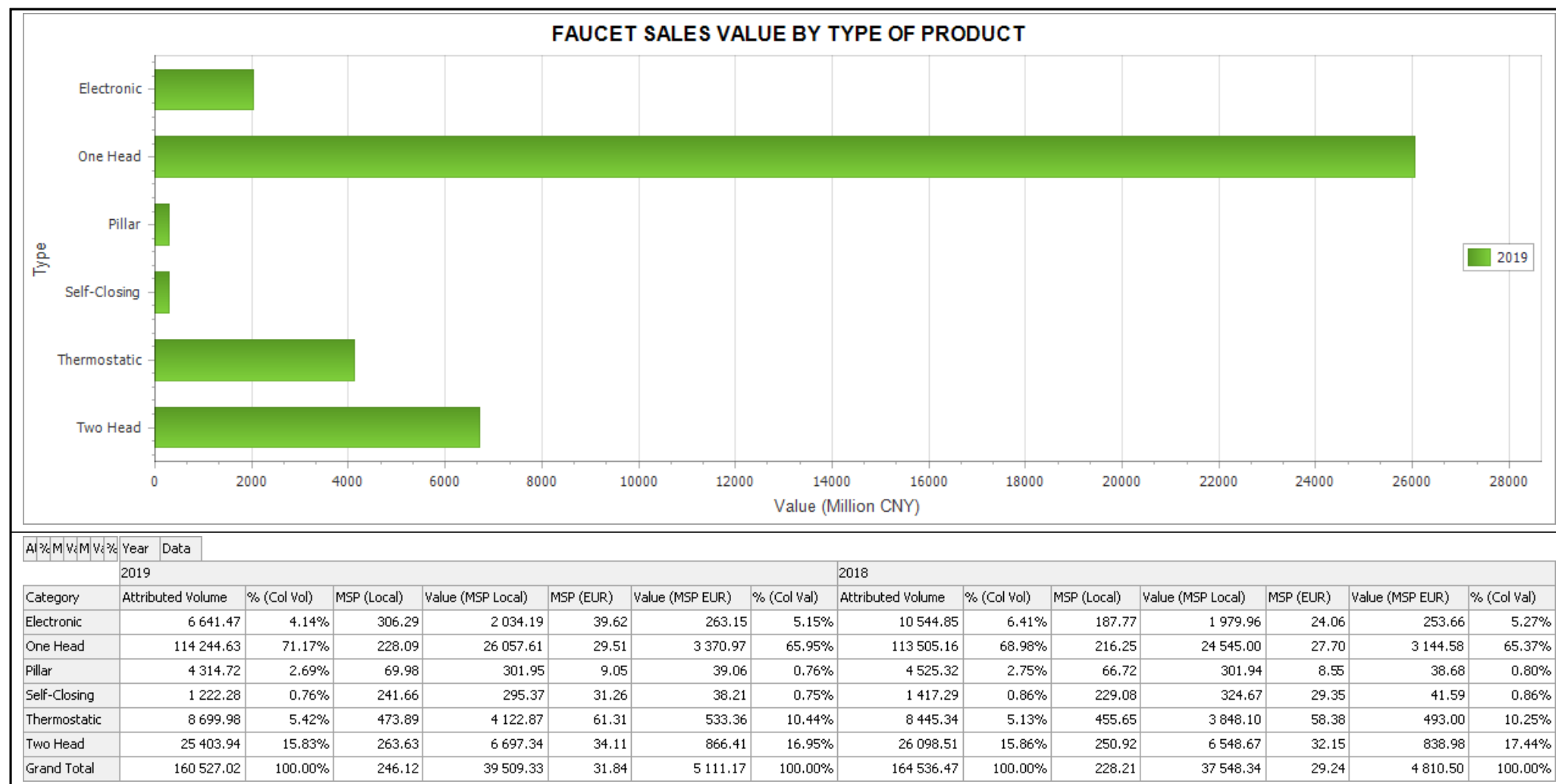
CN - ESTIMATED VALUES OF THE FAUCETS MARKET



Source: BRG based on industry interviews. Local Prices Shown in CNY. Values Shown in Millions (at MSP = Manufacturers' Selling Price).

3.3.2 Prices and Market Values by Type : Faucets

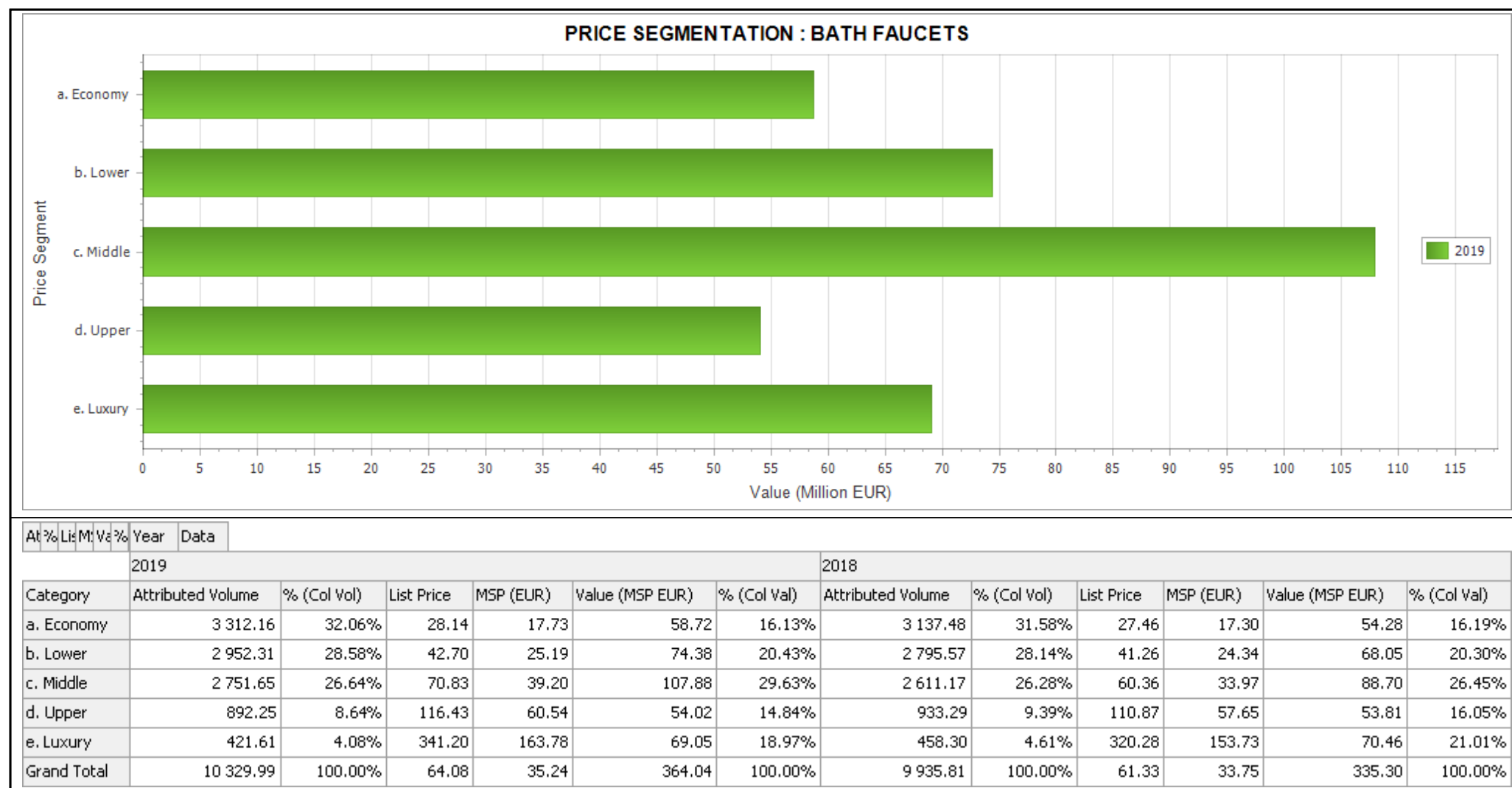
CN - ESTIMATED VALUES OF THE FAUCET MARKET BY TYPE OF PRODUCT



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. Local Prices Shown in CNY. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price).

3.3.3 Price Segmentation : Bath Faucets

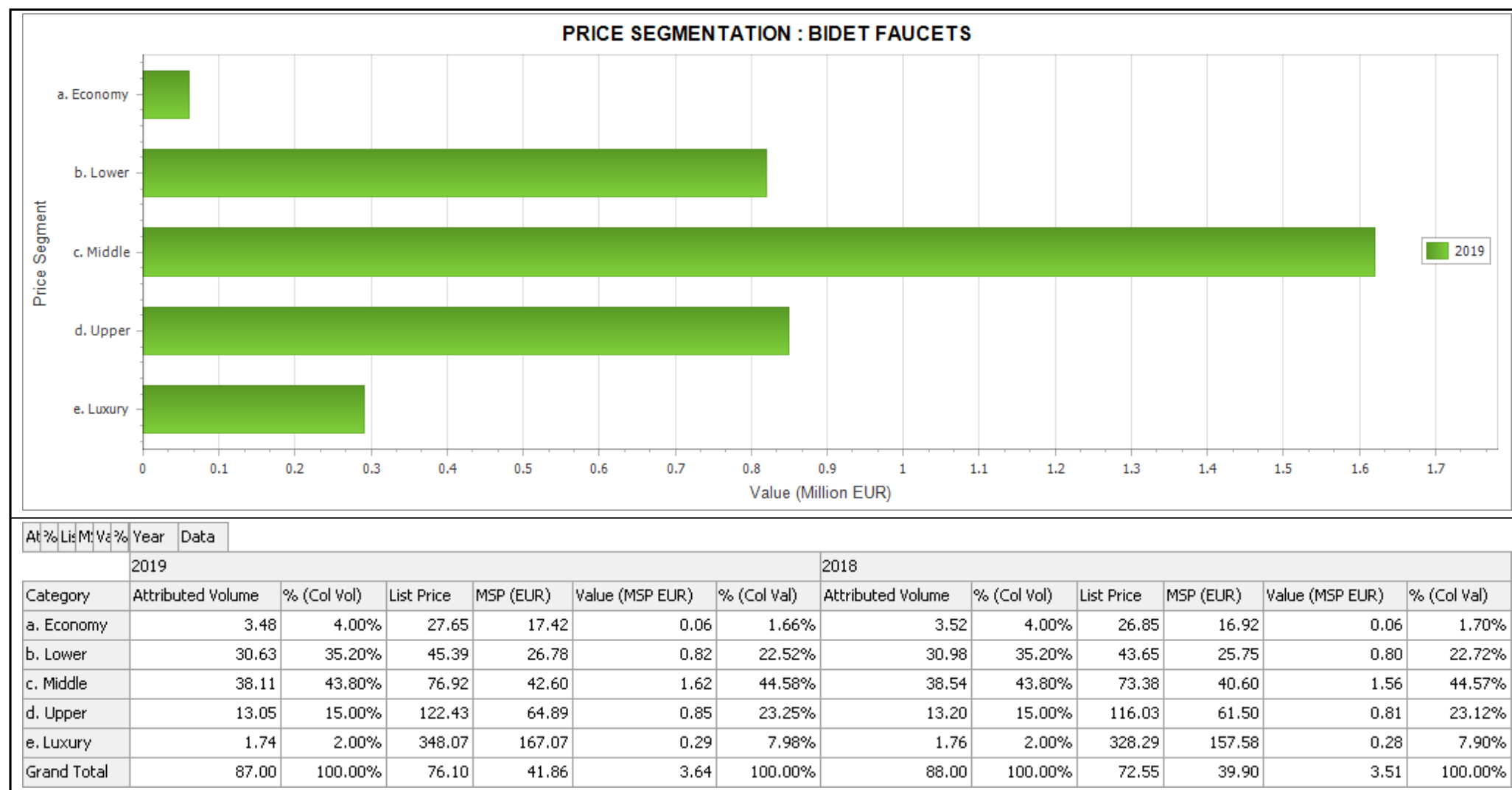
CN - ESTIMATED PRICE SEGMENTATION FOR BATH FAUCETS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€20, Lower €20-€30, Middle €30-€50, Upper €50-€70 and Luxury >€70

3.3.4 Price Segmentation : Bidet Faucets

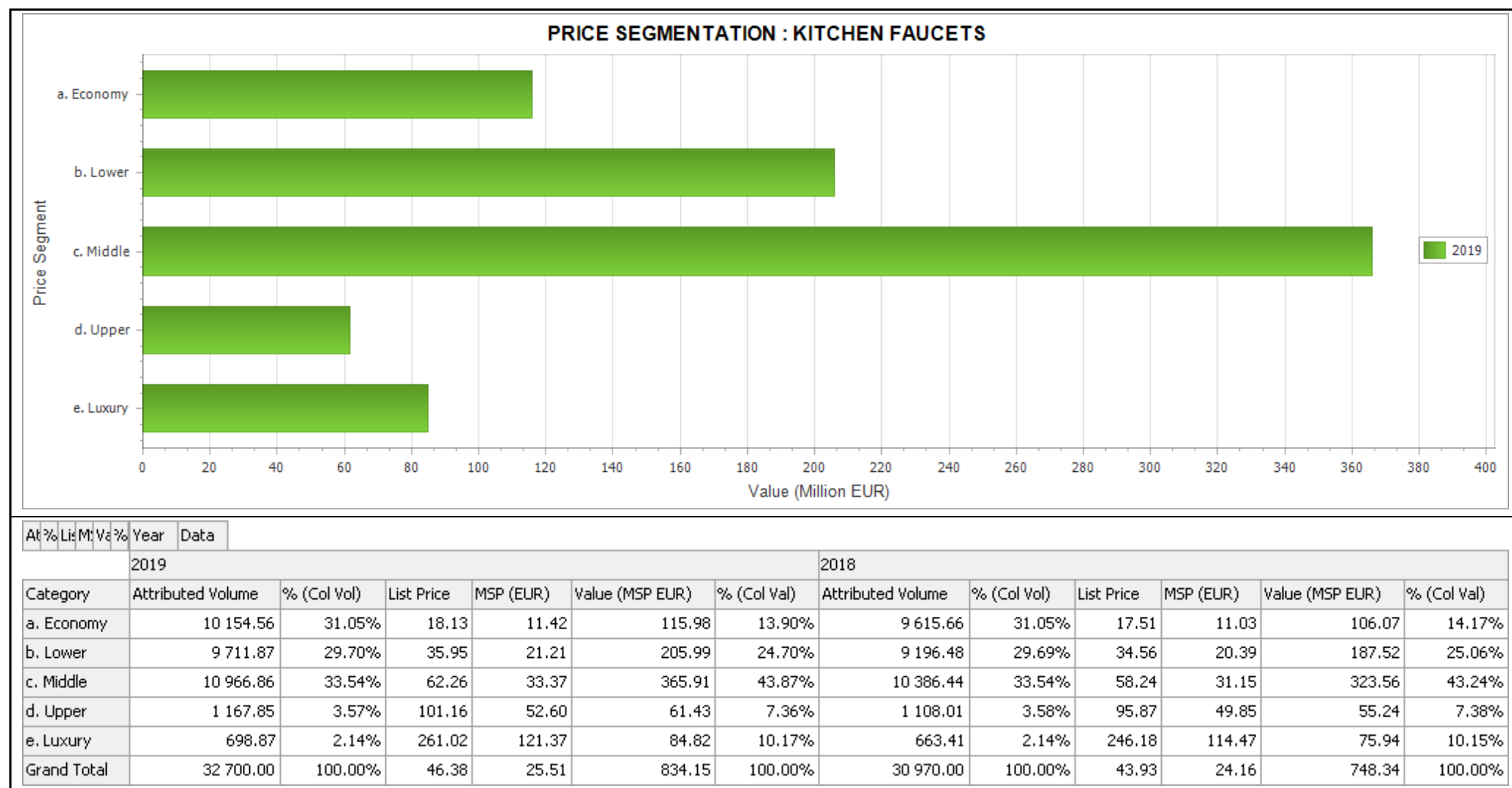
CN - ESTIMATED PRICE SEGMENTATION FOR BIDET FAUCETS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€20, Lower €20-€30, Middle €30-€50, Upper €50-€70 and Luxury >€70

3.3.5 Price Segmentation : Kitchen Faucets

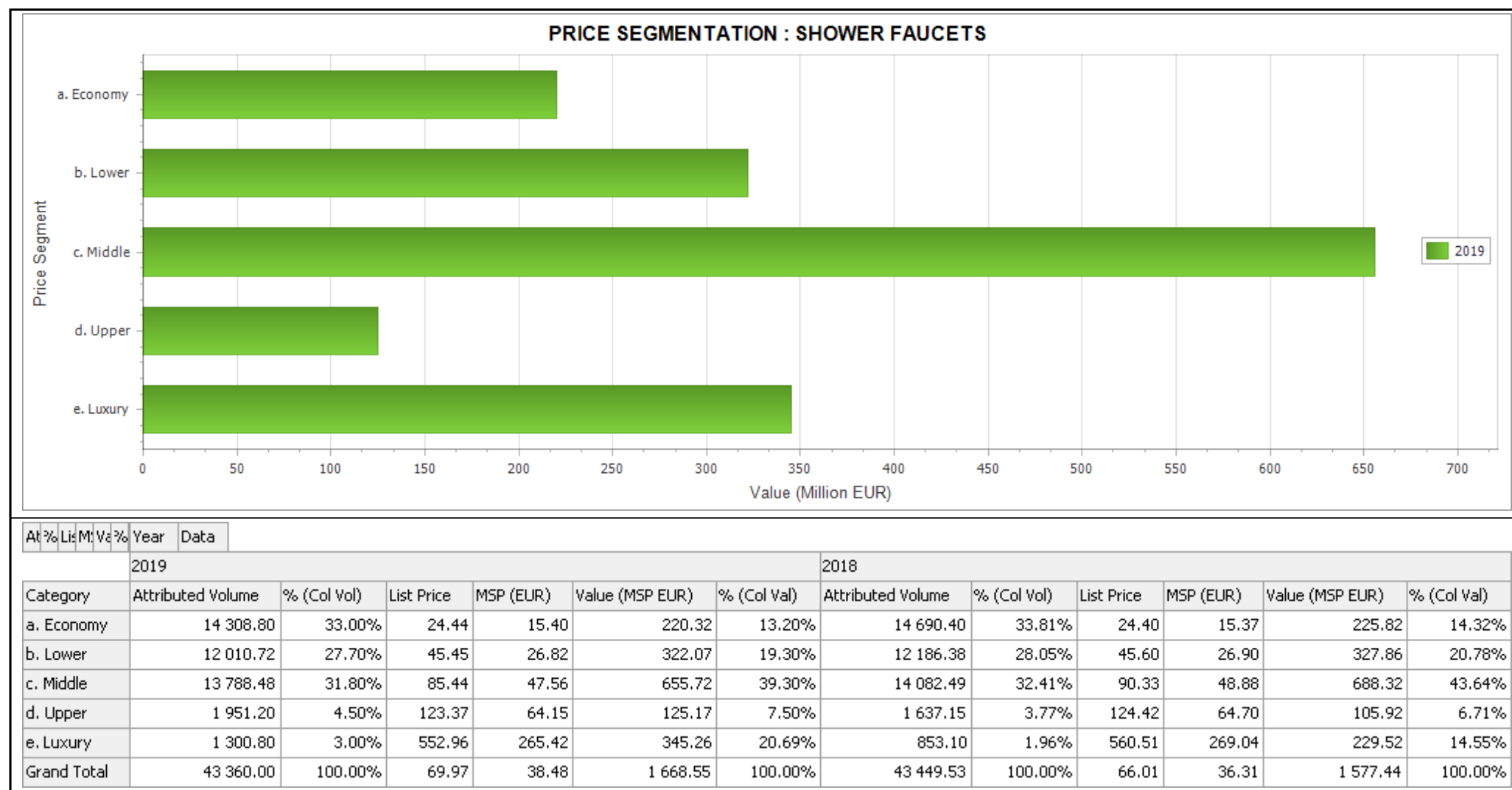
CN - ESTIMATED PRICE SEGMENTATION FOR KITCHEN FAUCETS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€15, Lower €15-€25, Middle €25-€40, Upper €40-€60 and Luxury >€60

3.3.6 Price Segmentation : Shower Faucets

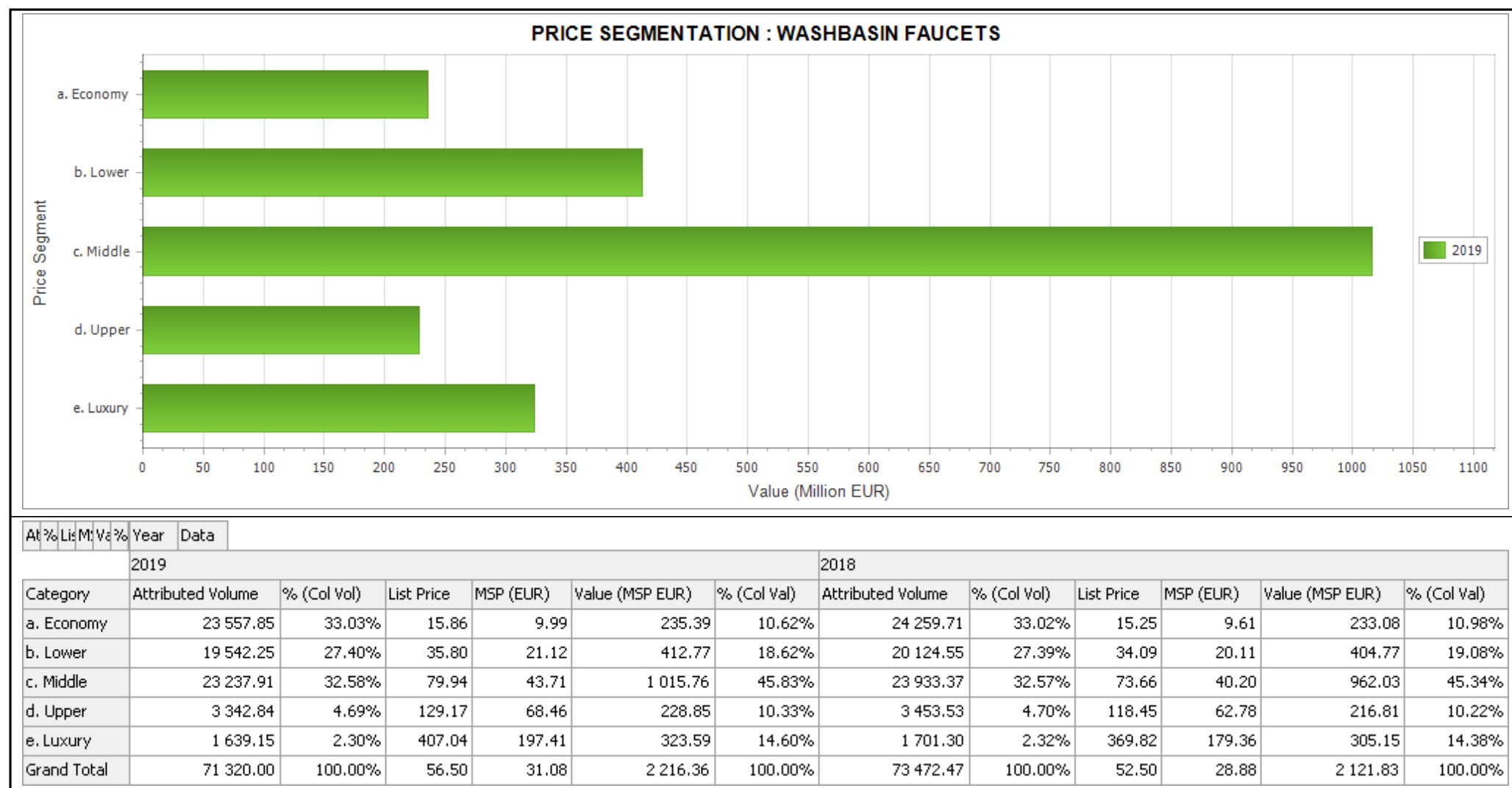
CN - ESTIMATED PRICE SEGMENTATION FOR SHOWER FAUCETS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€20, Lower €20-€30, Middle €30-€50, Upper €50-€80 and Luxury >€80

3.3.7 Price Segmentation : Washbasin Faucets

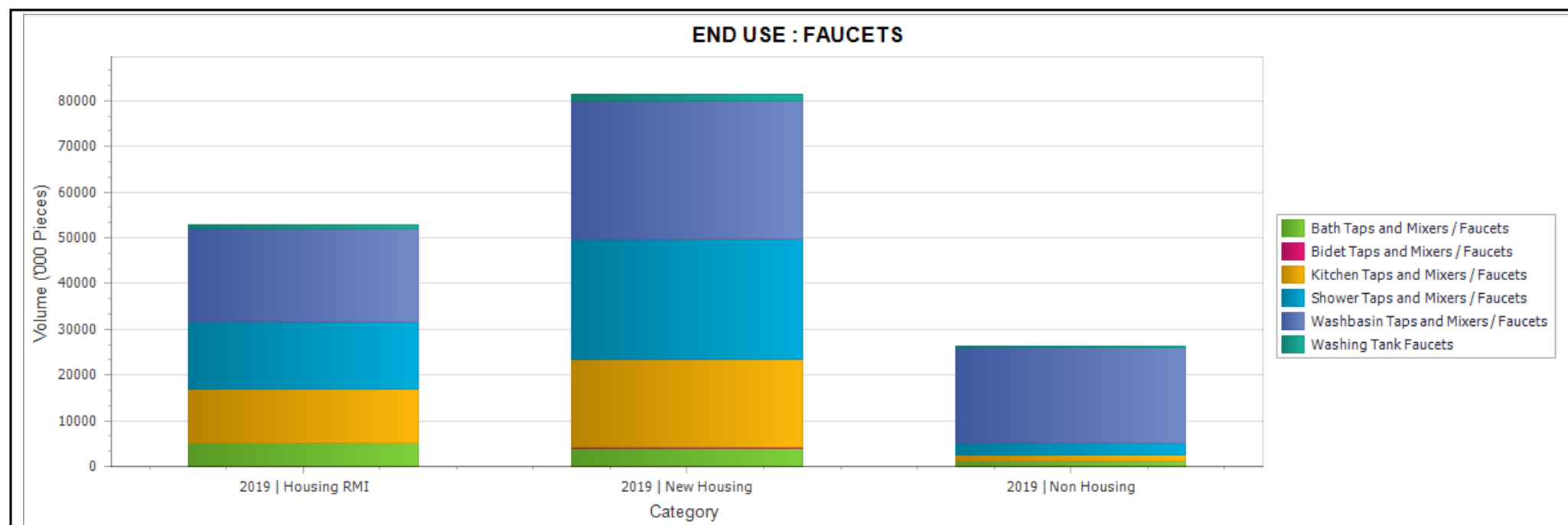
CN - ESTIMATED PRICE SEGMENTATION FOR WASHBASIN FAUCETS



Source: BRG based on industry interviews. Total may show slight variance compared to market data due to rounding. Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy <€15, Lower €15-€30, Middle €30-€50, Upper €50-€100 and Luxury >€100

3.4.1 End Use : Base Year

CN - SALES OF FAUCETS BY TYPE AND END USE SEGMENT ('000 PIECES)

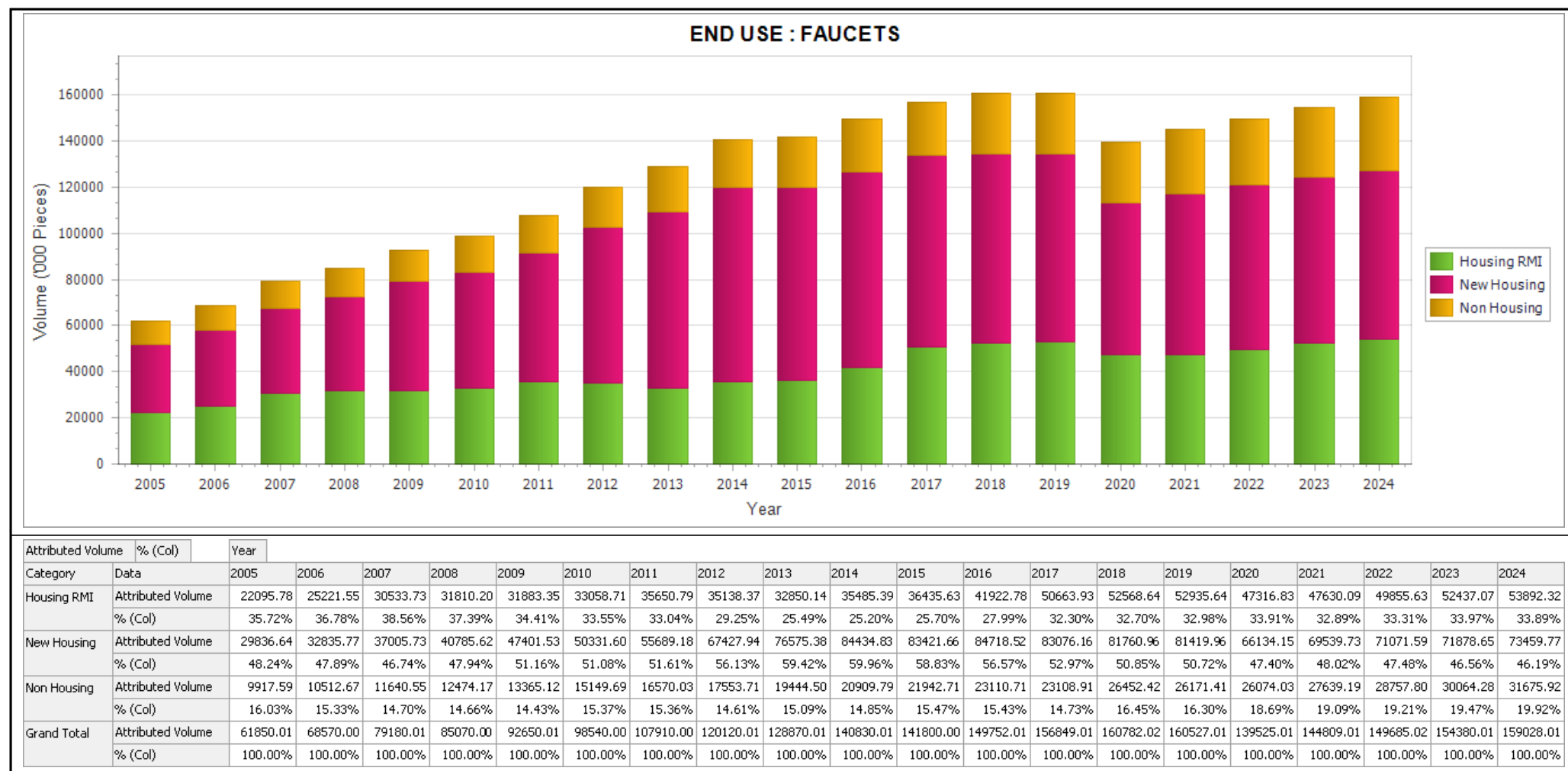


Attributed Volume	% (Col)	% (Row)	Year	Category	Data								
						2019					2019 Total		
						Housing RMI			New Housing				
Alt Product	Attributed Volume	% (Col)	% (Row)	Attributed Volume	% (Col)	% (Row)	Attributed Volume	% (Col)	% (Row)	Attributed Volume	% (Col)	% (Row)	
Bath Taps and Mixers / Faucets	5186.49	9.80%	50.21%	4080.38	5.01%	39.50%	1063.13	4.06%	10.29%	10330.00	6.44%	100.00%	
Bidet Taps and Mixers / Faucets	25.08	0.05%	28.82%	27.19	0.03%	31.25%	34.74	0.13%	39.93%	87.00	0.05%	100.00%	
Kitchen Taps and Mixers / Faucets	11863.59	22.41%	36.28%	19385.04	23.81%	59.28%	1451.38	5.55%	4.44%	32700.00	20.37%	100.00%	
Shower Taps and Mixers / Faucets	14535.89	27.46%	33.52%	26203.63	32.18%	60.43%	2620.49	10.01%	6.04%	43360.01	27.01%	100.00%	
Washbasin Taps and Mixers / Faucets	20263.35	38.28%	28.41%	30284.00	37.19%	42.46%	20772.65	79.37%	29.13%	71319.99	44.43%	100.00%	
Washing Tank Faucets	1061.25	2.00%	38.87%	1439.73	1.77%	52.74%	229.02	0.88%	8.39%	2730.00	1.70%	100.00%	
Grand Total	52935.64	100.00%	32.98%	81419.96	100.00%	50.72%	26171.41	100.00%	16.30%	160527.01	100.00%	100.00%	

Source: BRG based on industry interviews. Totals may show slight variance due to rounding. N.B. Includes Kitchen and Sanitary Taps & Mixers

3.4.2 End Use : Historical and Forecasts

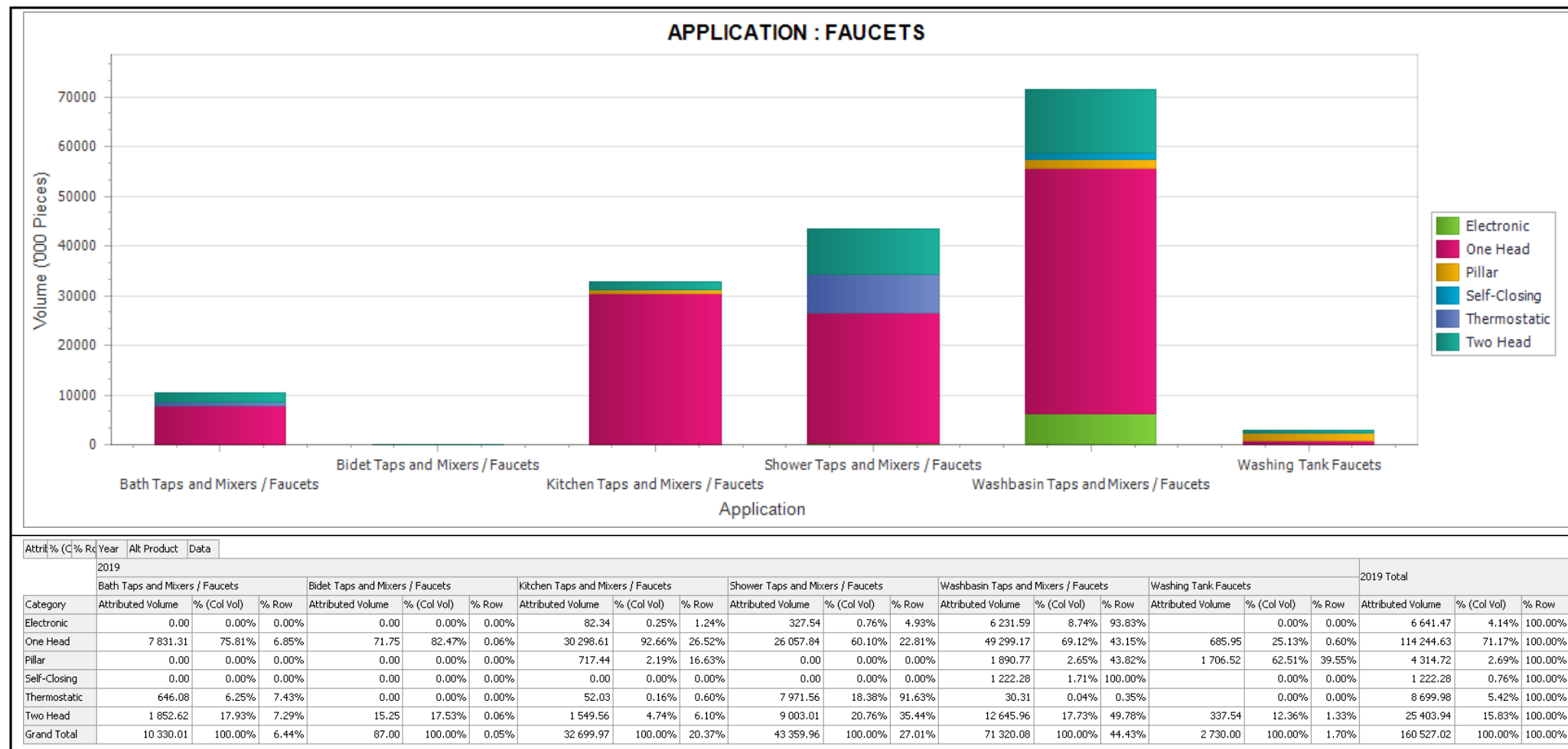
CN - SALES OF FAUCETS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding. N.B. Includes Both Kitchen and Sanitary Taps & Mixers

3.4.3 Sales by Type and Application : Faucets

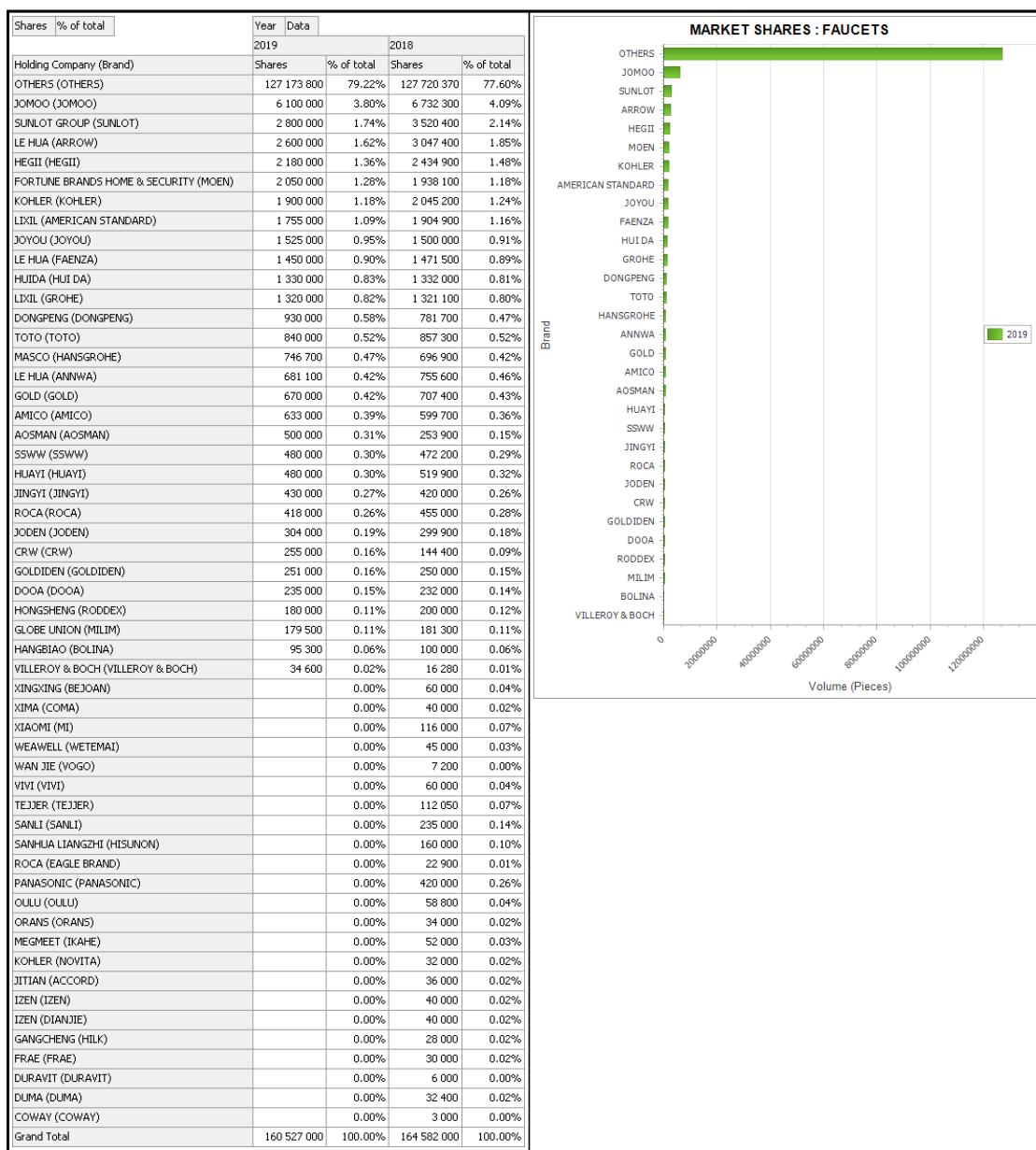
CN - FAUCETS SALES BY TYPE AND APPLICATION ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

3.5 Market Shares : Faucets

CN - ESTIMATED MARKET SHARES FOR FAUCETS



Source: BRG based on industry interviews. N.B. Includes Both Kitchen and Sanitary Taps & Mixers.

3.6.1 CN - IMPORTS OF FAUCETS 2019

Source Country	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
Germany	20 197	23.7%	909 270	23.5%	19 548	23.1%	846 926	23.7%
Japan	10 649	12.5%	644 316	16.7%	10 159	12.0%	549 749	15.4%
Italy	9 597	11.2%	282 970	7.3%	6 676	7.9%	186 458	5.2%
United States of America	6 448	7.6%	522 511	13.5%	8 379	9.9%	583 158	16.3%
Korea, Republic of	5 952	7.0%	189 475	4.9%	7 058	8.3%	186 312	5.2%
Taipei, Chinese	4 200	4.9%	126 969	3.3%	5 010	5.9%	161 743	4.5%
India	3 445	4.0%	76 398	2.0%	2 140	2.5%	49 860	1.4%
United Kingdom	2 348	2.7%	148 771	3.9%	2 056	2.4%	106 287	3.0%
Thailand	1 794	2.1%	42 804	1.1%	1 775	2.1%	46 435	1.3%
France	1 659	1.9%	156 680	4.1%	2 013	2.4%	145 123	4.1%
Finland	1 649	1.9%	48 819	1.3%	1 601	1.9%	39 447	1.1%
OTHERS	17 452	20.4%	714 020	18.5%	18 130	21.4%	671 070	18.8%
Total	85 390	100.0%	3 863 003	100.0%	84 545	100.0%	3 572 568	100.0%

Source : Trademap

3.6.2 CN - EXPORTS OF FAUCETS 2019

Destination	2019				2018			
	Volume		Value 000		Volume		Value 000	
	Tonnes	%	Euros	%	Tonnes	%	Euros	%
United States of America	230 215	18.3%	2 008 664	20.2%	292 662	23.2%	2 388 321	24.9%
Russian Federation	82 241	6.5%	527 867	5.3%	78 683	6.2%	466 231	4.9%
United Kingdom	47 364	3.8%	442 782	4.5%	46 413	3.7%	431 837	4.5%
Germany	39 582	3.1%	414 611	4.2%	35 367	2.8%	346 757	3.6%
Indonesia	38 371	3.1%	242 891	2.4%	27 941	2.2%	181 815	1.9%
United Arab Emirates	38 131	3.0%	252 918	2.5%	30 995	2.5%	179 546	1.9%
Spain	35 890	2.9%	265 727	2.7%	34 161	2.7%	242 733	2.5%
India	35 517	2.8%	235 156	2.4%	31 955	2.5%	202 435	2.1%
Korea, Republic of	33 636	2.7%	253 557	2.6%	34 720	2.8%	245 255	2.6%
Canada	30 246	2.4%	253 280	2.6%	33 110	2.6%	263 133	2.7%
Italy	27 951	2.2%	201 445	2.0%	27 212	2.2%	186 651	1.9%
OTHERS	617 788	49.2%	4 832 661	48.7%	588 153	46.6%	4 447 677	46.4%
Total	1 256 932	100.0%	9 931 559	100.0%	1 261 372	100.0%	9 582 391	100.0%

Source : Trademap

3.7.1 Route to Market: Kitchen Taps & Mixers / Faucets
CN - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS & MIXERS / FAUCETS: 2019

Manufacturer / Importer	Manufacturers / Importers					
	100.0%					
Wholesale Level	69.2%					
	Wholesalers / Merchants					
	69.2%					
Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	1.7%	3.0%	8.4%	20.5%	4.2%	9.5%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	4.7%		28.9%		13.7%	
</						

3.7.2 Route to Market: Kitchen Taps & Mixers / Faucets
CN - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS & MIXERS / FAUCETS: 2024

Manufacturer / Importer	Manufacturers / Importers					
	100.0%					
Wholesale Level	69.9%					
	Wholesalers / Merchants					
	69.9%					
Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	1.8%	3.2%	8.4%	21.2%	4.3%	9.7%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	5.0%		29.6%		14.0%	

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

3.7.3 Route to Market - Consolidated View : Kitchen Taps and Mixers / Faucets

CN - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS AND MIXERS / FAUCETS : 2019

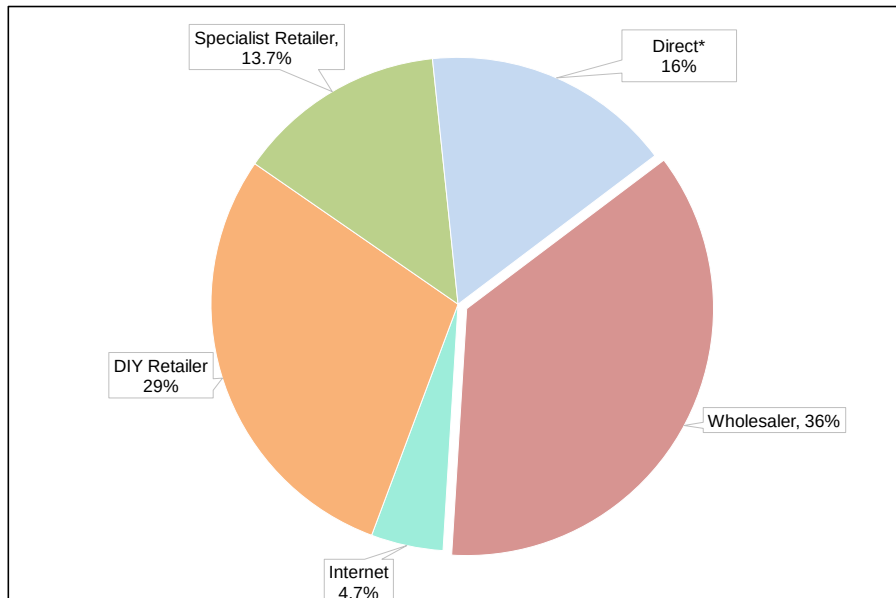
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
36.2%	4.7%	28.9%	13.7%	16.4%	55.7%	2.7%	39.9%	1.7%
100%					58.4%		41.6%	

3.7.4 Route to Market - Consolidated View : Kitchen Taps and Mixers / Faucets

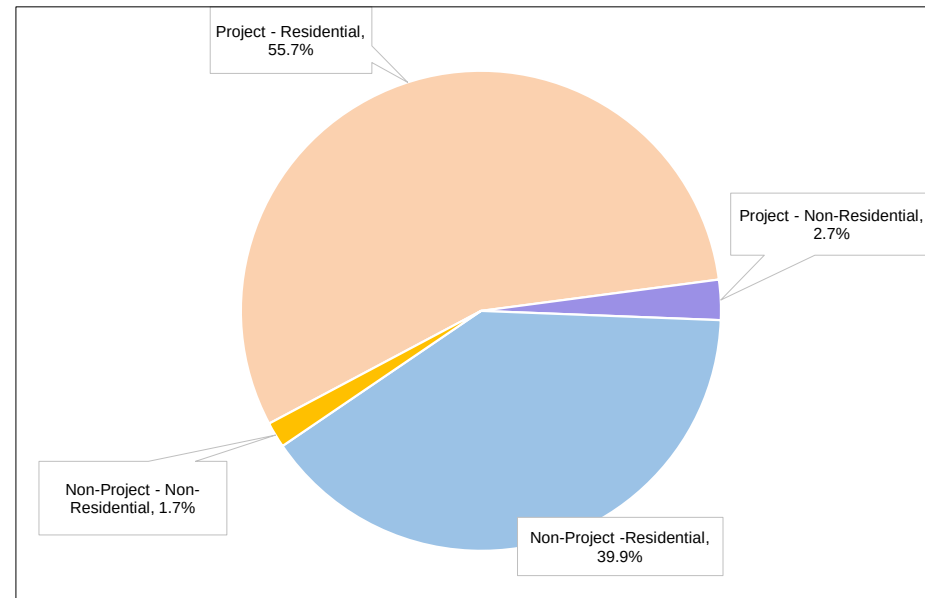
CN - ESTIMATED DISTRIBUTION FOR KITCHEN TAPS AND MIXERS / FAUCETS : 2024

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
35.8%	5.0%	29.6%	14.0%	15.6%	53.9%	3.1%	40.7%	2.3%
100%					57.0%		43.0%	

CN - SALES OF KITCHEN TAPS AND MIXERS / FAUCETS BY DISTRIBUTION CHANNEL: 2019



CN - SALES OF KITCHEN TAPS AND MIXERS / FAUCETS BY POINT OF DECISION AND END-USE SECTOR: 2019



Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

3.7.5 Route to Market: Sanitary Taps & Mixers / Faucets
CN - ESTIMATED DISTRIBUTION FOR SANITARY TAPS & MIXERS / FAUCETS: 2019

Manufacturer / Importer	Manufacturers / Importers							
	100.0%							
Wholesale Level	73.7%							
	Wholesalers / Merchants							
	73.7%							
Retail Level	Manufacturer		Wholesaler					
	1.7%		4.6%					
	Internet Retailers							
	6.3%							
	Manufacturer		Wholesaler					
	5.3%		24.0%					
	DIY Retailers							
	29.3%							
	Manufacturer		Wholesaler					
	3.1%		5.7%					
Specialist Retailers								
8.8%								
End Use Level	Manufacturer		Wholesaler		Internet	DIY	Specialist	
	11.1%		4.0%		1.0%	4.2%	2.2%	
	Installers							
	22.5%							
	Manufacturer		Wholesaler		Internet	DIY	Specialist	Installer
	0.8%		34.3%		4.9%	24.5%	4.5%	11.7%
	Residential							
	80.7%							
	New Build				Residential RMI			
	48.5%				32.1%			
Project		Non-Project		Project		Non-Project		
35.8%		12.7%		6.0%		26.1%		
Manufacturer		Wholesaler		Internet	DIY	Specialist	Installer	
4.3%		1.1%		0.4%	0.6%	2.1%	10.8%	
Non-Residential								
19.3%								
New Build				Non-Residential RMI				
11.5%				7.8%				
Project		Non-Project		Project		Non-Project		
9.2%		2.3%		4.5%		3.3%		

3.7.6 Route to Market: Sanitary Taps & Mixers / Faucets
CN - ESTIMATED DISTRIBUTION FOR SANITARY TAPS & MIXERS / FAUCETS: 2024

Manufacturer / Importer	Manufacturers / Importers					
	100.0%					
Wholesale Level	73.0%					
	Wholesalers / Merchants					
	73.0%					
Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	1.9%	5.0%	5.0%	24.0%	3.5%	6.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	6.9%		29.0%		9.5%	

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

3.7.7 Route to Market - Consolidated View : Sanitary Taps and Mixers / Faucets

CN - ESTIMATED DISTRIBUTION FOR SANITARY TAPS AND MIXERS / FAUCETS : 2019

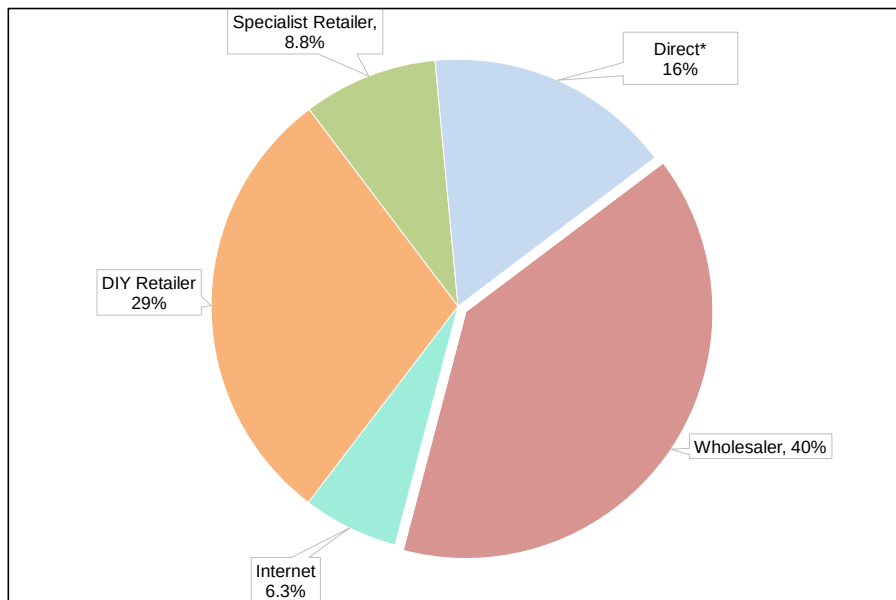
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
39.4%	6.3%	29.3%	8.8%	16.2%	41.8%	13.7%	38.9%	5.6%
100%					55.5%		44.5%	

3.7.8 Route to Market - Consolidated View : Sanitary Taps and Mixers / Faucets

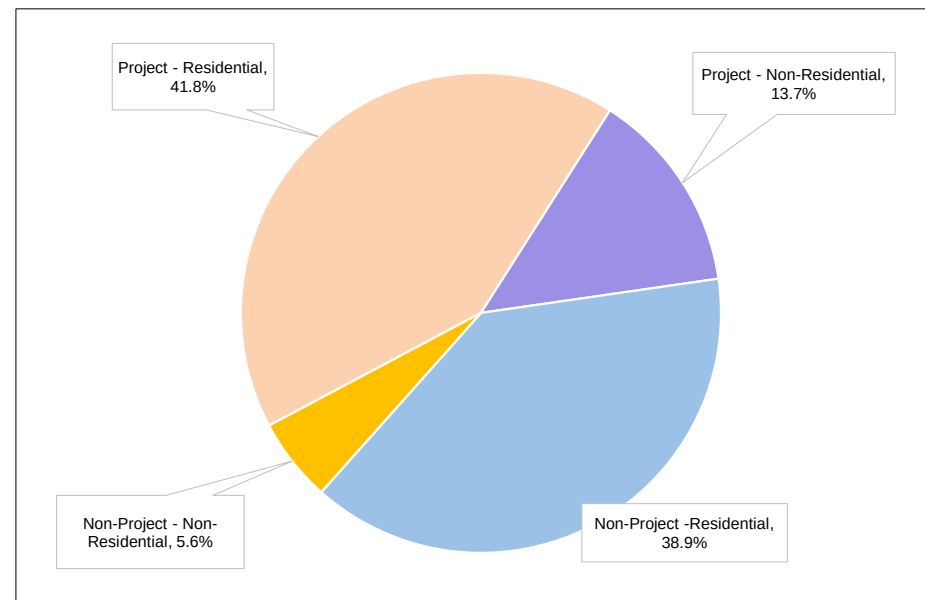
CN - ESTIMATED DISTRIBUTION FOR SANITARY TAPS AND MIXERS / FAUCETS : 2024

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
38.0%	6.9%	29.0%	9.5%	16.6%	36.9%	15.9%	39.4%	7.8%
100%					52.8%		47.2%	

CN - SALES OF SANITARY TAPS AND MIXERS / FAUCETS BY DISTRIBUTION CHANNEL: 2019



CN - SALES OF SANITARY TAPS AND MIXERS / FAUCETS BY POINT OF DECISION AND END-USE SECTOR: 2019



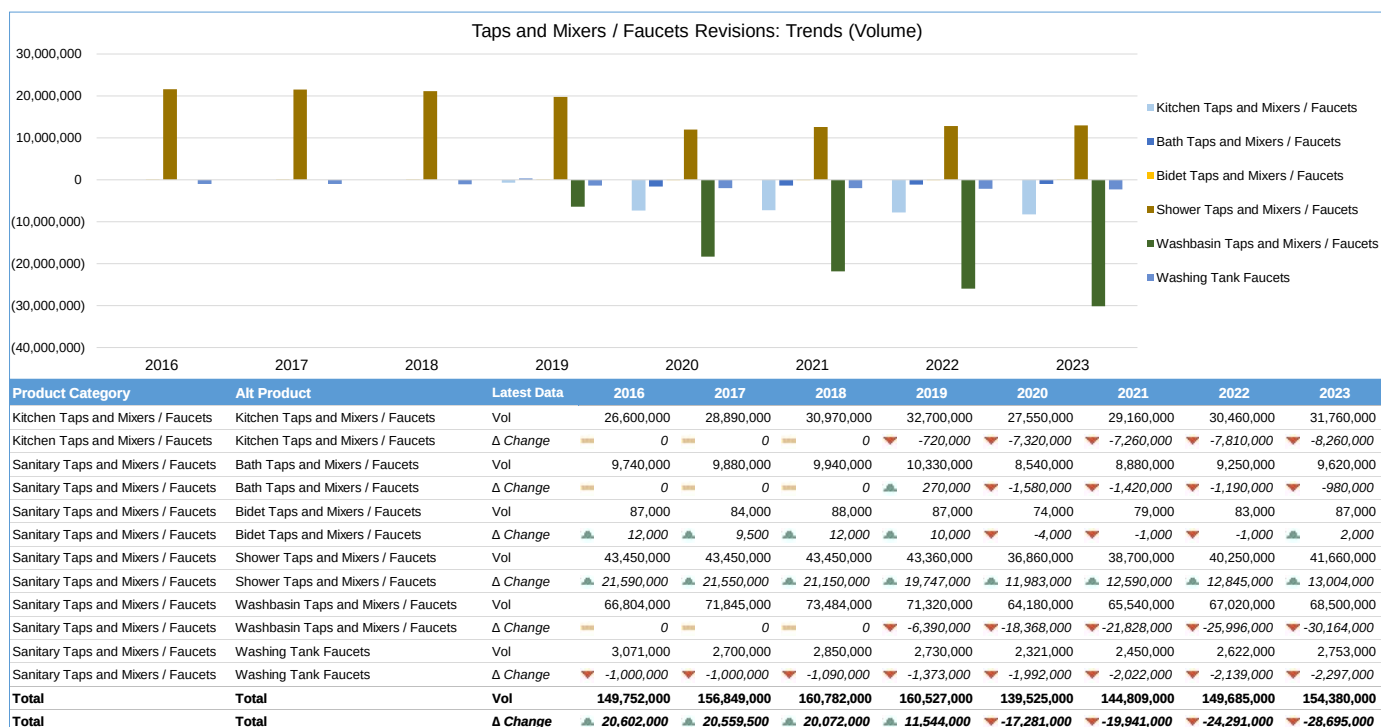
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

3.8 Revisions in the Chinese Taps and Mixers / Faucets Market (July 2020)

CN - REVISIONS IN THE CHINESE TAPS AND MIXERS / FAUCETS MARKET

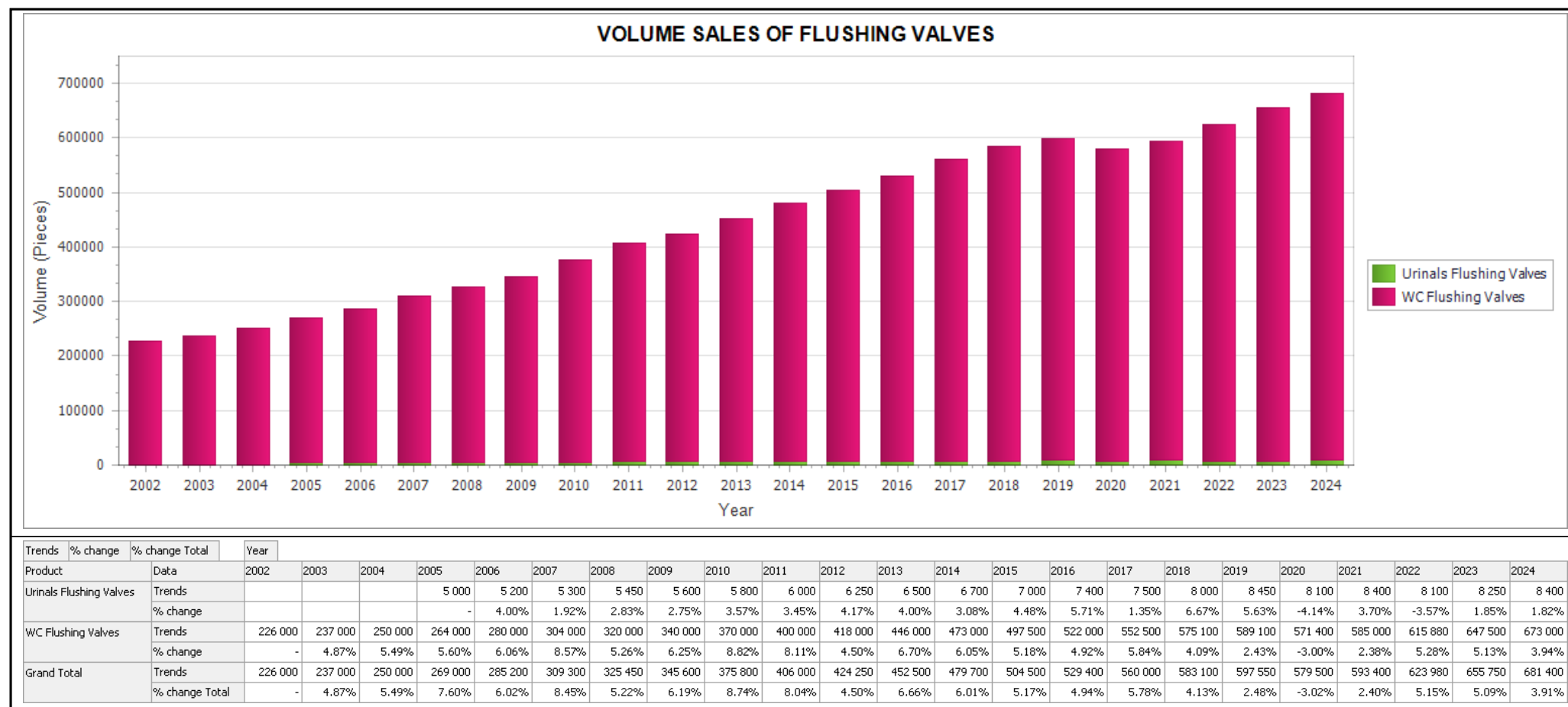
Data Comparison: July 2019



The shower faucet number has been revised up historically. There has been a rapid rise in completed family homes in China, and the majority will have a wet room with no shower pan. These wet rooms had been underestimated and hence the shower faucet volume sales were adjusted upwards.

3.9.1 Historical Trends and Forecasts : Flushing Valves

CN - SUMMARY OF TRENDS IN THE FLUSHING VALVE MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

CHINA

SHOWER WALL

THE EAST AND SOUTH EAST ASIAN BATHROOM PRODUCT MARKETS



Market Characteristics and Product Mix

Today, the Chinese shower wall market is evolving from the 'traditional shower enclosure and shower cubicle/tray' combination to more sophisticated showering solutions, such as using shower enclosures in combination with a tiled shower area ('wet shower rooms' or 'walk-in shower') since the latter is much easier to clean and provides floor level access. A small market for shower enclosures with shower trays still exists for older consumers, or designed bathrooms. Shower cubicles, today, are considered to be 'obsolete' and are increasingly replaced by shower enclosures in economy housing projects, or with hydrotherapy cubicles for the high-end segment of the market. Bath screens are not considered a necessity and the market is also hampered by the increasing popularity of showers over baths in China.

Product development has evolved from the previous simple, and functional, shower enclosures to safe, easy to clean and aesthetic products, in line with the improving production facilities and variety of designs available in China. Consumers are increasingly interested in the type of glass used, especially in regard to safety and easiness for cleaning, often demanding a combination of tempered, laminated glass and self-cleaning glass.

The design of shower enclosures also tends to be artistic and high-tech to appeal to the trend of full bathroom renovations. The transparency of the glass, strength of metal frames and fittings, shape and door types of shower enclosures are also considered by consumers when purchasing. Demand for shower enclosures is becoming increasingly more personalized, and manufacturers tend to provide more customized products and services to the market. The frame and fitting materials tend to be anti-corrosion, durable and aesthetic; hence, thicker, strong and smooth frames and fitting materials are preferred in the market. Recently, titanium-aluminum alloys and stainless-steel frames started to gain popularity with the upgrading of bathrooms and improvement of income levels. Frames and other components, such as handles and fittings, are also more artistic, colorful and have high-quality metallic touches. Silver is the traditional color of frames, while gold, rose gold, black, pink, red and yellow frames have become increasingly more popular in recent years due to a growing demand for luxury and personalized bathroom decoration in the market. The ember-yellow glass is considered luxury and stylish for the high-end market.

Plastic shower wall products are relatively rare since these are not considered to be durable, are difficult to clean and not attractive and, thus, are increasingly replaced by glass shower wall products.

Recent Market Trends

The Chinese shower wall market maintained its growth cycle in 2019 because of the growing demands from the fully renovated housing sectors, despite the

decrease in new rough cast / shell housing completions, increased home redecoration as personal income grew and rapid sales to the hotel renovation market. Sales of shower enclosures, the largest segment in this market, grew by approximately 9% which offset the segment size contractions of between 15 and 17% for bath screens and hydrotherapy and conventional shower cubicles.

Shower enclosures sold without trays, such as the shower enclosures used in wet shower rooms, continued to be the main products sold on the market, while shower cubicles and bath screens continued to be disfavored by many consumers. The overall shower enclosure market continued to move towards more sophisticated, luxury and aesthetics products, such as ones inlaid with gold, copper or diamond, carved patterns and glass fitted with pictures and patterns.

Round ('Japanese fan shape') and sliding doors remained the most popular shape because these require less space compared to other types. 'Diamond', or five-angled, shaped models have recently gained popularity as they are considered to be more personalized and modern. Shower enclosures using hinged doors are mainly found in hotels, although the installation of these has recently started to increase in some high-end housing with a large bathroom space. The folding and pivoting door models are seldom used in China.

The demand for the bath screens and the shower cubicles is expected to dwindle and these products are expected to become obsolete.

Prices and Market Values

The average MSP of shower wall products in local currency increased in 2019, with the price of shower enclosures growing at a slightly higher pace than shower cubicles and bath screens. Increases in the price of raw materials, components, glass, packaging, labor and transportation as attributed to the overall MSP rise. Environmental regulations were also tightened and inspection on manufacturers became more frequent in 2019, which made many shower wall products manufacturers and raw material and components suppliers increase prices to accommodate the disruption.

New Housing

In 2019, the sales of shower enclosures to the new housing sector continued to increase mainly because of the rapid growth in the fully renovated housing sector. More and more regional governments such as Beijing, Shanghai, Jiangsu, Shandong, Zhejiang, Guangdong, and even Anhui promoted the construction of fully renovated housings, not only in tier 1 cities but also in more tier 3 to tier 4 cities.

The sales of shower cubicles and bath screens continued to decline due to less demand in the market.

With unlikely changes of the controlling policies on the property market, and continuous penetration of fully renovated housings into more regions such as tier 3 to tier 4 cities, it is expected that the sales of shower enclosures will continue to grow, and the sales of shower cubicles and bath screens will continue to decline quickly in the next few years post-pandemic.

Housing RMI

The installation of shower enclosures in housing RMI is predominantly seen in the tier 1 cities. In addition, the number of shower wall products reaching the end of their lifespan, often installed 15 to 20 years ago, also resulted in higher demand for shower wall products. Indeed, there is also an increasing demand for like-for-like replacement of particularly shower enclosures in existing bathrooms.

Shower wall products sold for housing RMI purposes tend to be to the middle-to-upper price segment, rather than the economy price segment, due to rising incomes, the general need for improving living conditions and the growing acceptance of wet shower rooms.

The installation of shower cubicles and bath screens in housing RMI continued to be small and is declining due to the obsolete nature of the products and declining installation of baths in residential sectors.

The like-for-like replacement and displacement of baths to shower enclosures in wet rooms will continue to grow and influence the RMI market. The controlling policies on property market are unlikely to change, which will make the sales of shower enclosures for housing RMI purposes to shift from the tier 1 cities to tier 3 to tier 4 cities and even rural areas where there are no controlling policies in place.

Non-Housing

In 2019, the number of new build hotels continued to decline, however, the renovation and refurbishment of existing hotels as many low star-rating hotels were upgraded to high star-rating hotels due to the boom of tourism industry, and expansion of large hotel chains through merger and acquisition. The majority of the star-rating hotels are equipped with shower enclosures in China. In addition, sales of shower enclosures installed in rural hospitality accommodation buildings (e.g. Nong Jia Le, Happy Farmhouse), gyms and spas also grew, with the latter the result of the growth in leisure and rural travelling.

BRG expects that sales of shower enclosures will continue to grow in the non-housing segments, such as for hotels, rural holiday villages, and guesthouses, post COVID-19 crisis due to the increases in leisure time and income of Chinese consumers in the next few years.

Supply Structure

The shower wall market remains very fragmented in China with over 2,000 local and foreign owned manufacturers, mostly located in Zhongshan, Xiaoshan and Foshan, where Zhongshan is where most of the medium to high end product manufactures are concentrated, while Xiaoshan has a large number of small manufacturers producing economy to low end products. There are many large manufacturers in Foshan, which focus on exports. The majority of the shower wall products are supplied by a large number of local suppliers, while several international companies such as KOHLER, HÜPPE, AMERICAN STANDARD, and ROCA GROUP supply to the medium to upper end market.

In 2019, the shares in the shower enclosure market changed slightly in comparison to last year, with the top 10 large suppliers led by LENS, JOMOO, ARROW, SALLY, DELICACY, FRAE, KOHLER, FAENZA, ORANS, ROSERY and HEGII. SENLIA was replaced by HEGII and is now out of the top 10 list since its market share decreased as the company changed focus from the domestic market to exports.

There are also a large export market and OEM production for shower enclosures in China, mainly from specialized suppliers such as ROSERY and SALLY.

The activities of major shower wall suppliers in 2019 are:

DELICACY started the production of intelligent shower enclosures in Zhongshan in 2019 with an annual capacity of 1 million pieces.

ANNWA invested RMB 550 million to set up a production facility in Gaoming, Foshan to produce ceramic sanitary ware, shower enclosures and bathroom furniture, which is expected to be completed by 2021.

DONGPENG started the production of its intelligent home creative industrial park in Yongchuan, Chongqing in November 2019. The industrial park has 10 tile lines, 3 lines of ceramic sanitary wares (including intelligent WCs), 1 line of bathroom cabinets, 1 line of shower enclosures, 1 line of faucets, and a R&D center and logistic center.

Forecasts

Given the exceptional circumstances related to the spread of the COVID-19 virus and measures in place for its containment, forecasts for 2020 and beyond are based on the latest information available at the time of publication. BRG has endeavored to factor in the latest developments – see Macroeconomic data in Table A.3 and supporting summary table at the end of the Background section.

The COVID-19 pandemic has widely affected the sales of all shower wall products in China due to the cessation of both production and construction in the first quarter. With the resumption of production and construction/RMI activities, it is estimated that the market will decline by under 15% in 2020 as there are still pockets of reoccurring virus infections particularly in the bigger cities curtailing

building starts.

The market is estimated to pick up however in 2021 due to the increased penetration of fully decorated new housings, the construction of government housing, and the expansion of the market from tier 1 and tier 2 cities to tier 3 to tier 4 cities and rural areas, which will continue to boost the sales through the reporting period.

The outlook after the pandemic of the Chinese shower wall product market is positive. The major drivers would be further growing urbanization, growing demands in Tier 2 to Tier 4 cities and rural markets, the government's program on West Development and construction of New Countryside. The resumption of the government's controlling policies on the property market may affect sales of shower enclosures in the new housing sector; however, the growing number of fully renovated dwelling constructions should offset the slow growth in new dwelling completions in the next few years.

The awareness of dry bathrooms and demands for wet rooms will continue to grow in both large cities and medium to small cities in China, which will continue to encourage the installation of shower enclosures.

The rising number of young consumers and the middle class will continue to encourage the sales of high quality and innovative products for bathroom upgrading. There will be more and more like-for-like replacement and displacement of bathrooms for shower enclosures in the next few years. These will continue to boost the growth of sales and installation in the housing RMI segment.

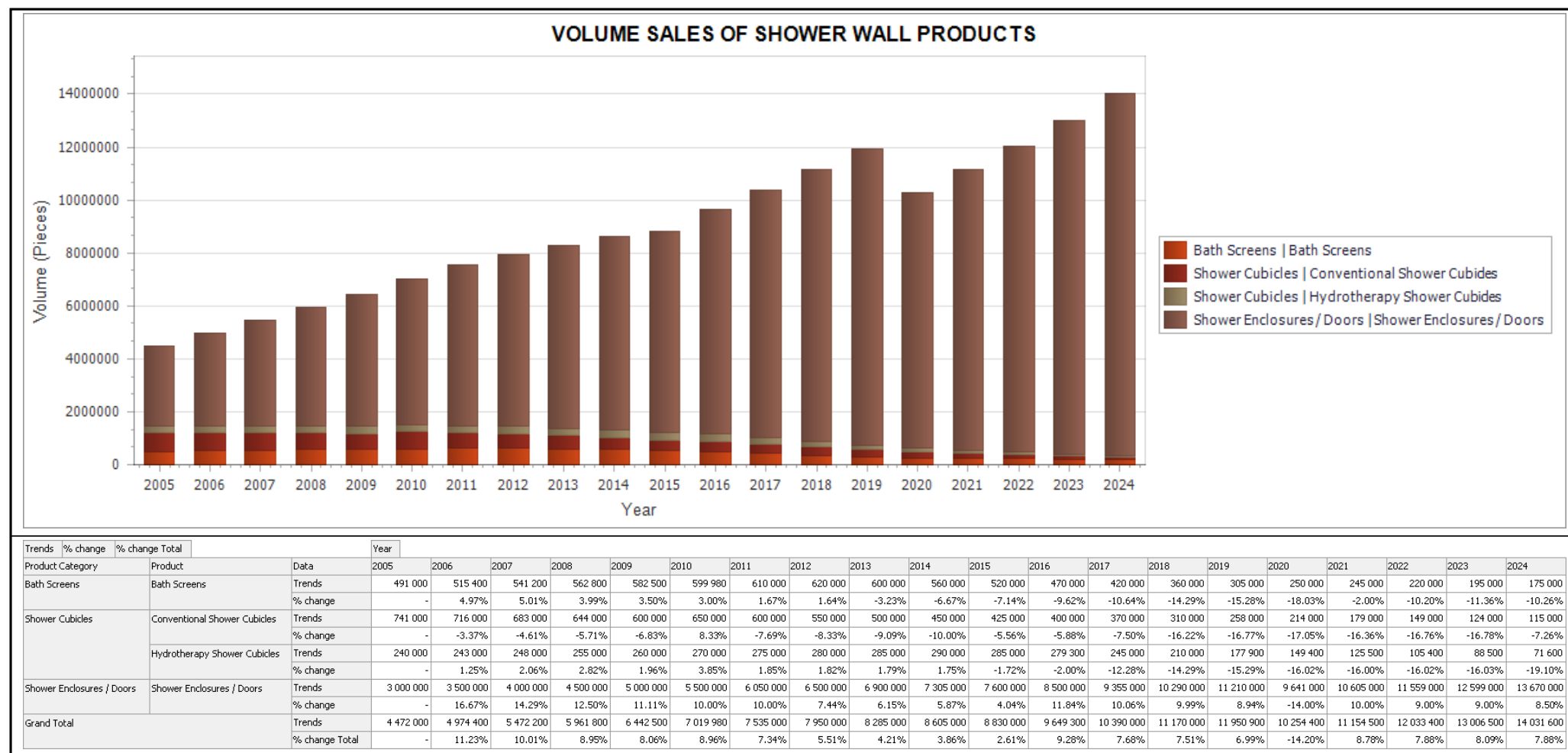
The sales and installation of shower enclosures in the non-housing segments, mainly hotels and other accommodation segments will continue to grow with the booming of tourism and more leisure time of young consumers.

The sales in certain regions such as Beijing, Hebei, Shanghai, Hangzhou, and Guangdong, etc, will grow quickly with the construction of some large products such as Beijing New Administration Centre, Xiong'an, Beijing Winter Olympic Game Venues 2022, Shanghai Zhangjiang Tech Industrial Park, Hangzhou Asian Game Venue 2022, and Free Trade Zones.

However, the environmental regulations will continue to be tight, which will affect the production and supplies of shower enclosures, raw materials and components. Other costs will also continue to be high, such as labor, packaging and transportation. Thus, the price and sales of shower enclosures will continue to be affected over the next few years.

4.2 Historical Trends and Forecasts : Shower Wall

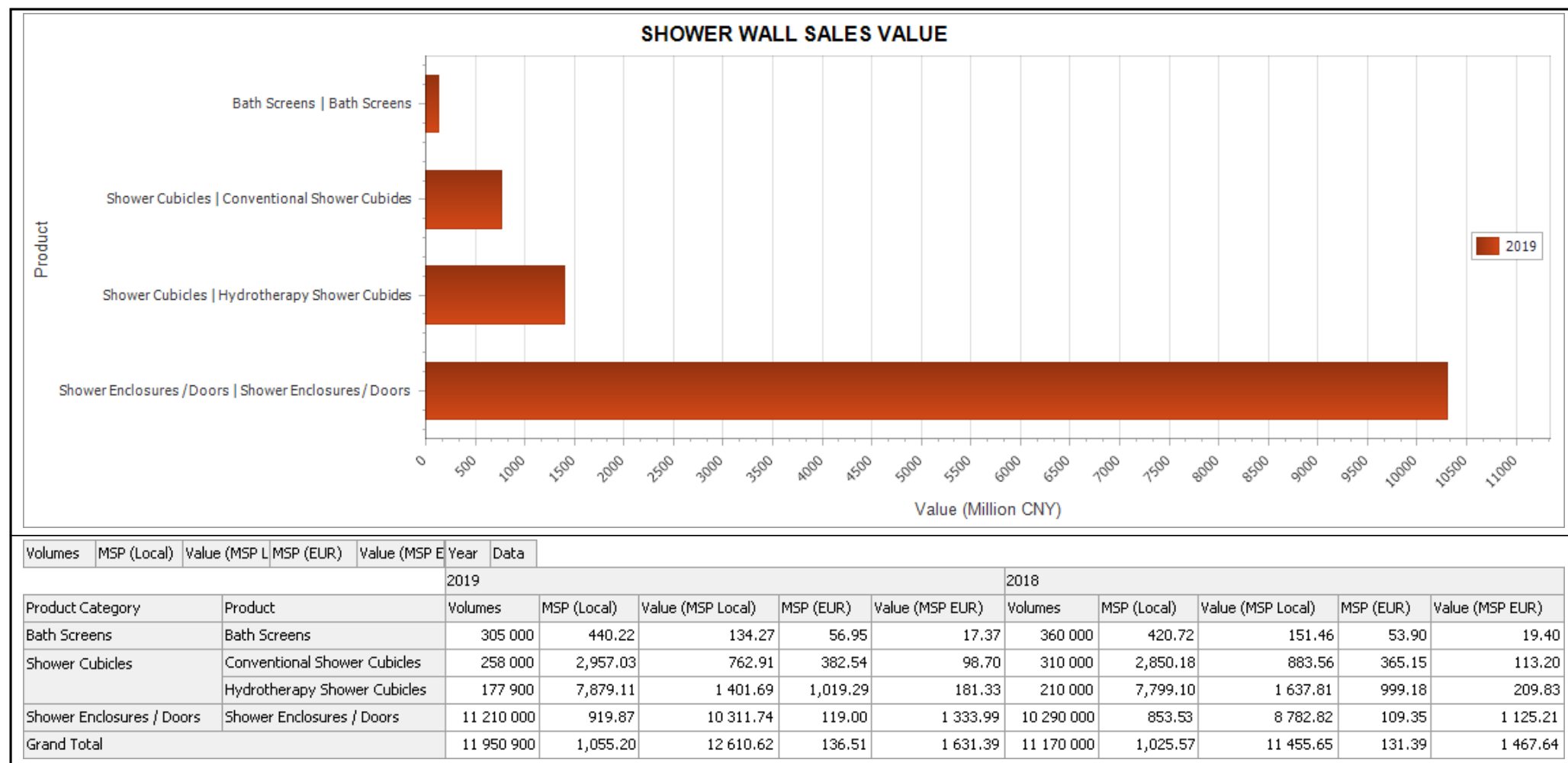
CN - SUMMARY OF TRENDS IN THE SHOWER WALL MARKET : HISTORICAL AND FORECASTS



Source: BRG based on industry interviews.

4.3.1 Prices and Market Values : Shower Wall

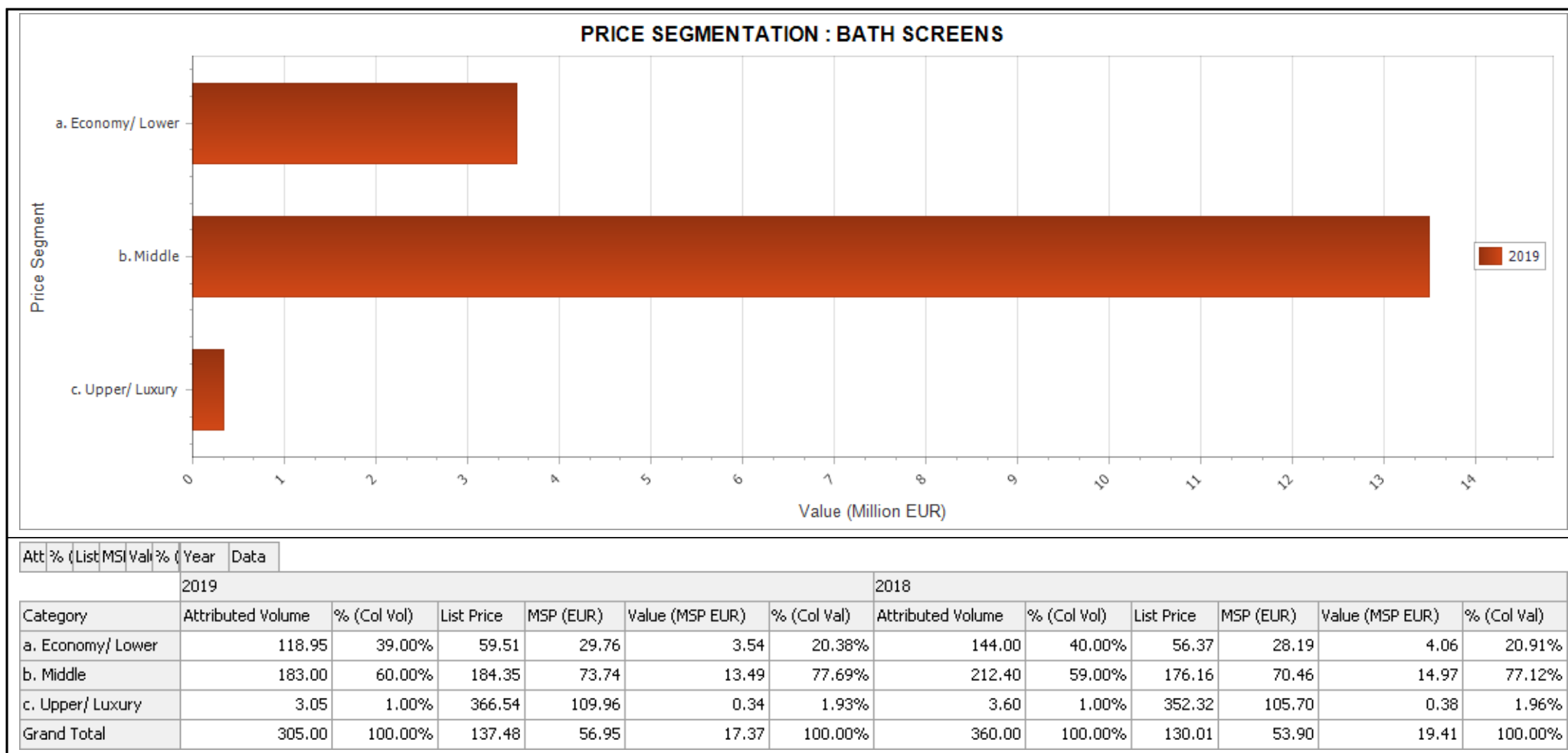
CN - ESTIMATED VALUES OF THE SHOWER WALL MARKET



Source: BRG based on industry interviews. Local Prices Shown in CNY. Values Shown in Millions (at MSP = Manufacturers' Selling Price).

4.3.2 Price Segmentation : Bath Screens

CN - ESTIMATED PRICE SEGMENTATION FOR BATH SCREENS

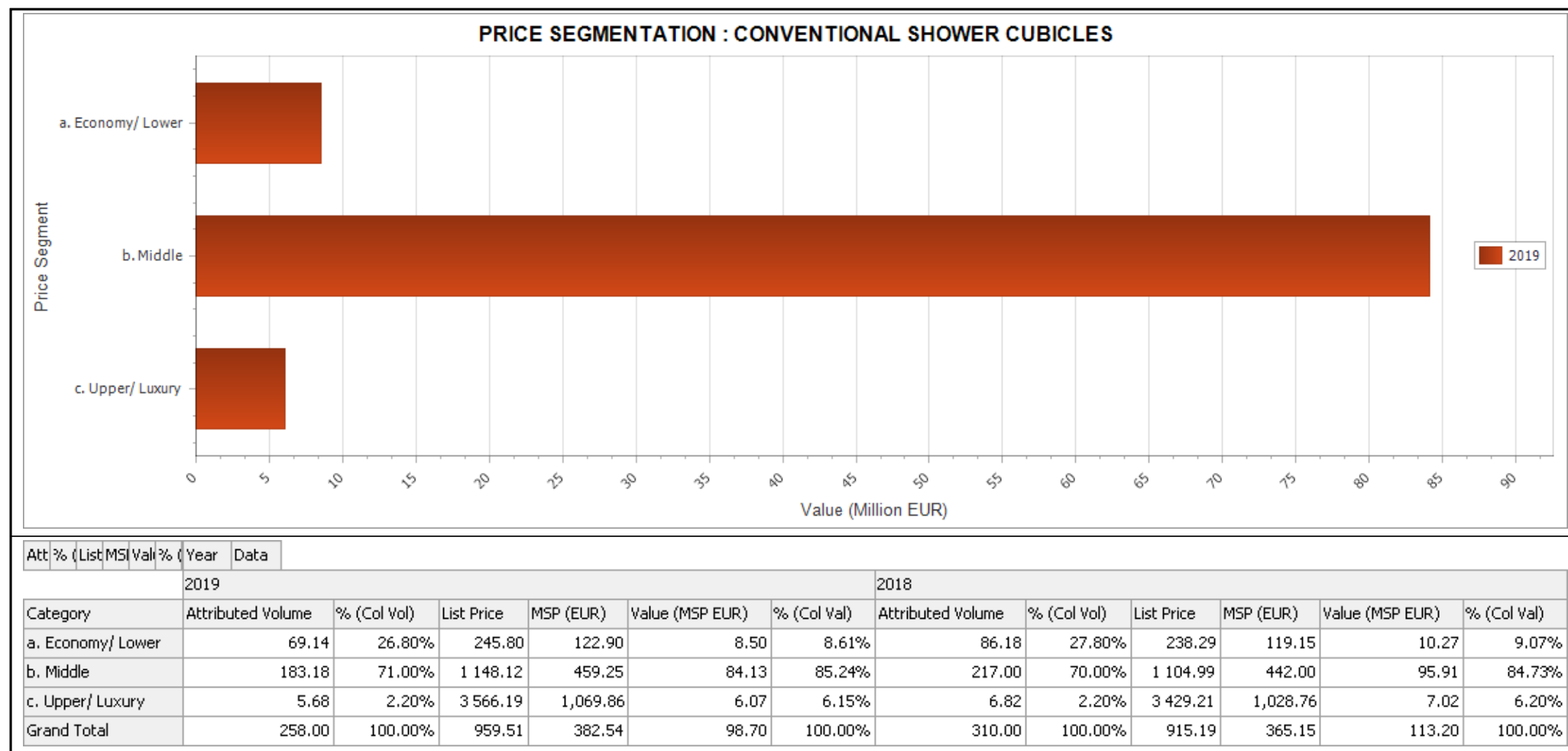


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€60, Middle €60-€100, Upper/ Luxury >€100

4.3.3 Price Segmentation : Conventional Shower Cubicles

CN - ESTIMATED PRICE SEGMENTATION FOR CONVENTIONAL SHOWER CUBICLES

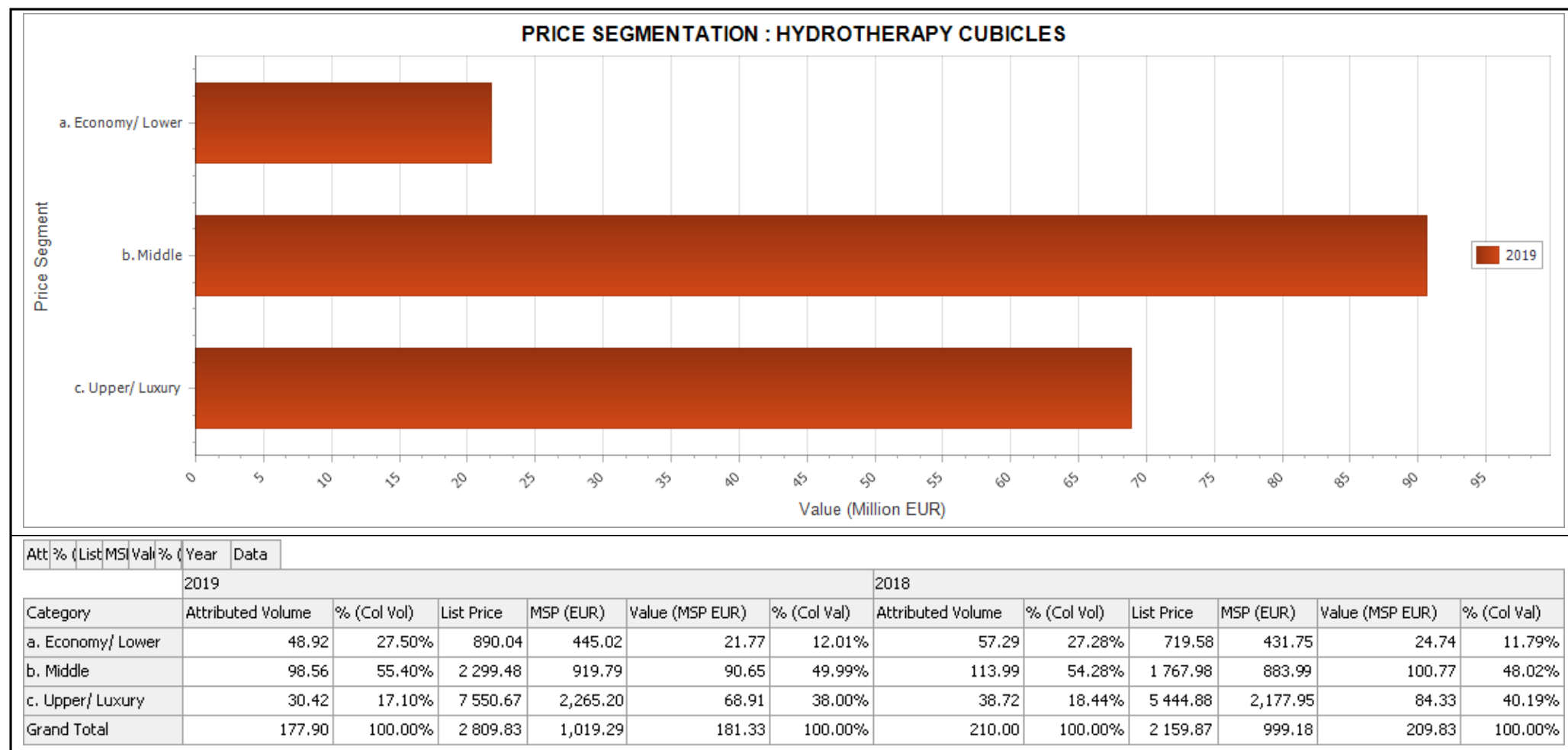


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€200, Middle €200-€750, Upper/ Luxury >€750

4.3.4 Price Segmentation : Hydrotherapy Shower Cubicles

CN - ESTIMATED PRICE SEGMENTATION FOR HYDROTHERAPY CUBICLES

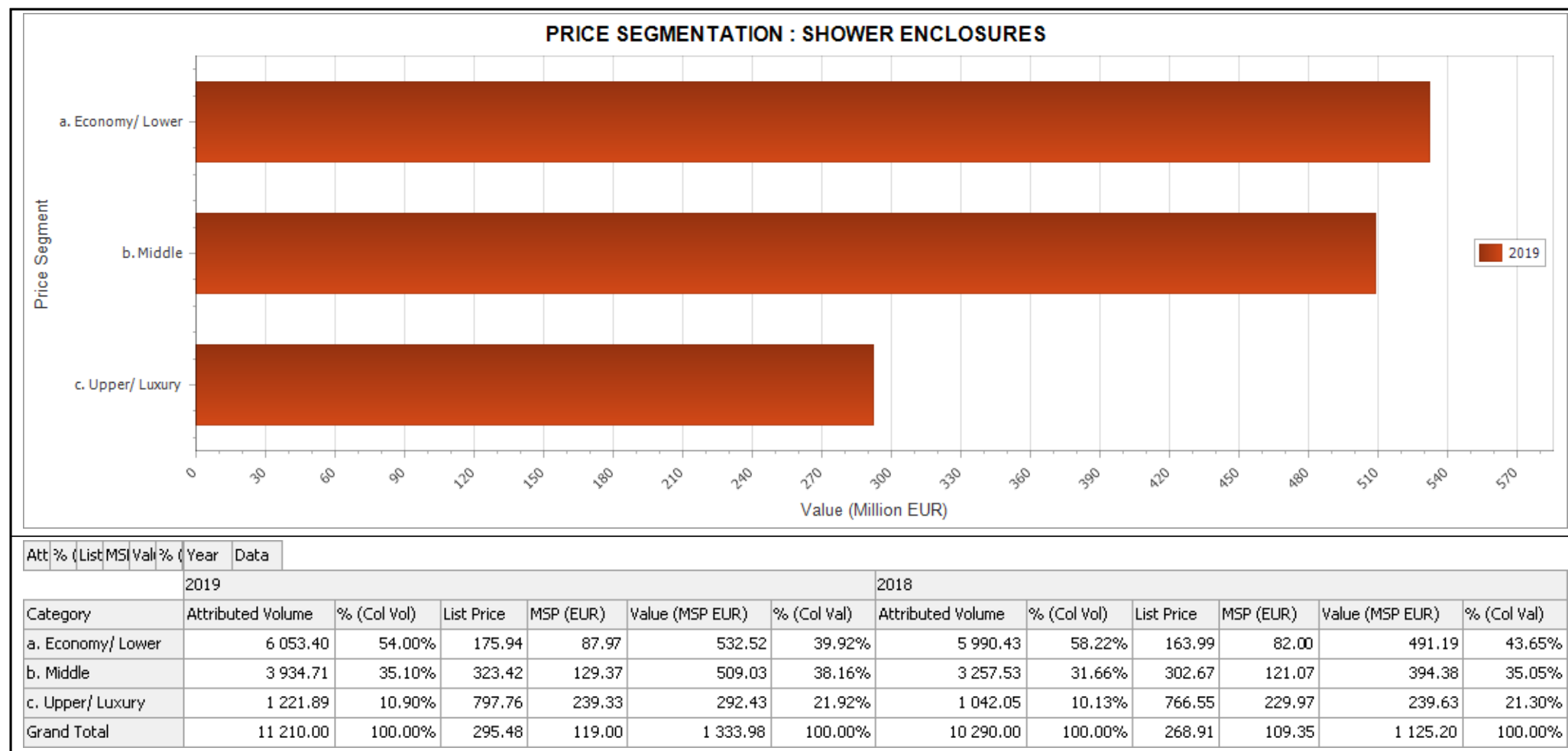


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€500, Middle €500-€1500, Upper/ Luxury >€1500

4.3.5 Price Segmentation : Shower Enclosures

CN - ESTIMATED PRICE SEGMENTATION FOR SHOWER ENCLOSURES

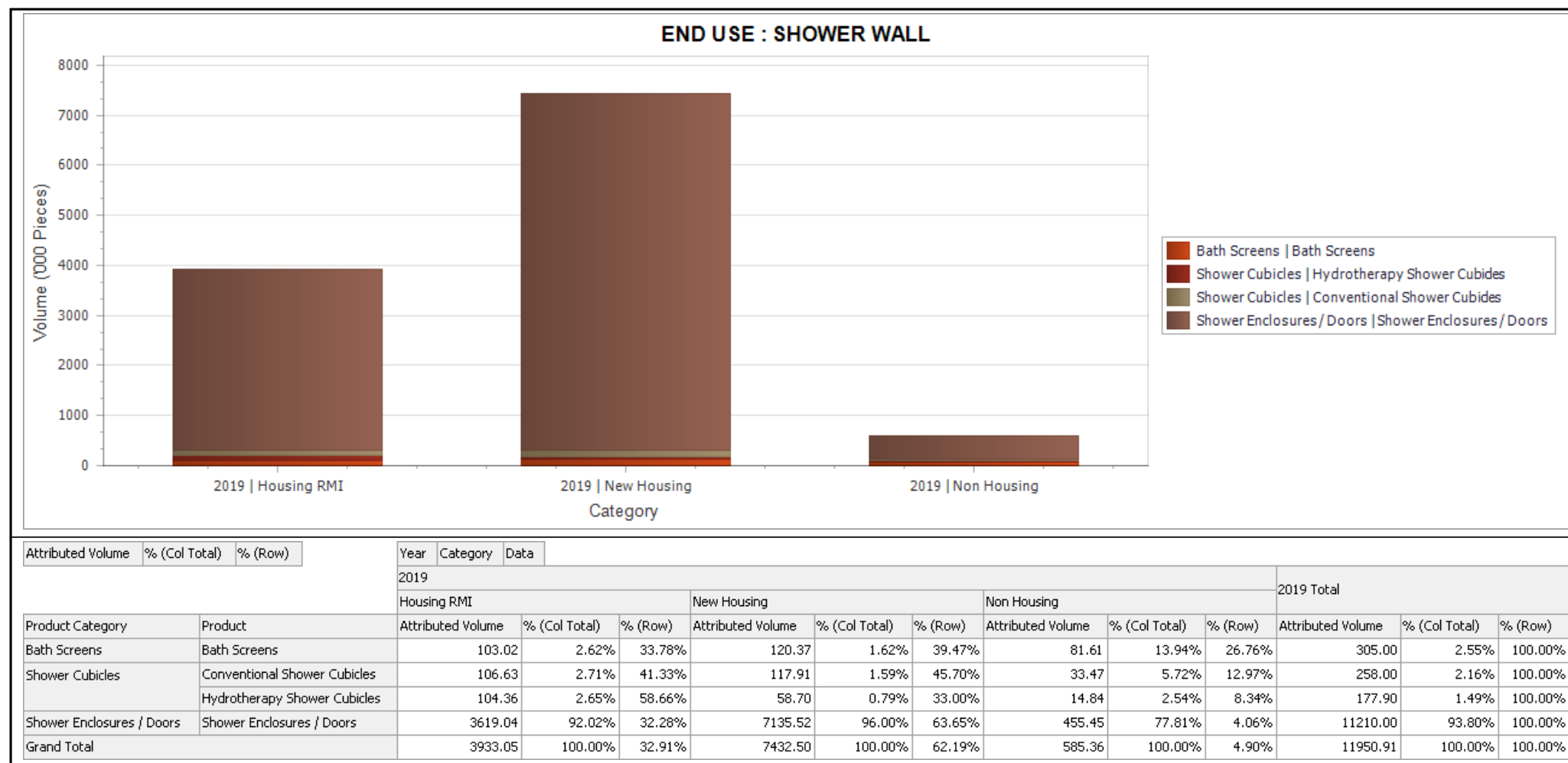


Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

Volumes Shown in Thousands and Values Shown in Millions (at MSP = Manufacturers' Selling Price). Economy/ Lower <€100, Middle €100-€200, Upper/ Luxury >€200

4.4.1 End Use : Base Year

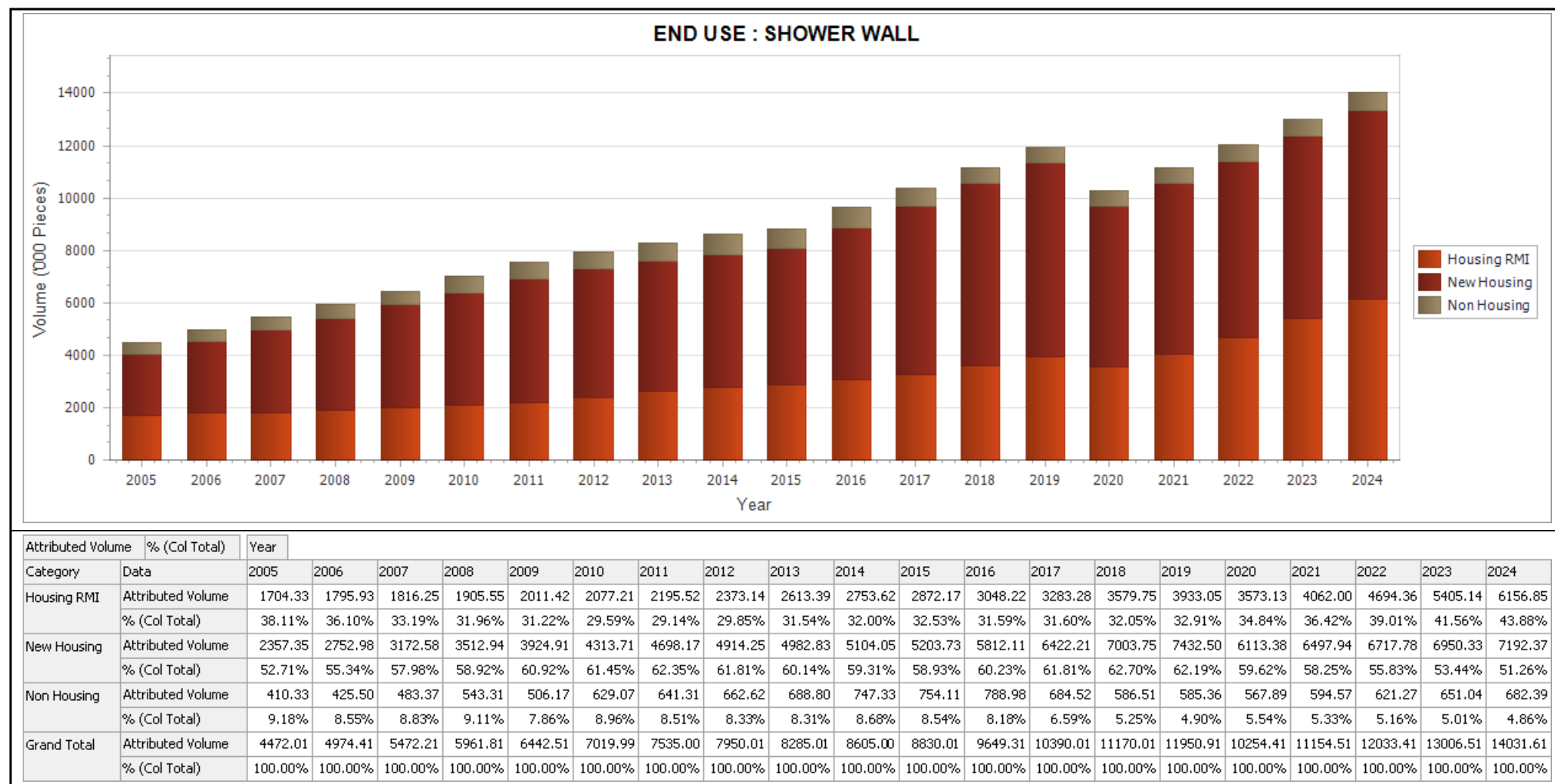
CN - SALES OF SHOWER WALL BY TYPE AND END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.4.2 End Use : Historical and Forecasts

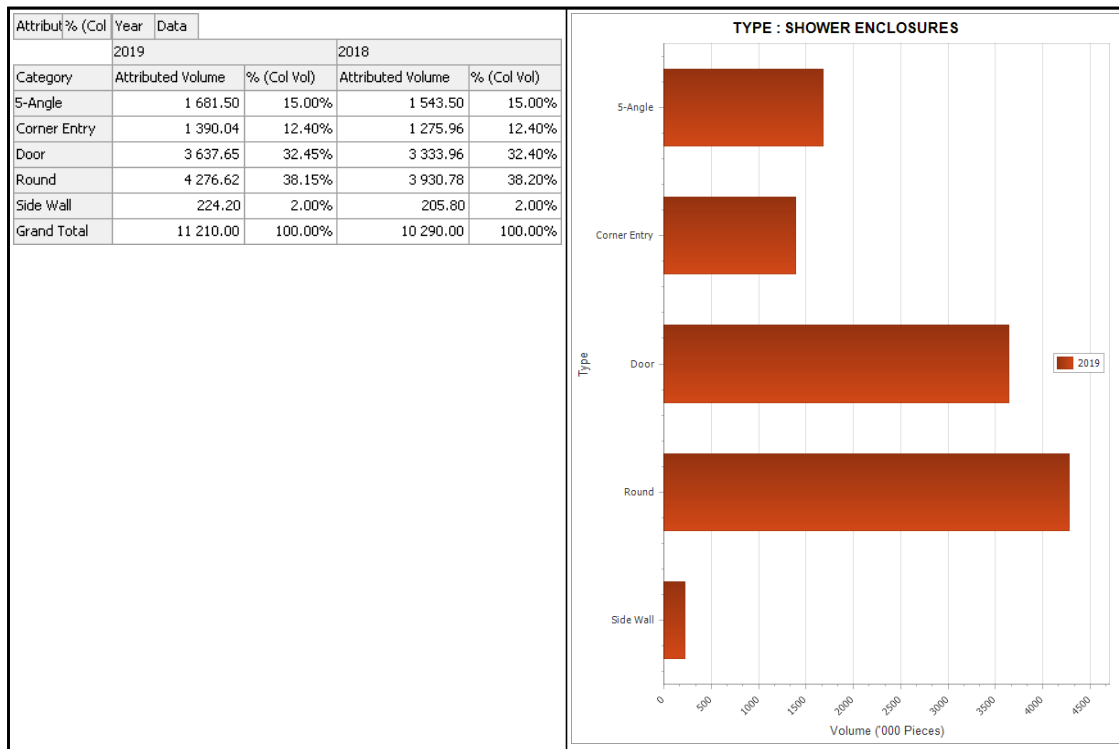
CN - SALES OF SHOWER WALL PRODUCTS BY END USE SEGMENT ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.4.3 Sales by Type : Shower Enclosures

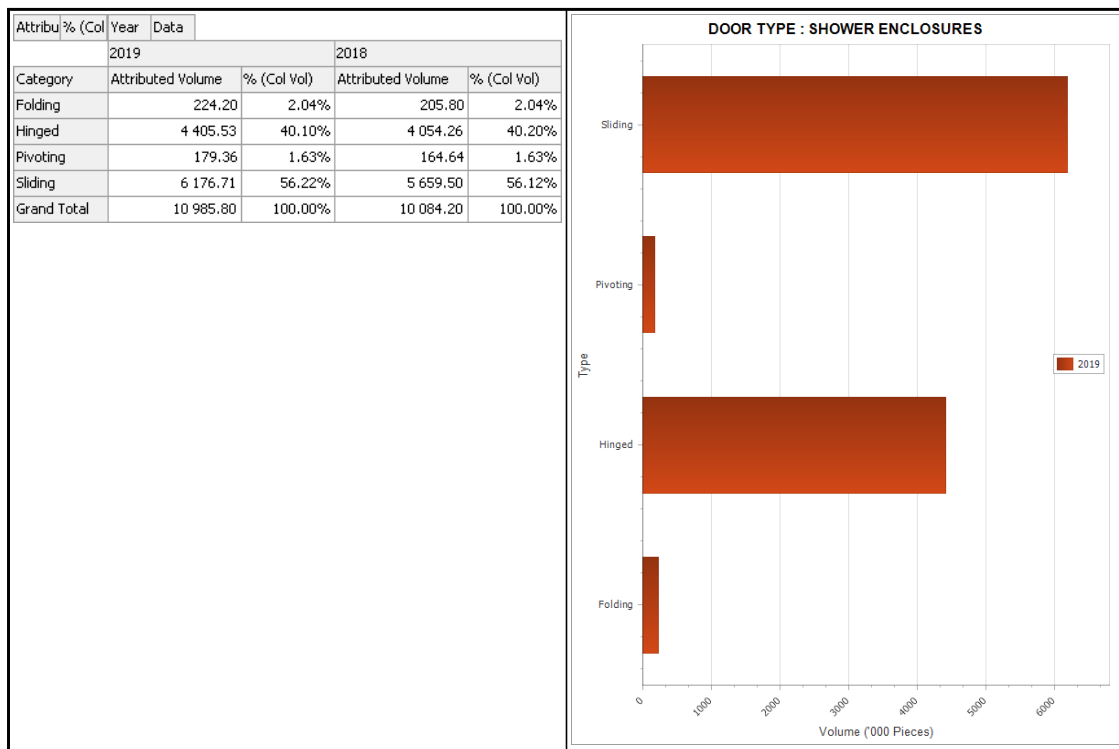
CN - SHOWER ENCLOSURE SALES BY TYPE ('000 PIECES)



Source: BRG based on industry interviews

4.4.4 Sales by Type of Door : Shower Enclosures

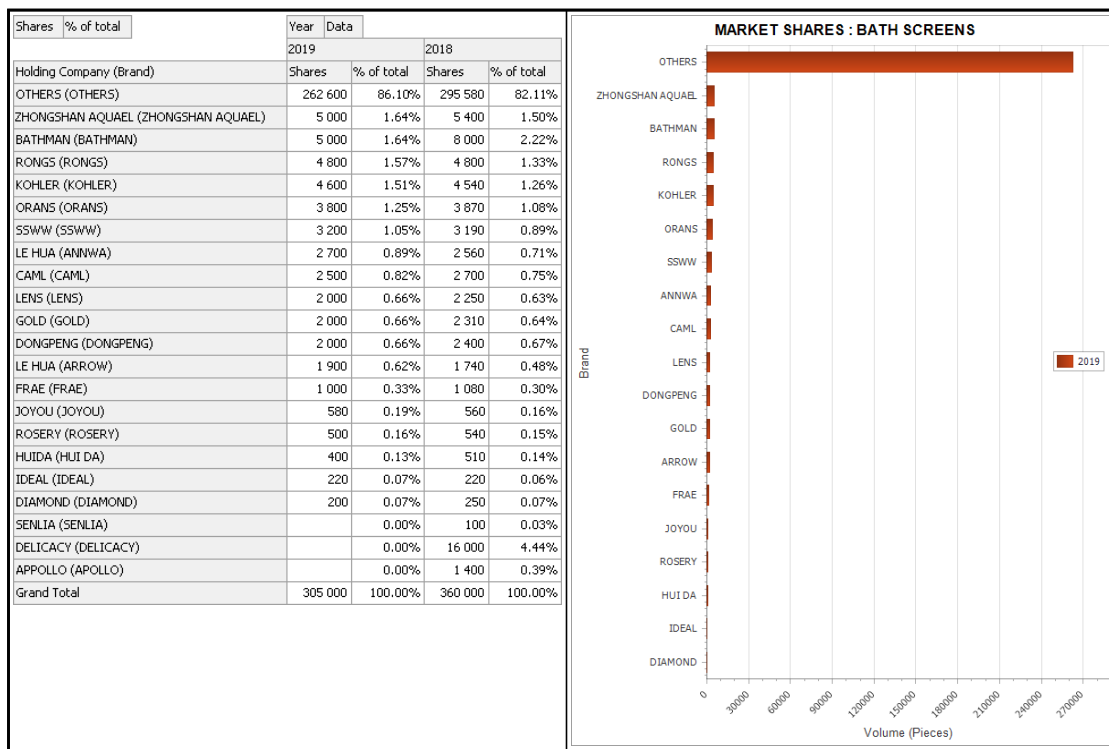
CN - SHOWER ENCLOSURE SALES BY DOOR TYPE ('000 PIECES)



Source: BRG based on industry interviews. Totals may show slight variance compared to market data due to rounding.

4.5.1 Market Shares : Bath Screens

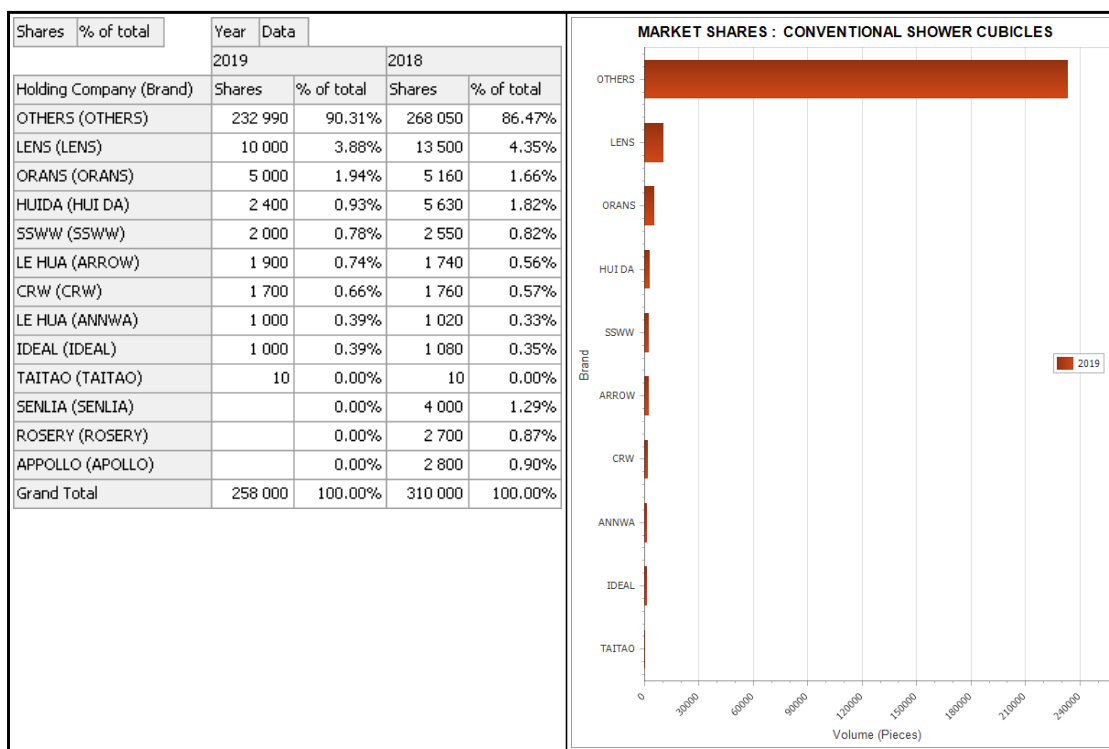
CN - ESTIMATED MARKET SHARES FOR BATH SCREENS



Source: BRG based on industry interviews.

4.5.2 Market Shares : Conventional Shower Cubicles

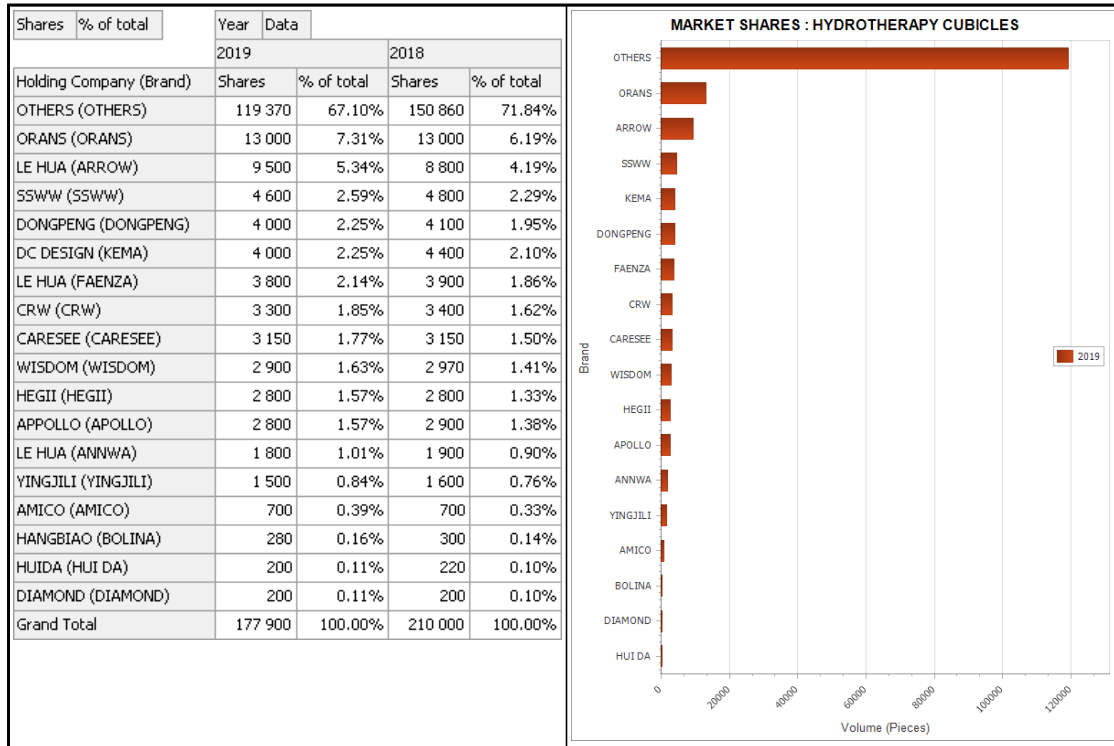
CN - ESTIMATED MARKET SHARES FOR CONVENTIONAL SHOWER CUBICLES



Source: BRG based on industry interviews.

4.5.3 Market Shares : Hydrotherapy Shower Cubicles

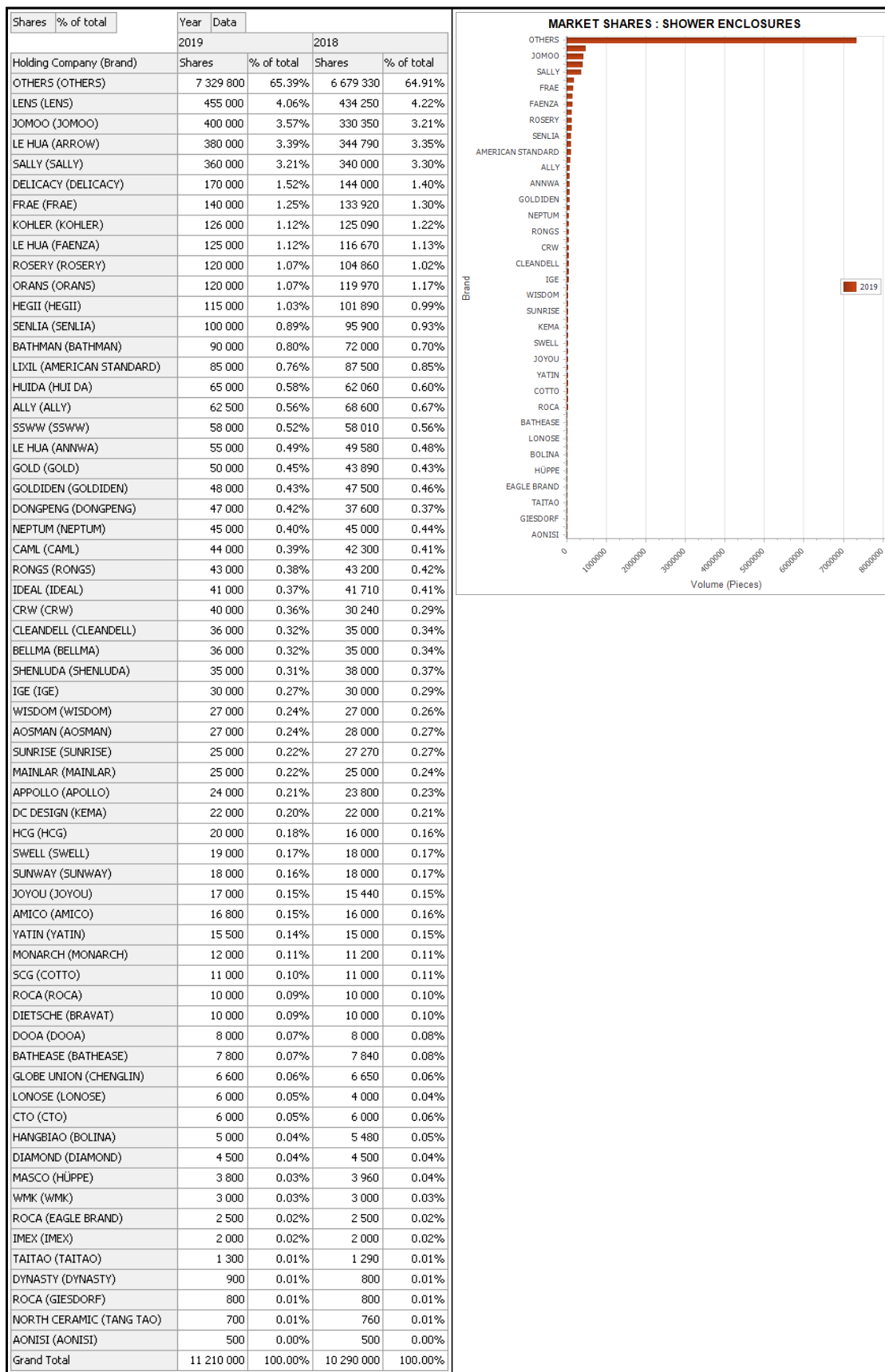
CN - ESTIMATED MARKET SHARES FOR HYDROTHERAPY CUBICLES



Source: BRG based on industry interviews.

4.5.4 Market Shares : Shower Enclosures

CN - ESTIMATED MARKET SHARES FOR SHOWER ENCLOSURES



Source: BRG based on industry interviews.

4.6.1 Route to Market: Shower Wall
CN - ESTIMATED DISTRIBUTION FOR SHOWER WALL: 2019

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	77.4%
	Wholesalers / Merchants
	77.4%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	1.4%	1.6%	6.8%	8.1%	3.3%	4.0%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	3.0%		14.9%		7.3%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
10.1%	5.0%	0.2%	3.0%	3.0%
Installers				
21.3%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.2%	56.7%	2.7%	11.6%	4.0%	19.9%
	Residential					
	95.1%					

4.6.2 Route to Market: Shower Wall
CN - ESTIMATED DISTRIBUTION FOR SHOWER WALL: 2024

Manufacturer / Importer	Manufacturers / Importers
	100.0%

Wholesale Level	77.0%
	Wholesalers / Merchants
	77.0%

Retail Level	Manufacturer	Wholesaler	Manufacturer	Wholesaler	Manufacturer	Wholesaler
	1.7%	1.9%	7.0%	8.3%	3.1%	3.7%
	Internet Retailers		DIY Retailers		Specialist Retailers	
	3.6%		15.3%		6.8%	

Manufacturer	Wholesaler	Internet	DIY	Specialist
10.0%	5.0%	0.5%	3.2%	2.8%
Installers				
21.5%				

End Use Level	Manufacturer	Wholesaler	Internet	DIY	Specialist	Installer
	0.5%	56.2%	2.9%	11.7%	3.8%	20.0%
	Residential					
	95.1%					

Source: BRG Based on Industry Interviews. - Totals may vary due to roundings

4.6.3 Route to Market - Consolidated View : Shower Wall

CN - ESTIMATED DISTRIBUTION FOR SHOWER WALL : 2019

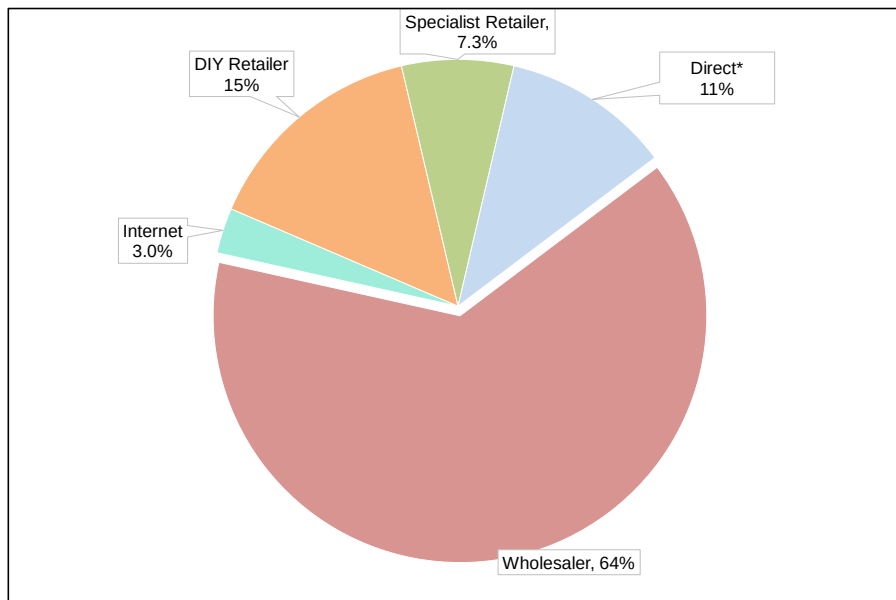
Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
63.7%	3.0%	14.9%	7.3%	11.1%	57.9%	3.8%	37.2%	1.1%
100%					61.7%		38.3%	

4.6.4 Route to Market - Consolidated View : Shower Wall

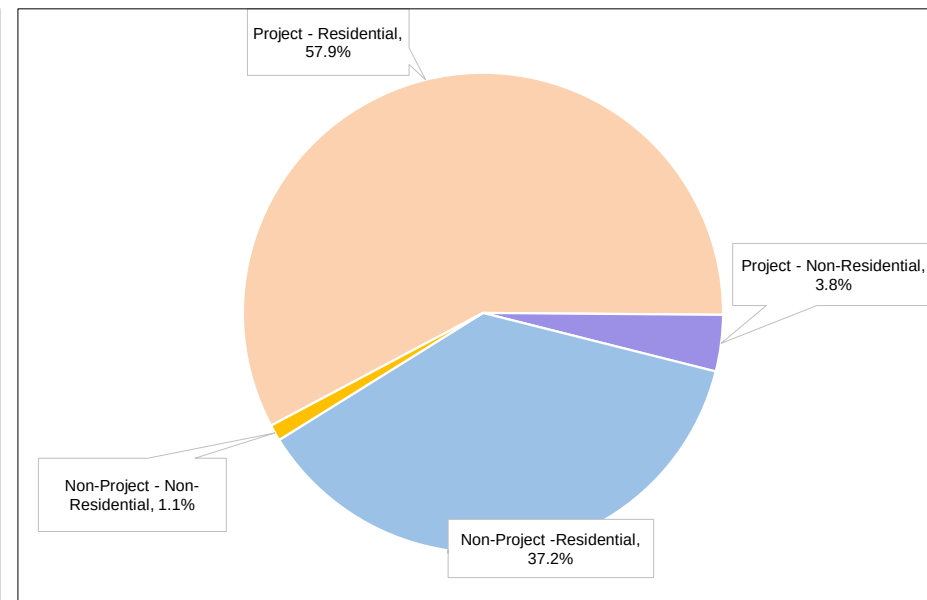
CN - ESTIMATED DISTRIBUTION FOR SHOWER WALL : 2024

Route to Market - End Use Level					Project		Non-Project	
Wholesaler	Internet	DIY Retailer	Specialist Retailer	Direct*	Residential	Non-Residential	Residential	Non-Residential
63.1%	3.6%	15.3%	6.8%	11.2%	50.6%	4.0%	44.5%	0.9%
100%					54.6%		45.4%	

CN - SALES OF SHOWER WALL BY DISTRIBUTION CHANNEL: 2019



CN - SALES OF SHOWER WALL BY POINT OF DECISION AND END-USE SECTOR: 2019



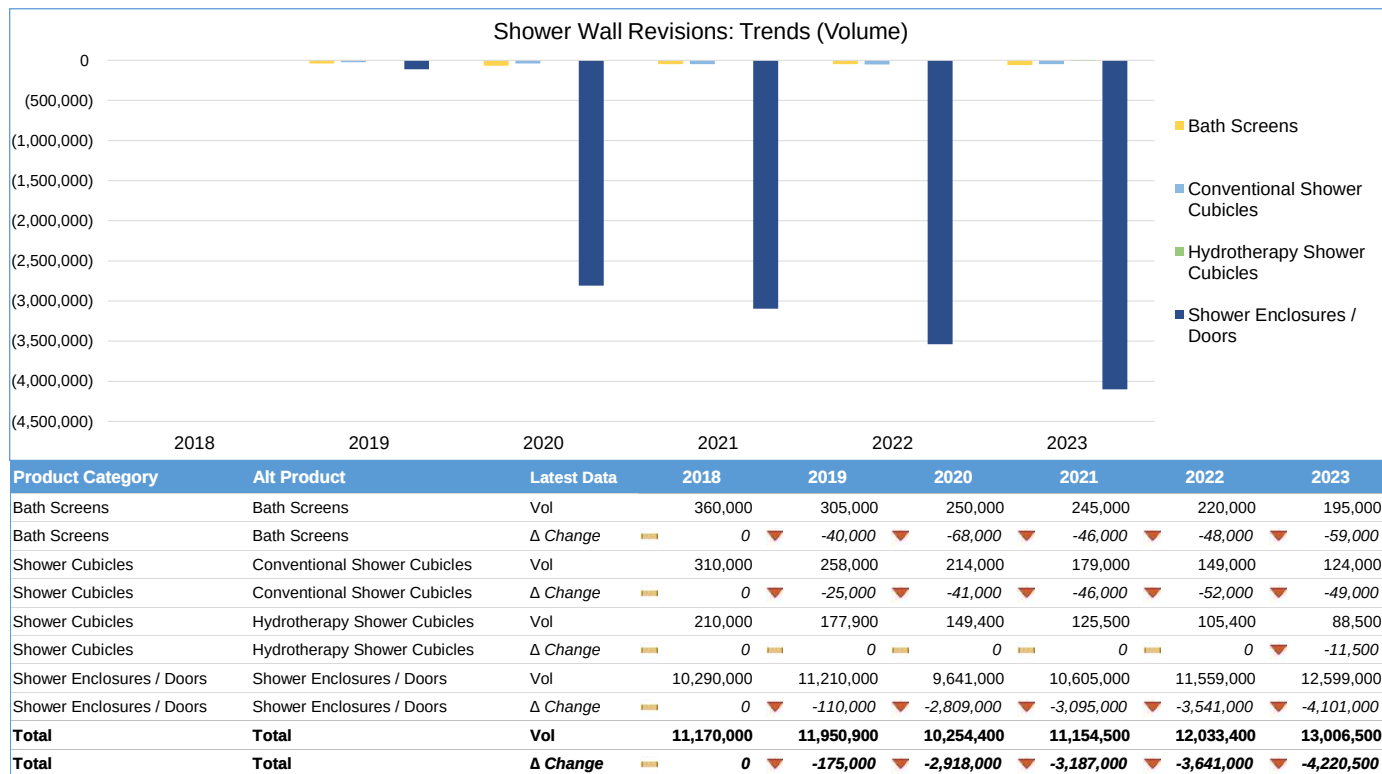
Source: BRG Based on Industry Interviews.

*Direct includes sales going directly from manufacturers to installers, contractors and private customers.

4.7 Revisions in the Chinese Shower Wall Market (July 2020)

CN - REVISIONS IN THE CHINESE SHOWER WALL MARKET

Data Comparison: July 2019



There are no historical changes for shower wall market sizes. Base year volume sales were adjusted downwards as new housing construction output in 2019 was lower than initially predicted.